

Relay Catalysis for Selective Aerobic Oxidative Esterification of Primary Alcohols with Methanol

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Supporting Information

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General Information. NMR spectra were taken with an Agilent-400 spectrometer (400 MHz for ^1H NMR, 100 MHz for ^{13}C NMR, 376 MHz for ^{19}F NMR). All reactions were carried out in 50 mL flask unless otherwise mentioned. $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was purchased from Energy Chemical; TEMPO (98%) was purchased from Shanghai Darui Fine Chemical Co., Ltd.; BiCl_3 was purchased from Alfa Aesar; DCE and MeOH were purchased from Sinopharm Chemical Reagent Co., Ltd and Shanghai Titan Scientific Co., Ltd, respectively, and used directly as received without further treatment. All the temperatures were referred to the oil baths or ice-water bath (0 °C) used. The HRMS analysis was performed with Agilent 7200B GC/Q-TOF.

TEMPO: 2,2,6,6-Tetramethylpiperidinoxy

4-NHAc-TEMPO: 4-(Acetylamino)-2,2,6,6-tetramethylpiperidinoxy

4-OMe-TEMPO: 4-Methoxy -2,2,6,6-tetramethylpiperidinoxy

4-OH-TEMPO: 4-Hydroxy)-2,2,6,6-tetramethylpiperidinoxy

ABNO: 9-Azabicyclo[3.3.1]nonane *N*-oxyl

Identification of the slowly oxidizing alcohols

To a 50 mL flask were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.06 mmol), TEMPO (0.05 mmol), BiCl_3 (0.1 mmol), **1a** (1.0 mmol), **2** (5.0 mmol), and DCE (3 mL) sequentially under air atmosphere. The reaction was then carried out at 50 °C under the atmosphere of O_2 from a balloon (about 2.5 L) for 48 h. The resulting mixture was cooled to 0 °C with an ice-water bath for 10 min, then warmed to rt. CH_2Br_2 (35 μL) was added as the internal standard. About 0.1 mL of the resulting mixture was taken and filtrated through a disposable syringe filter (PTFE, 13 mm, 0.45 μm) eluted with CDCl_3 (0.5 mL), and analyzed via ^1H NMR. Then the resulting mixture was collected and filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL), concentrated, and analyzed via ^1H NMR with CH_2Br_2 (35 μL) as the internal standard. According to the comparison of two ^1H NMR spectra, the yields of products that have low boiling point were calculated.

Table S1. Identification of the slowly oxidizing alcohols^a

(A) Identification of the slowly oxidizing alcohols.

identify the slowly oxidizing alcohols

Entry	$\text{R}'\text{-CH}_2\text{OH}$	Recovery of 1a	NMR yield					
			3a	8	10	6	5a	11
1	BnOH (2a)	0	69%	86% (8a)	3% (10a)	0	10%	0
2	<i>n</i> -BuOH (2b)	11%	51%	32% (8b)	14% (10b)	19% ^b (6ab + 5a + 11ba)		
3	<i>n</i> -PrOH (2c)	9%	51%	14% (8c)	12% (10c)	8% (6ac)	12% ^c (5a + 11ca)	
4	EtOH (2d)	0	63%	12% (8d)	14% (10d)	14% (6ad)	9%	8% (11da)
5	MeOH (2e)	0	9%	ND ^d	9% (10e)	83% (6ae)	3%	1% (11ea)
6	<i>i</i> -PrOH (2f)	49%	24%	45% ^e (8f)	-	0	3%	-

^a The reaction was conducted with **1a** (1.0 mmol), $\text{R}'\text{CH}_2\text{OH}$ (5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (6 mol%), TEMPO (5 mol%), and BiCl_3 (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O_2 balloon in a 50 mL flask. The NMR yield and recovery were determined by ^1H NMR analysis using dibromomethane as the internal standard. ^b Combined yield of **6ab**, **5a** and **11ba** based on alcohol **1a**. ^c Combined yield of **5a** and **11ca** based on alcohol **1a**. ^d ND. = not determined. ^e NMR yield of acetone **8f**.

Optimization for the oxidative methyl esterification of alcohols

To a 25 mL or 50 mL flask were added Fe(NO₃)₃•9H₂O, TEMPO, Lewis acid, **1a**, MeOH, and DCE sequentially in open air. The reaction was then carried out at the specified temperature under the atmosphere of O₂ from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC. The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL), concentrated, and analyzed via ¹H NMR with CH₂Br₂ (35 μL) as the internal standard.

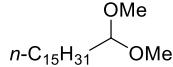
Table S2. Optimization of the amount of catalyst^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH		$\xrightarrow[\text{TEMPO (y mol\%)}]{\text{Fe(NO}_3)_3\cdot 9\text{H}_2\text{O (x mol\%)} \\ \text{BiCl}_3 \text{ (z mol\%)} \\ \text{Air + pure O}_2 \text{ balloon} \\ \text{DCE, 50 }^\circ\text{C, 40 h}}}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO} + \text{HCOO}n\text{-C}_{16}\text{H}_{33} + n\text{-C}_{15}\text{H}_{31}\text{COOMe}$			
1a (1.0 mmol)	2e (5.0 equiv)			3a	11ea	6ae	
				$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$	+	$n\text{-C}_{15}\text{H}_{31}\text{C}(\text{OMe})_2$	
				5a		7ae	

Entry	x/y/z	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	5/5/10	0	20%	1%	65%	4%	8%
2	6/5/10	0	17%	1%	74%	2%	6%
3	7/5/10	0	13%	1%	76%	4%	9%
4	8/5/10	0	14%	1%	77%	2%	4%
5	9/5/10	0	13%	1%	74%	2%	4%
6	10/5/10	0	18%	1%	66%	4%	6%
7	10/10/10	0	17%	2%	61%	6%	7%
8	8/3/10	12%	34%	1%	8%	4%	20%
9	8/7/10	0	14%	1%	77%	2%	2%
10	6/5/8	0	18%	1%	66%	4%	6%
11	6/5/6	0	22%	1%	64%	2%	8%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (x mol%), TEMPO (y mol%), and BiCl₃ (z mol%) in 4 mL of DCE at 50 °C for 40 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S3. Optimization of the amount of MeOH^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH			$\xrightarrow[\text{DCE, 50 } ^\circ\text{C, 40 h}]{\text{BiCl}_3 \text{ (10 mol\%)} \\ \text{TEMPO (5 mol\%)} \\ \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$		
1a (1.0 mmol)			2e (<i>x</i> equiv)		3a 11ea 6ae		
			Air + pure O ₂ balloon		$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ + 		
			DCE, 50 °C, 40 h		5a 7ae		
Entry	<i>x</i>	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	3	0	26%	1%	65%	4%	3%
2	4	0	22%	1%	71%	4%	5%
3	5	0	14%	1%	74%	4%	8%
4	6	0	15%	1%	75%	4%	7%
5	7	0	15%	1%	74%	0	10%
6	8	0	16%	1%	67%	2%	14%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (x mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 4 mL of DCE at 50 °C for 40 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S4. Optimization of the concentration^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH				$\xrightarrow[\text{DCE (x mL), 50 } ^\circ\text{C, t h}]{\text{BiCl}_3 \text{ (10 mol\%)} \\ \text{TEMPO (5 mol\%)} \\ \text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}}$		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$		
1a (1.0 mmol)		2e (5.0 equiv)		Air + pure O ₂ balloon	3a	11ea	6ae	
					$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$			
					5a			7ae
Entry	x	t	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	2	40 h	0	27%	0	36%	4%	16%
2	3	40 h	0	15%	1%	75%	4%	6%
3	3	48 h	0	8%	1%	84%	4%	5%
4	3.5	40 h	0	15%	1%	76%	4%	6%
5	4	40 h	0	14%	1%	74%	4%	8%
6	5	40 h	0	22%	1%	63%	4%	12%
7	6	40 h	0	28%	1%	53%	4%	17%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in x mL of DCE at 50 °C for t h or 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S5. Optimization of the reaction temperature^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH		BiCl ₃ (10 mol%) TEMPO (5 mol%) Fe(NO ₃) ₃ ·9H ₂ O (6 mol%)		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$			
1a (1.0 mmol)	2e (5.0 equiv)	Air + pure O ₂ balloon DCE, T °C, 48 h		3a	11ea	6ae	
				$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$	+	$n\text{-C}_{15}\text{H}_{31}\text{COOMe}$	
				5a		7ae	
Entry	T	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	25	0	8%	1%	52%	4%	25%
2	35	0	7%	1%	76%	4%	9%
3	40	0	10%	1%	80%	4%	5%
4	45	0	11%	1%	81%	4%	7%
5	50	0	8%	1%	84%	4%	5%
6	55	0	23%	1%	60%	4%	7%
7	60	0	31%	1%	56%	4%	5%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃·9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 3 mL of DCE at **T** °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S6. Optimization of the solvent^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH		BiCl ₃ (10 mol%) TEMPO (5 mol%) Fe(NO ₃) ₃ ·9H ₂ O (6 mol%)		$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$			
1a (1.0 mmol)	2e (5.0 equiv)	Air + pure O ₂ balloon solvent , 50 °C, 48 h		3a	11ea	6ae	
				$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$	+	$n\text{-C}_{15}\text{H}_{31}\text{COOMe}$	
				5a		7ae	
Entry	solvent	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	DCE	0	8%	1%	84%	4%	5%
2 ^b	DCM	4%	38%	2%	9%	4%	40%
3	CHCl ₃	0	43%	1%	25%	0	31%
4	CH ₃ CN	10%	27%	1%	9%	2%	23%
5	THF	48%	22%	0	0	0	0
6	Toluene	0	30%	2%	44%	0	15%
7	1,4-dioxane	0	20%	2%	21%	0	1%

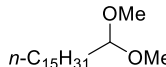
^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃·9H₂O (6 mol%), TEMPO (5 mol%), and BiCl₃ (10 mol%) in 3 mL of **solvent** at 50 °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard. ^b The reaction was conducted at 30 °C.

Table S7. Optimization of the nitroxyl^a

$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (5.0 equiv)		$\xrightarrow[\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}]{\text{BiCl}_3 \text{ (10 mol\%)}nitroxyl \text{ (5 mol\%)}}$ Air + pure O ₂ balloon DCE, 50 °C, 48 h	$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae	$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{C}(\text{OMe})_2$ 5a 7ae			
Entry	nitroxyl	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1	TEMPO	0	8%	1%	84%	4%	5%
2	4-NHAc-TEMPO	0	42%	0	21%	0	18%
3	4-OMe-TEMPO	0	11%	1%	81%	2%	3%
4	4-OH-TEMPO	0	33%	1%	40%	4%	18%
5	ABNO	3%	44%	2%	8%	0	22%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), **nitroxyl** (5 mol%), and BiCl₃ (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O₂ balloon in a 25 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

Table S8. Optimization of the Lewis acid^a

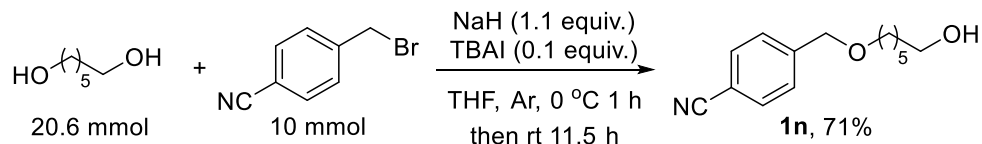
$n\text{-C}_{15}\text{H}_{31}\text{CH}_2\text{OH}$ + MeOH 1a (1.0 mmol) 2e (5.0 equiv)		$\xrightarrow[\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O (6 mol\%)}]{\text{LA (10 mol\%)}TEMPO (5 mol\%)}$ Air + pure O ₂ balloon DCE, 50 °C, 48 h	$n\text{-C}_{15}\text{H}_{31}\text{CHO}$ + $\text{HCOO}n\text{-C}_{16}\text{H}_{33}$ + $n\text{-C}_{15}\text{H}_{31}\text{COOMe}$ 3a 11ea 6ae	$n\text{-C}_{15}\text{H}_{31}\text{COO}n\text{-C}_{16}\text{H}_{33}$ +  5a 7ae			
Entry	LA	Recovery of 1a	NMR Y of 3a	NMR Y of 11ea	NMR Y of 6ae	NMR Y of 5a	NMR Y of 7ae
1^b	BiCl₃	0	9%	1%	83% (73%)	2%	3%
2	FeCl ₃	0	33%	0	31%	4%	22%
3	InCl ₃	0	29%	15%	35%	8%	7%
4	InBr ₃	0	10%	6%	57%	8%	11%
5	BiBr ₃	26%	16%	1%	2%	2%	16%

^a The reaction was conducted with **1a** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃•9H₂O (6 mol%), TEMPO (5 mol%), and **Lewis acid** (10 mol%) in 3 mL of DCE at 50 °C for 48 h with an O₂ balloon in a 50 mL flask. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard. ^b 5% of *n*-C₁₅H₃₁COOH **4a** was isolated.

Experimental details and analytical data

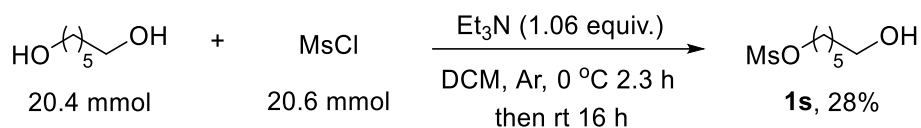
1. Preparation of alcohols

(1) Preparation of 4-(((6-hydroxyhexyl)oxy)methyl)benzonitrile (**1n**, yyb-7-005)



To a flask was added 1,6-hexanediol (2.4879 g, 20.6 mmol, 98% purity). Then the flask was degassed and refilled with argon for three times, the anhydrous THF (40 mL) was added and cooled to 0 °C with an ice-water bath before the addition of NaH (445.7 mg, 11 mmol, 60% in oil). After stirring for 1 h at the same temperature, 4-cyanobenzyl bromide (2.0057 g, 10 mmol, 98% purity) and TBAI (Tetrabutylammonium iodide, 373.9 mg, 1.0 mmol) were added. The resulting mixture was warmed to rt, stirred for 11.5 h, quenched with a saturated aqueous solution of NH₄Cl, and extracted with ethyl acetate (30 mL x 3). The combined organic layer was washed with brine and dried over anhydrous Na₂SO₄. After filtration and evaporation, the crude product was purified by column chromatography on silica gel to afford **1n** (1.6552 g, 71%) [eluent: petroleum ether/ethyl acetate = 10/1 (~330 mL) to 3/1 (~400 mL), then 2/1 (~600 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl₃) δ = 7.63 (d, *J* = 8.0 Hz, 2 H, Ar-H), 7.45 (d, *J* = 8.4 Hz, 2 H, Ar-H), 4.55 (s, 2 H, OCH₂), 3.62 (t, *J* = 6.4 Hz, 2 H, OCH₂), 3.50 (t, *J* = 6.6 Hz, 2 H, OCH₂), 2.18 (br, 1 H, OH), 1.65 (quint, *J* = 6.8 Hz, 2 H, CH₂), 1.57 (quint, *J* = 6.8 Hz, 2 H, CH₂), 1.47-1.32 (m, 4 H, 2 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 144.2, 132.0, 127.5, 118.7, 110.9, 71.6, 70.8, 62.4, 32.4, 29.4, 25.8, 25.4; **IR** (neat, cm⁻¹): ν = 2934, 2858, 2228, 1610, 1408, 1361, 1172, 1096; **MS** (70 eV, EI) *m/z* (%): 233 (M⁺, 13.1), 116 (100); **HRMS** (EI) calcd *m/z* for C₁₄H₁₉NO₂ [M]⁺: 233.1410, Found: 233.1410.

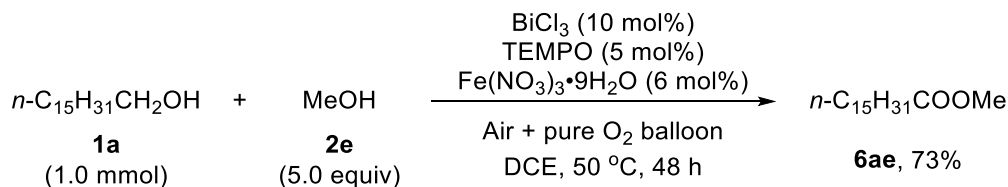
(2) Preparation of 6-hydroxyhexyl methanesulfonate¹ (**1s**, yyb-6-079)



To a flask were added 1,6-hexanediol (2.4841 g, 20.4 mmol, 97% purity), Et₃N (3.0 mL, d = 0.728 g/mL, 21.6 mmol), and DCM (40 mL). Then MsCl (1.6 mL, d = 1.478 g/mL, 20.6 mmol) was added dropwise within 20 min at 0 °C with an ice-water bath. After being stirred for 2 h at the same temperature, the resulting mixture was warmed to rt and stirred for 16 h. The solvent was removed under reduced pressure and the residue was purified by chromatography on silica gel to afford **1s** (1.1072 g, 28%) [first round eluent: petroleum ether/ethyl acetate = 5/1 (~300 mL) to 3/1 (~400 mL), then 2/1 (~600 mL) to 1/1 (~600 mL) to afford pure **1s** (141.5 mg) and impure **1s**; second round eluent: DCM/methol = 200/1 (~600 mL) to 100/1 (~400 mL), then 20/1 (~420 mL) to afford pure **1s** (0.9657 g)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃) δ = 4.23 (t, *J* = 6.6 Hz, 2 H, OCH₂), 3.62 (t, *J* = 6.4 Hz, 2 H, OCH₂), 3.01 (s, 3 H, CH₃), 2.22 (s, 1 H, OH), 1.77 (quint, *J* = 6.6 Hz, 2 H, CH₂), 1.57 (quint, *J* = 6.6 Hz, 2 H, CH₂), 1.50-1.34 (m, 4 H, 2 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 70.0, 62.2, 37.1, 32.2, 28.9, 24.99, 24.97; IR (neat, cm⁻¹): ν = 2934, 2864, 1468, 1340, 1262, 1169, 1052, 1022; MS (ESI) *m/z*: 219 (M+Na)⁺, 197 (M+H)⁺; HRMS (ESI) calcd *m/z* for C₇H₁₇O₄S [M+H]⁺: 197.0842, found 197.0843.

2. Preparation of carboxylic acid esters

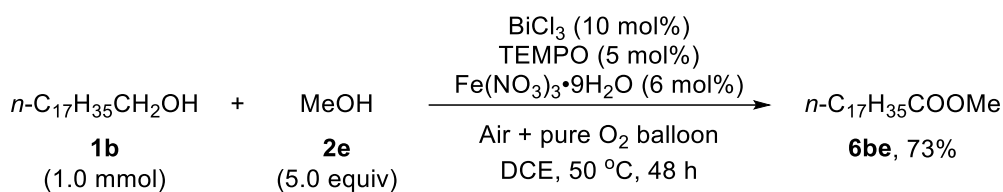
(1) Preparation of methyl palmitate (**6ae**, yyb-8-102)



Typical Procedure I: To a 50 mL flask were added Fe(NO₃)₃•9H₂O (24.9 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), **1a** (243.1 mg, 1.0 mmol), MeOH (202 μL, 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 °C under the atmosphere of O₂ from a balloon (about

2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6ae** was formed in 83% NMR yield based on ^1H NMR analysis of the crude product with CH_2Br_2 (35 μL) as the internal standard. After evaporation, MeOH (2 mL) and a saturated NaHSO_3 aqueous solution (1 mL) were added to the residue. The resulting mixture was stirred vigorously at room temperature for 1 h, dried with anhydrous Na_2SO_4 , filtrated through a short column of silica gel (1 cm) eluted with DCM (3 x 15 mL), and concentrated. After evaporation, the residue was purified by chromatography on silica gel to afford **6ae**² (198.3 mg, 73%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), 2/1 (~150 mL), then petroleum ether/ethyl acetate = 5/1 (~300 mL)] as light yellow solid (Low melting point solid, unable to recrystallize to measure melting point). ^1H NMR (400 MHz, CDCl_3): δ = 3.66 (s, 3 H, OCH_3), 2.30 (t, J = 7.4 Hz, 2 H, CH_2), 1.67-1.55 (m, 2 H, CH_2), 1.38-1.15 (m, 24 H, 12 x CH_2), 0.88 (t, J = 6.8 Hz, 3 H, CH_3); ^{13}C NMR (100 MHz, CDCl_3): δ = 174.2, 51.3, 34.0, 31.9, 29.65, 29.63, 29.62, 29.55, 29.4, 29.3, 29.2, 29.1, 24.9, 22.6, 14.0; IR (neat, cm^{-1}): ν = 2950, 2916, 2849, 1739, 1465, 1436, 1377, 1195, 1164; MS (70 eV, EI) m/z (%): 270 (M^+ , 19.23), 74 (100).

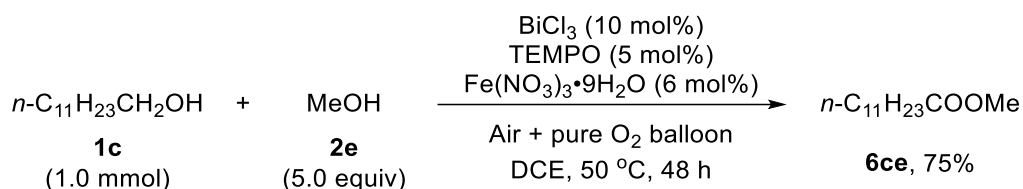
(2) Preparation of methyl stearate (**6be**, QAN-8-169)



Following Typical Procedure I, the reaction of **1b** (270.5mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.6 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6be**² (217.8 mg, 73%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~450 mL)] as white solid: m.p. 42.1-43.5 $^\circ\text{C}$ (We were not able to the recrystallize from the solvents tested, the m.p. of **6be** was determined by using the solid right after evaporation of the eluent). ^1H NMR (400 MHz, CDCl_3) δ = 3.66 (s, 3 H, OCH_3), 2.30 (t, J = 7.6 Hz, 2 H, CH_2), 1.65-1.58 (m, 2 H, CH_2), 1.32-1.17 (m, 28 H, 14 x CH_2), 0.88

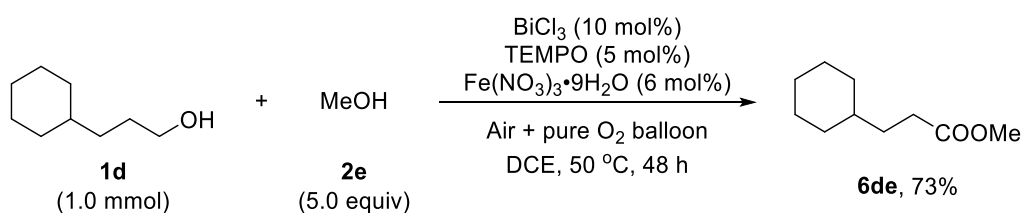
(t, $J = 6.4$ Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃) $\delta = 174.3, 51.3, 34.1, 31.9, 29.7, 29.6, 29.4, 29.3, 29.2, 29.1, 24.9, 22.7, 14.1$; IR (neat, cm⁻¹): $\nu = 2916, 2848, 1738, 1463, 1435, 1380, 1331, 1254, 1213, 1194, 1169, 1105$; MS (70 eV, EI) m/z (%): 298 (M⁺, 43.97), 74 (100).

(3) Preparation of methyl dodecanoate (**6ce**, lj-8-081)



Following Typical Procedure I, the reaction of **1c** (186.4 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.6 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.6 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6ce**² (160.7 mg, 75%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as oil. ¹H NMR (400 MHz, CDCl₃) $\delta = 3.66$ (s, 3 H, OCH₃), 2.30 (t, $J = 7.6$ Hz, 2 H, CH₂), 1.70-1.57 (m, 2 H, CH₂), 1.45-1.15 (m, 16 H, 8 x CH₂), 0.88 (t, $J = 6.6$ Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃) $\delta = 174.2, 51.3, 34.1, 31.9, 29.7, 29.6, 29.4, 29.3, 29.2, 29.1, 24.9, 22.6, 14.0$; IR (neat, cm⁻¹): $\nu = 2924, 2855, 1742, 1463, 1438, 1363, 1236, 1198, 1171, 1116$; MS (70 eV, EI) m/z (%): 214 (M⁺, 4.71), 74 (100).

(4) Preparation of methyl 3-cyclohexylpropanoate (**6de**, yyb-7-092)



Typical Procedure II: To a 50 mL flask were added Fe(NO₃)₃•9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.4 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), **1d** (141.7 mg, 1.0 mmol), MeOH (202 μ L, 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 °C under the atmosphere of O₂ from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6de** was formed in 83% NMR yield based on ¹H NMR

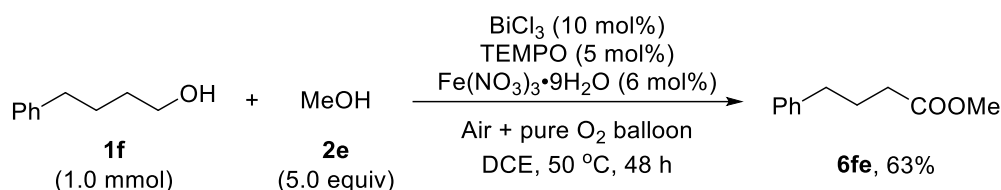
analysis of the crude product with CH₂Br₂ (35 μ L) as the internal standard. After evaporation, THF (2 mL) and 3 M HCl (0.5 mL) were added to the residue with vigorous stirring at room temperature for 2 h. To the resulting mixture was added a saturated aqueous solution of NaHCO₃ slowly until pH ~8. Then a saturated NaHSO₃ aqueous solution (2 mL) was added to the residue with vigorous stirring at room temperature for 2 h. The resulting mixture was extracted with DCM (15 mL $\times$ 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford **6de**³ (122.0 mg, 73%) [eluent: petroleum ether/ethyl acetate = 100/1 (~300 mL) to 50/1 (~100 mL)] as yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 2.32 (t, J = 7.8 Hz, 2 H, CH₂), 1.77-1.59 (m, 5 H), 1.52 (dd, J_1 = 15.4 Hz, J_2 = 7.0 Hz, 2 H, CH₂), 1.30-1.10 (m, 4 H), 0.96-0.82 (m, 2 H); ¹³C NMR (100 MHz, CDCl₃): δ = 174.5, 51.3, 37.1, 32.9, 32.3, 31.6, 26.4, 26.1; IR (neat, cm⁻¹): ν = 2922, 2851, 1739, 1449, 1436, 1366, 1195, 1165; MS (70 eV, EI) m/z (%): 170 (M⁺, 1.01), 97 (100).

(5) Preparation of methyl 3-phenylpropanoate (**6ee**, lj-8-082)



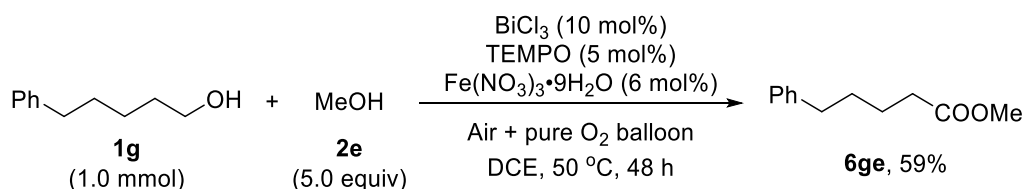
Following Typical Procedure I, the reaction of **1e** (136.3 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.6 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6ee**⁴ (105.1 mg, 64%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.32-7.25 (m, 2 H, Ar-H), 7.24-7.12 (m, 3 H, Ar-H), 3.67 (s, 3 H, OCH₃), 2.95 (t, J = 7.8 Hz, 2 H, CH₂), 2.64 (t, J = 7.8 Hz, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.3, 140.5, 128.5, 128.2, 126.2, 51.5, 35.7, 30.9; IR (neat, cm⁻¹): ν = 3026, 2950, 2834, 1736, 1496, 1443, 1364, 1290, 1255, 1164, 1075, 1055, 1028; MS (70 eV, EI) m/z (%): 164 (M⁺, 38.9), 104 (100).

(6) Preparation of methyl 4-phenylbutanoate (**6fe**, yyb-6-060)



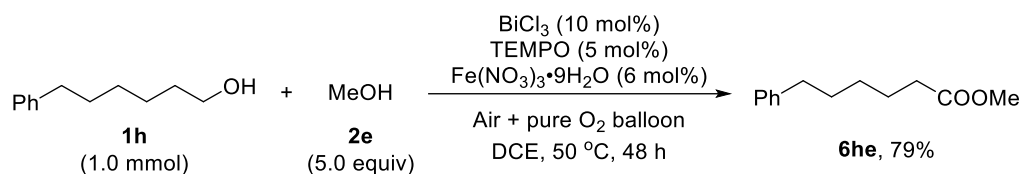
Following Typical Procedure I, the reaction of **1f** (149.5 mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.9 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6fe**³ (115.6 mg, 63%, 96% purity) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 7.26 (t, J = 7.4 Hz, 2 H, Ar-H), 7.21-7.10 (m, 3 H, Ar-H), 3.64 (s, 3 H, OCH_3), 2.64 (t, J = 7.6 Hz, 2 H, CH_2), 2.31 (t, J = 7.6 Hz, 2 H, CH_2), 1.95 (quint, J = 7.5 Hz, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 173.8, 141.3, 128.4, 128.3, 125.9, 51.3, 35.0, 33.3, 26.4; IR (neat, cm^{-1}): ν = 3027, 2946, 2864, 1735, 1496, 1442, 1366, 1246, 1168, 1146; MS (70 eV, EI) m/z (%): 178 (M^+ , 42.91), 104 (100).

(7) Preparation of methyl 5-phenylpentanoate (**6ge**, yyb-6-061)



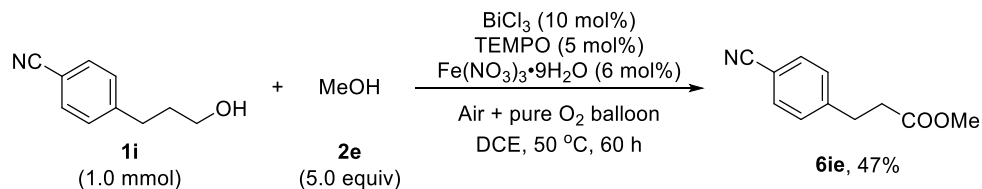
Following Typical Procedure I, the reaction of **1g** (167.5 mg, 1.0 mmol, 98% purity), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (32.1 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6ge**³ (111.8 mg, 59%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 7.26 (t, J = 7.4 Hz, 2 H, Ar-H), 7.20-7.12 (m, 3 H, Ar-H), 3.65 (s, 3 H, OCH_3), 2.62 (t, J = 6.6 Hz, 2 H, CH_2), 2.32 (t, J = 6.6 Hz, 2 H, CH_2), 1.74-1.55 (m, 4 H, 2 x CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 174.0, 142.1, 128.32, 128.26, 125.7, 51.4, 35.5, 33.9, 30.8, 24.5; IR (neat, cm^{-1}): ν = 3026, 2943, 2861, 1736, 1495, 1436, 1360, 1198, 1142; MS (70 eV, EI) m/z (%): 192 (M^+ , 3.56), 91 (100).

(8) Preparation of methyl 6-phenylhexanoate (**6he**, yyb-6-096)



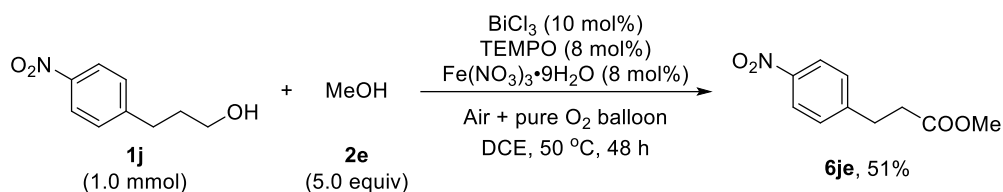
Following Typical Procedure I, the reaction of **1h** (182.8 mg, 97% purity, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.5 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6he**⁵ (162.6 mg, 79%) [eluent: petroleum ether/DCM = 15/1 (~320 mL) to 5/1 (~480 mL), then 3/1 (~400 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.25 (t, J = 7.2 Hz, 2 H, Ar-H), 7.20-7.09 (m, 3 H, Ar-H), 3.64 (s, 3 H, OCH_3), 2.59 (t, J = 7.6 Hz, 2 H, CH_2), 2.28 (t, J = 7.6 Hz, 2 H, CH_2), 1.71-1.56 (m, 4 H, 2 x CH_2), 1.35 (quint, J = 7.6 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 174.0, 142.4, 128.3, 128.2, 125.6, 51.3, 35.6, 33.9, 30.9, 28.6, 24.7; **IR** (neat, cm^{-1}): ν = 3026, 2930, 2857, 1736, 1495, 1436, 1362, 1198, 1170; **MS** (70 eV, EI) m/z (%): 206 (M^+ , 9.32), 91 (100).

(9) Preparation of methyl 3-(4-cyanophenyl)propanoate (**6ie**, yyb-7-024)



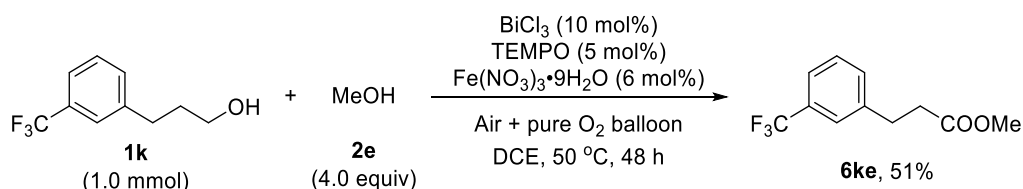
Following Typical Procedure II, the reaction of **1i** (161.1 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.6 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6ie**⁶ (89.7 mg, 47%) [eluent: petroleum ether/ethyl acetate = 10/1 (~270 mL) to 8/1 (~450 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.58 (d, J = 8.4 Hz, 2 H, Ar-H), 7.32 (d, J = 8.4 Hz, 2 H, Ar-H), 3.67 (s, 3 H, OCH_3), 3.02 (t, J = 7.6 Hz, 2 H, CH_2), 2.66 (t, J = 7.6 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 172.5, 146.0, 132.2, 129.1, 118.8, 110.1, 51.6, 34.7, 30.7; **IR** (neat, cm^{-1}): ν = 2952, 2227, 1733, 1608, 1506, 1437, 1365, 1174, 1159; **MS** (70 eV, EI) m/z (%): 189 (M^+ , 21.63), 129 (100).

(10) Preparation of methyl 3-(4-nitrophenyl)propanoate (**6je**, yyb-7-049)



Following Typical Procedure II, the reaction of **1j** (181.3 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.5 mg, 0.08 mmol), TEMPO (13.0 mg, 0.08 mmol), BiCl_3 (31.3 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6je**⁷ (106.4 mg, 51%) [eluent: petroleum ether/ethyl acetate = 7/1 (~400 mL) to 5/1 (~120 mL)] as light yellow solid; m.p. 73.7-74.2 °C (petroleum ether/DCM). ¹H NMR (400 MHz, CDCl_3): δ = 8.15 (d, J = 8.4 Hz, 2 H, Ar-H), 7.38 (d, J = 8.4 Hz, 2 H, Ar-H), 3.68 (s, 3 H, OCH_3), 3.07 (t, J = 7.4 Hz, 2 H, CH_2), 2.69 (t, J = 7.4 Hz, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 172.5, 148.2, 146.6, 129.1, 123.7, 51.7, 34.7, 30.5; IR (neat, cm^{-1}): ν = 3113, 3083, 1727, 1607, 1514, 1430, 1346, 1191, 1169; MS (70 eV, EI) m/z (%): 209 (M^+ , 26.18), 149 (100).

(11) Preparation of methyl 3-(3-(trifluoromethyl)phenyl)propanoate (**6ke**, yyb-7-072)



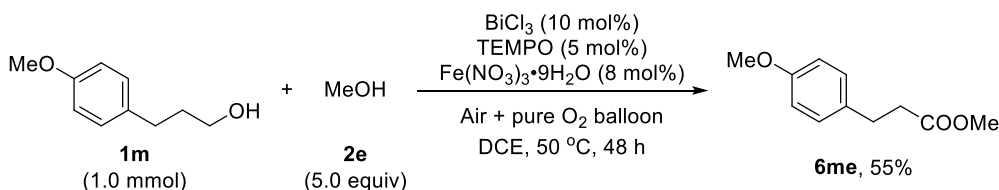
Following Typical Procedure II, the reaction of **1k** (204.1 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (162 μL , 4 mmol) in DCE (4 mL) afforded **6ke**⁶ (121.8 mg, 51%, 98% purity) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 7.51-7.44 (m, 2 H, Ar-H), 7.42-7.34 (m, 2 H, Ar-H), 3.67 (s, 3 H, OCH_3), 3.01 (t, J = 7.8 Hz, 2 H, CH_2), 2.66 (t, J = 7.6 Hz, 2 H, CH_2); ¹³C NMR (100 MHz, CDCl_3): δ = 172.8, 141.4, 131.7, 130.8 (q, J = 31.9 Hz), 128.9, 125.0 (q, J = 3.7 Hz), 124.1 (q, J = 270.4 Hz), 123.2 (q, J = 3.7 Hz), 51.6, 35.3, 30.6; ¹⁹F NMR (376 MHz, CDCl_3): δ = -63.1; IR (neat, cm^{-1}): ν = 1737, 1439, 1365, 1327, 1199, 1159, 1118, 1073; MS (70 eV, EI) m/z (%): 232 (M^+ , 44.2), 172 (100).

(12) Preparation of methyl 3-(4-chlorophenyl)propanoate (**6le**, yyb-7-091)



Following Typical Procedure II, the reaction of **1l** (169.9 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.7 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6le**⁴ (103.2 mg, 52%) [eluent: petroleum ether/ethyl acetate = 25/1 (~260 mL) to 20/1 (~100 mL)] as yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.30-7.20 (m, 2 H, Ar-H), 7.17-7.07 (m, 2 H, Ar-H), 3.66 (s, 3 H, OCH_3), 2.91 (t, J = 7.6 Hz, 2 H, CH_2), 2.60 (t, J = 7.8 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.0, 138.9, 132.0, 129.6, 128.5, 51.6, 35.4, 30.2; **IR** (neat, cm^{-1}): ν = 2951, 1734, 1492, 1436, 1364, 1195, 1157, 1092; **MS** (70 eV, EI) m/z (%): 200 ($\text{M}^{(37}\text{Cl})^+$, 9.7), 198 ($\text{M}^{(35}\text{Cl})^+$, 29.17), 138 (100).

(13) Preparation of methyl 3-(4-methoxyphenyl)propanoate (**6me**, yyb-7-063)



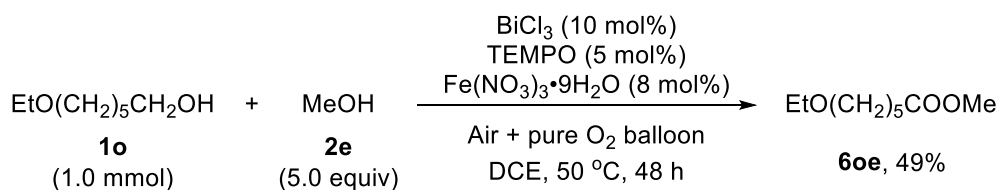
Following Typical Procedure II, the reaction of **1m** (175.5 mg, 95% purity, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.3 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (32.0 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (4 mL) afforded **6me**⁶ (107.3 mg, 55%) [eluent: petroleum ether/ethyl acetate = 15/1 (~320 mL) to 10/1 (~110 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.10 (d, J = 8.8 Hz, 2 H, Ar-H), 6.82 (d, J = 8.4 Hz, 2 H, Ar-H), 3.76 (s, 3 H, OCH_3), 3.65 (s, 3 H, OCH_3), 2.88 (t, J = 7.8 Hz, 2 H, CH_2), 2.59 (t, J = 7.8 Hz, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.3, 158.0, 132.5, 129.1, 113.8, 55.1, 51.4, 35.9, 30.0; **IR** (neat, cm^{-1}): ν = 3011, 2929, 1729, 1610, 1511, 1436, 1373, 1176, 1157; **MS** (70 eV, EI) m/z (%): 194 (M^+ , 21.47), 121 (100).

(14) Preparation of methyl 6-(4-cyanobenzyloxy)hexanoate (**6ne**, yyb-7-015)



Following Typical Procedure II, the reaction of **1n** (233.0 mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.4 mg, 0.06 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6ne** (129.9 mg, 50%) [eluent: petroleum ether/ ethyl acetate = 10/1 (~330 mL) to 5/1 (~240 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.63 (d, J = 8.0 Hz, 2 H, Ar-H), 7.44 (d, J = 8.0 Hz, 2 H, Ar-H), 4.54 (s, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 3.50 (t, J = 6.6 Hz, 2 H, OCH_2), 2.33 (t, J = 7.6 Hz, 2 H, CH_2), 1.71-1.59 (m, 4 H, 2 x CH_2), 1.48-1.35 (m, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.9, 144.2, 132.0, 127.5, 118.7, 111.0, 71.7, 70.6, 51.3, 33.8, 29.2, 25.6, 24.5; **IR** (neat, cm^{-1}): ν = 2947, 2863, 2223, 1738, 1609, 1448, 1371, 1236, 1170, 1099; **MS** (70 eV, EI) m/z (%): 261 (M^+ , 3.52), 116 (100); **HRMS** (EI) calcd m/z for $\text{C}_{15}\text{H}_{19}\text{NO}_3$ [$\text{M}]^+$: 261.1359, Found: 261.1363.

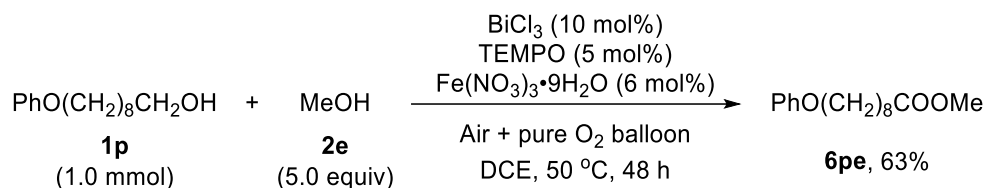
(15) Preparation of methyl 6-ethoxyhexanoate (**6oe**, yyb-6-155)



Following Typical Procedure II, the reaction of **1o** (146.0 mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (32.6 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6oe** (88.6 mg, 49%, 96% purity) [eluent: petroleum ether/ethyl acetate = 20/1 (~210 mL) to 10/1 (~220 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 3.66 (s, 3 H, OCH_3), 3.46 (q, J = 6.9 Hz, 2 H, OCH_2), 3.41 (t, J = 6.6 Hz, 2 H, OCH_2), 2.32 (t, J = 7.6 Hz, 2 H, CH_2), 1.73-1.52 (m, 4 H, 2 x CH_2), 1.46-1.32 (m, 2 H, CH_2), 1.19 (t, J = 7.0 Hz, 3 H, CH_3); **¹³C NMR** (100 MHz, CDCl_3): δ = 174.0, 70.2, 66.0, 51.3, 33.9, 29.3, 25.7, 24.7, 15.0; **IR** (neat, cm^{-1}): ν = 2975, 2940, 2860, 1738, 1437, 1376, 1201, 1167, 1109; **MS** (ESI) m/z : 197 ($\text{M}+\text{Na}^+$), 175 ($\text{M}+\text{H}^+$); **HRMS** (ESI) calcd m/z for $\text{C}_9\text{H}_{19}\text{O}_3$ [$\text{M}+\text{H}]^+$:

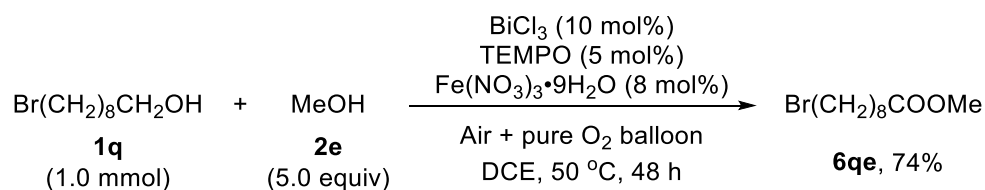
175.1329, found 175.1330.

(16) Preparation of methyl 9-phenoxy-nonanoate (6pe, yyb-6-172)



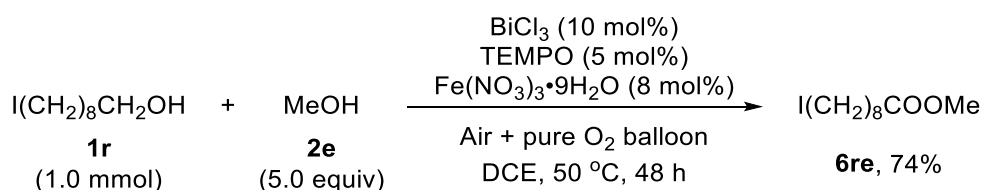
Typical Procedure III: To a 50 mL flask were added $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.5 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl_3 (31.9 mg, 0.1 mmol), **1p** (236.7 mg, 1.0 mmol), MeOH (202 μL , 5 mmol), and DCE (3 mL) sequentially in open air. The reaction was then carried out at 50 $^\circ\text{C}$ under the atmosphere of O_2 from a balloon (about 2.5 L) until completion of the reaction as monitored by TLC (48 h). The resulting mixture was filtrated through a short column of silica gel eluted with diethyl ether (3 x 25 mL) and concentrated. **6pe** was formed in 73% NMR yield based on ^1H NMR analysis of the crude product with CH_2Br_2 (35 μL) as the internal standard. After evaporation, the residue was purified by chromatography on silica gel to afford **6pe** (166.4 mg, 63%) [first round eluent: petroleum ether/ethyl acetate = 40/1 (~610 mL); second round eluent: petroleum ether/ethyl acetate = 50/1 (~510 mL)] as light yellow oil. ^1H NMR (400 MHz, CDCl_3): δ = 7.32-7.20 (m, 2 H, Ar-H), 6.96-6.83 (m, 3 H, Ar-H), 3.93 (t, J = 6.6 Hz, 2 H, OCH_2), 3.65 (s, 3 H, OCH_3), 2.29 (t, J = 7.6 Hz, 2 H, CH_2), 1.76 (quint, J = 7.0 Hz, 2 H, CH_2), 1.62 (quint, J = 7.3 Hz, 2 H, CH_2), 1.52-1.40 (m, 2 H, CH_2), 1.38-1.27 (m, 6 H, 3 x CH_2); ^{13}C NMR (100 MHz, CDCl_3): δ = 174.1, 159.0, 129.3, 120.3, 114.4, 67.7, 51.3, 33.9, 29.2, 29.1, 29.0, 25.9, 24.8; IR (neat, cm^{-1}): ν = 2930, 2856, 1737, 1599, 1496, 1470, 1436, 1242, 1198, 1169; MS (ESI) m/z : 287 ($\text{M}+\text{Na}$) $^+$, 265 ($\text{M}+\text{H}$) $^+$; HRMS (ESI) calcd m/z for $\text{C}_{16}\text{H}_{25}\text{O}_3$ [$\text{M}+\text{H}$] $^+$: 265.1798, found 265.1796.

(17) Preparation of methyl 9-bromononanoate (6qe, yyb-6-085)



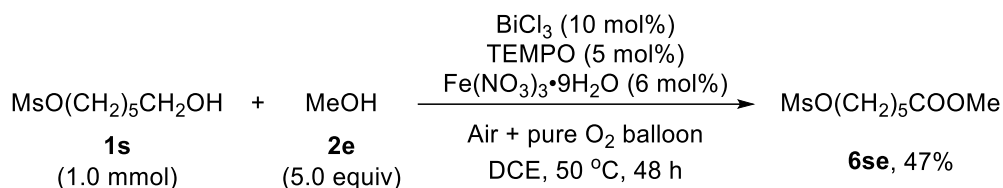
Following Typical Procedure I, the reaction of **1q** (223.5 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.3 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6qe**⁸ (189.5 mg, 74%, 98% purity) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 3.40 (t, *J* = 6.8 Hz, 2 H, CH₂), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.85 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.68-1.56 (m, 2 H, CH₂), 1.48-1.37 (m, 2 H, CH₂), 1.37-1.27 (m, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.0, 51.3, 33.9, 33.7, 32.7, 28.91, 28.87, 28.4, 28.0, 24.7; IR (neat, cm⁻¹): ν = 2930, 2855, 1737, 1458, 1437, 1362, 1197, 1170; MS (ESI) *m/z*: 275 (M(⁸¹Br)+Na)⁺, 273 (M(⁷⁹Br)+Na)⁺.

(18) Preparation of methyl 9-iodononanoate (**6re**, yyb-6-088)



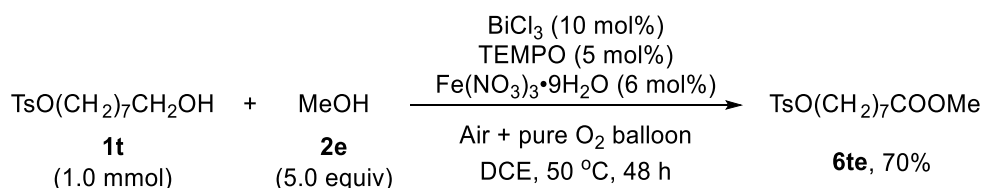
Following Typical Procedure I, the reaction of **1r** (269.9 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (32.1 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6re**⁹ (200.1 mg and 23.2 mg (90% purity), 74%) [eluent: petroleum ether/DCM = 10/1 (~330 mL) to 4/1 (~250 mL), then 3/1 (~200 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 3.18 (t, *J* = 7.0 Hz, 2 H, CH₂), 2.30 (t, *J* = 7.6 Hz, 2 H, CH₂), 1.82 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.68-1.56 (m, 2 H, CH₂), 1.45-1.35 (m, 2 H, CH₂), 1.35-1.24 (m, 6 H, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.1, 51.4, 34.0, 33.4, 30.3, 28.9, 28.2, 24.8, 7.1; IR (neat, cm⁻¹): ν = 2926, 2853, 1736, 1459, 1434, 1361, 1195, 1169; MS (ESI) *m/z*: 321 (M+Na)⁺, 299 (M+H)⁺.

(19) Preparation of methyl 6-((methanesulfonyl)oxy)hexanoate (6se, yyb-6-156)



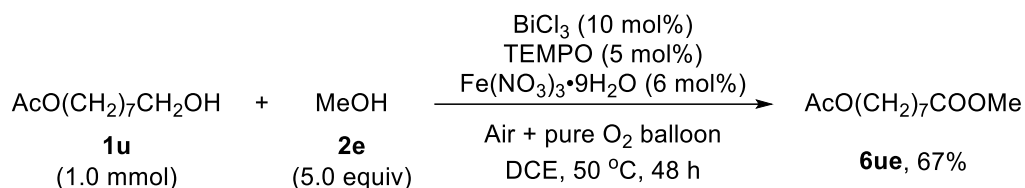
Following Typical Procedure II, the reaction of **1s** (196.7 mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.6 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl_3 (31.2 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6se**¹⁰ (106.3 mg, 47%) [eluent: petroleum ether/ethyl acetate = 3/1 (~200 mL) to 2/1 (~300 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl_3): δ = 4.23 (t, J = 6.4 Hz, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 3.01 (s, 3 H, CH_3), 2.34 (t, J = 7.4 Hz, 2 H, CH_2), 1.82-1.73 (m, 2 H, CH_2), 1.72-1.62 (m, 2 H, CH_2), 1.50-1.39 (m, 2 H, CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 173.7, 69.6, 51.4, 37.1, 33.5, 28.6, 24.8, 24.1; **IR** (neat, cm^{-1}): ν = 2935, 2859, 1732, 1438, 1349, 1169, 1104; **MS** (ESI) m/z : 247 ($\text{M}+\text{Na}$)⁺, 225 ($\text{M}+\text{H}$)⁺.

(20) Preparation of methyl 8-(tosyloxy)octanoate (6te, QAN-8-170)



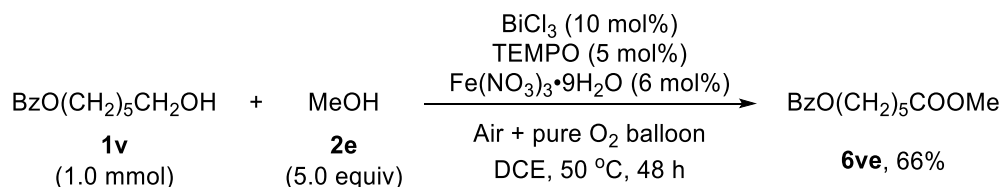
Following Typical Procedure I, the reaction of **1t** (300.5mg, 1.0 mmol), $\text{Fe(NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.5 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.8 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6te** (230.1 mg, 70%) [eluent: petroleum ether/ ethyl acetate = 15/1 (~300 mL) to 10/1 (~400 mL)] as oil. **¹H NMR** (400 MHz, CDCl_3): δ = 7.79 (d, J = 8.0 Hz, 2 H, Ar-H), 7.35 (d, J = 7.6 Hz, 2 H, Ar-H), 4.01 (t, J = 6.4 Hz, 2 H, CH_2), 3.66 (s, 3 H, OCH_3), 2.45 (s, 3 H, CH_3), 2.28 (d, J = 7.6 Hz, 2 H, CH_2), 1.68-1.50 (m, 4 H, 2 x CH_2), 1.34-1.18 (m, 6 H, 3 x CH_2); **¹³C NMR** (100 MHz, CDCl_3): δ = 174.0, 144.6, 133.2, 129.7, 127.8, 70.5, 51.3, 33.8, 28.7, 28.6, 28.4, 25.0, 24.6, 21.5; **IR** (neat, cm^{-1}): ν = 2939, 2910, 2868, 1731, 1598, 1466, 1434, 1347, 1311, 1248, 1215, 1173, 1098, 1071, 1048; **MS** (70 eV, EI) m/z (%): 328 (M^+ , 7.42), 91 (100); **HRMS** (EI) calcd m/z for $\text{C}_{16}\text{H}_{24}\text{O}_5\text{S}$ [M]⁺: 328.1339, Found: 328.1339.

(21) Preparation of methyl 8-acetyloxyoctanoate (**6ue**, yyb-7-076)



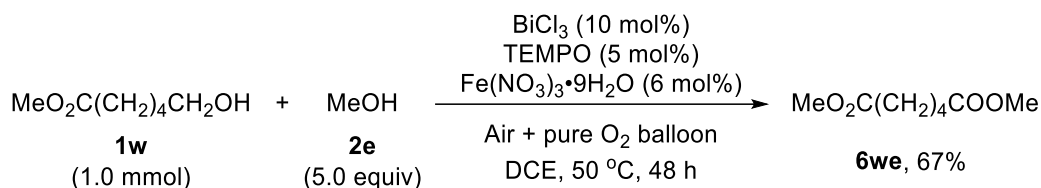
Following Typical Procedure II, the reaction of **1u** (189.1 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.3 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.4 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ue** (144.7 mg, 67%) [eluent: petroleum ether/ethyl acetate = 15/1 (~480 mL) to 10/1 (~110 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 4.05 (t, *J* = 6.8 Hz, 2 H, OCH₂), 3.67 (s, 3 H, OCH₃), 2.31 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.04 (s, 3 H, CH₃), 1.70-1.56 (m, 4 H, 2 x CH₂), 1.43-1.28 (m, 6 H, 3 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.0, 171.0, 64.3, 51.3, 33.8, 28.8, 28.7, 28.4, 25.6, 24.7, 20.8; IR (neat, cm⁻¹): ν = 2937, 2858, 1735, 1437, 1365, 1234, 1170, 1034; MS (ESI) *m/z*: 239 (M+Na)⁺, 217 (M+H)⁺; HRMS (ESI) calcd *m/z* for C₁₁H₂₁O₄ [M+H]⁺: 217.1434, found 217.1429.

(22) Preparation of 6-methoxy-6-oxohexyl benzoate (**6ve**, yyb-7-077)



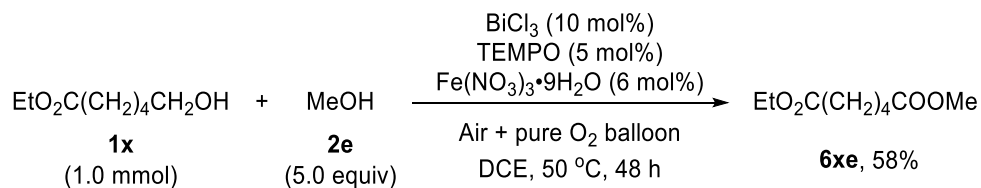
Following Typical Procedure II, the reaction of **1v** (220.1 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.5 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl₃ (31.0 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ve** (164.3 mg, 66%) [eluent: petroleum ether/ethyl acetate = 20/1 (~520 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 8.04 (d, *J* = 7.2 Hz, 2 H, Ar-H), 7.54 (t, *J* = 7.4 Hz, 1 H, Ar-H), 7.43 (t, *J* = 7.6 Hz, 2 H, Ar-H), 4.32 (t, *J* = 6.6 Hz, 2 H, OCH₂), 3.66 (s, 3 H, OCH₃), 2.34 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.79 (quint, *J* = 7.1 Hz, 2 H, CH₂), 1.71 (quint, *J* = 7.7 Hz, 2 H, CH₂), 1.54-1.42 (m, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.7, 166.4, 132.7, 130.2, 129.3, 128.2, 64.5, 51.3, 33.7, 28.2, 25.4, 24.4; IR (neat, cm⁻¹): ν = 2951, 1735, 1715, 1451, 1436, 1271, 1171, 1114; MS (70 eV, EI) *m/z* (%): 250 (M⁺, 1.09), 105 (100); HRMS (EI) calcd *m/z* for C₁₄H₁₈O₄ [M]⁺: 250.1200, Found: 250.1204.

(23) Preparation of dimethyl adipate (**6we**, wj-3-163)



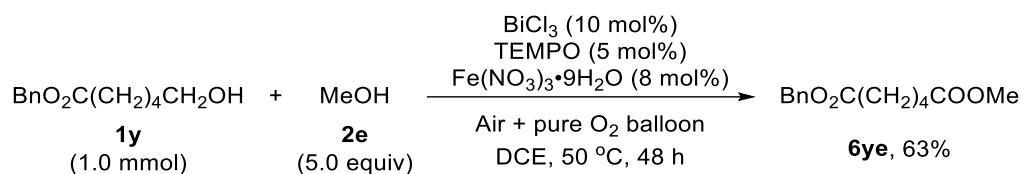
Following Typical Procedure I, the reaction of **1w** (146.3 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.4 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl_3 (31.4 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6we**¹¹ (116.3 mg, 67%) [eluent: petroleum ether/DCM = 2/1 (~300 mL) to 1/1 (~400 mL), then DCM (400 mL)] as oil. ¹H NMR (400 MHz, CDCl_3): δ = 3.56 (s, 6 H, 2 x OCH_3), 2.27-2.13 (m, 4 H, 2 x CH_2), 1.60-1.47 (m, 4 H, 2 x CH_2); ¹³C NMR (100 MHz, CDCl_3) δ = 173.7, 51.5, 33.6, 24.3; IR (neat, cm^{-1}) ν = 2955, 1734, 1437, 1367, 1248, 1197, 1171, 1083; MS (70 eV, EI) m/z (%): 143 ($(\text{M}-\text{OMe})^+$, 73.02), 114 (100).

(24) Preparation of ethyl methyl adipate (**6xe**, yyb-7-039)



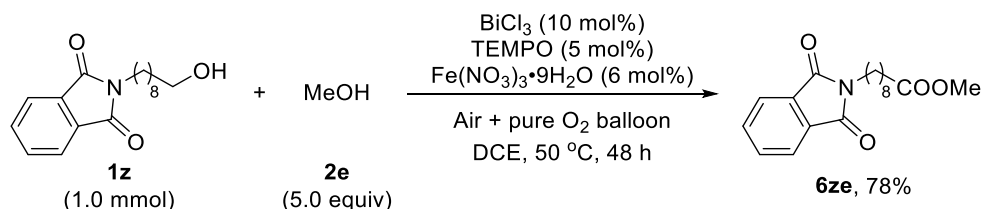
Following Typical Procedure II, the reaction of **1x** (160.7 mg, 1.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (24.8 mg, 0.06 mmol), TEMPO (8.0 mg, 0.05 mmol), BiCl_3 (32.0 mg, 0.1 mmol), and MeOH (202 μL , 5 mmol) in DCE (3 mL) afforded **6xe**¹² (110.3 mg, 58%) [eluent: petroleum ether/ethyl acetate = 20/1 (~420 mL) to 10/1 (~330 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl_3): δ = 4.13 (q, J = 7.1 Hz, 2 H, OCH_2), 3.67 (s, 3 H, OCH_3), 2.40-2.26 (m, 4 H, 2 x CH_2), 1.73-1.58 (m, 4 H, 2 x CH_2), 1.26 (t, J = 7.2 Hz, 3 H, CH_3); ¹³C NMR (100 MHz, CDCl_3) δ = 173.7, 173.2, 60.2, 51.4, 33.8, 33.5, 24.3, 24.2, 14.1; IR (neat, cm^{-1}) ν = 2981, 2945, 2873, 1731, 1438, 1372, 1242, 1172, 1143; MS (ESI) m/z : 211 ($(\text{M}+\text{Na})^+$, 189 ($(\text{M}+\text{H})^+$.

(25) Preparation of benzyl methyl adipate (**6ye**, yyb-7-036)



Following Typical Procedure II, the reaction of **1y** (223.8 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (32.3 mg, 0.08 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (31.6 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ye**¹³ (157.9 mg, 63%) [eluent: petroleum ether/ethyl acetate = 25/1 (~260 mL) to 20/1 (~630 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.44-7.25 (m, 5 H, Ar-H), 5.11 (s, 2 H, OCH₂), 3.65 (s, 3 H, OCH₃), 2.37 (t, *J* = 7.0 Hz, 2 H, CH₂), 2.32 (t, *J* = 7.0 Hz, 2 H, CH₂), 1.76-1.58 (m, 4 H, 2 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.6, 173.0, 135.9, 128.4, 128.1, 66.0, 51.4, 33.7, 33.5, 24.2; IR (neat, cm⁻¹): ν = 2951, 1731, 1455, 1438, 1381, 1358, 1164, 1140; MS (ESI) *m/z*: 273 (M+Na)⁺, 251 (M+H)⁺.

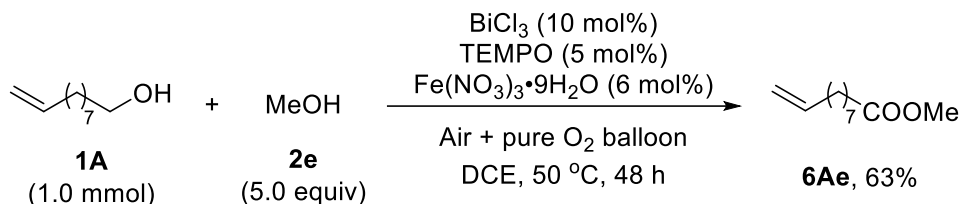
(26) Preparation of methyl 9-(1,3-dioxoisindolin-2-yl)nonanoate (**6ze**, yyb-6-197)



Following Typical Procedure I, the reaction of **1z** (290.2 mg, 1.0 mmol), Fe(NO₃)₃•9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6ze** (248.2 mg, 78%) [eluent: petroleum ether/DCM = 3/1 (~200 mL) to 1/1 (~400 mL), then 1/2 (~300 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.91-7.78 (m, 2 H, Ar-H), 7.77-7.65 (m, 2 H, Ar-H), 3.76-3.55 (m, 5 H, OCH₃ and NCH₂), 2.29 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.78-1.55 (m, 4 H, 2 x CH₂), 1.40-1.19 (m, 8 H, 4 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 173.9, 168.1, 133.6, 131.9, 122.9, 51.1, 37.7, 33.7, 28.8, 28.74, 28.69, 28.3, 26.5, 24.6; IR (neat, cm⁻¹): ν = 2928, 2855, 1736, 1707, 1466, 1436, 1395, 1170, 1060; MS (70 eV, EI) *m/z* (%): 317 (M⁺, 9.61), 160 (100); HRMS (EI) calcd *m/z* for

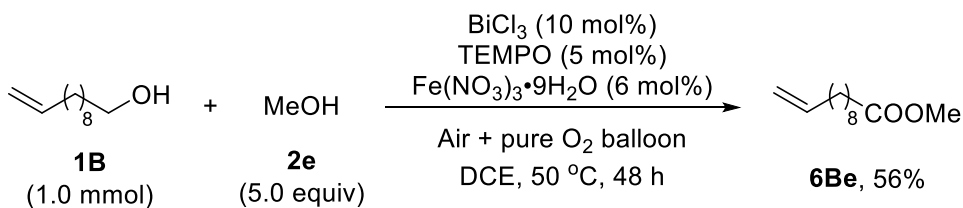
C₁₈H₂₃NO₄ [M]⁺: 317.1622, Found: 317.1623.

(27) Preparation of methyl dec-9-enoate (6Ae, yyb-7-075)



Following Typical Procedure II, the reaction of **1A** (165.1 mg, 95% purity, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Ae**¹⁴ (118.6 mg, 63%, 98% purity) [eluent: petroleum ether/ethyl acetate = 60/1 (~300 mL) to 50/1 (~250 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 5.87-5.73 (m, 1 H, =CH), 5.03-4.95 (m, 1 H, one proton of =CH₂), 4.95-4.88 (m, 1 H, one proton of =CH₂), 3.66 (s, 3 H, OCH₃), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.04 (q, *J* = 6.9 Hz, 2 H, CH₂), 1.67-1.57 (m, 2 H, CH₂), 1.42-1.25 (m, 8 H, 4 x CH₂); ¹³C NMR (100 MHz, CDCl₃): δ = 174.2, 139.0, 114.1, 51.3, 34.0, 33.7, 29.0, 28.83, 28.76, 24.9; IR (neat, cm⁻¹): ν = 3080, 2927, 2855, 1740, 1463, 1436, 1361, 1198, 1169; MS (ESI) *m/z*: 185 (M+H)⁺.

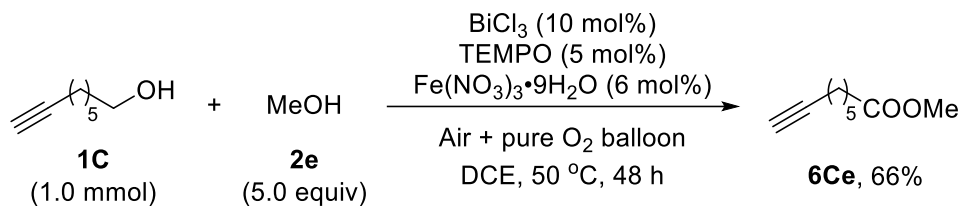
(28) Preparation of methyl undec-10-enoate (6Be, yyb-6-147)



Following Typical Procedure II, the reaction of **1B** (171.3 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.4 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Be**¹⁴ (115.9 mg, 56%, 96% purity) [eluent: petroleum ether/ethyl acetate = 50/1 (~250 mL)] as yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 5.87-5.73 (m, 1 H, =CH), 5.04-4.95 (m, 1 H, one proton of =CH₂), 4.95-4.88 (m, 1 H, one proton of =CH₂), 3.66 (s, 3 H, OCH₃), 2.30 (t,

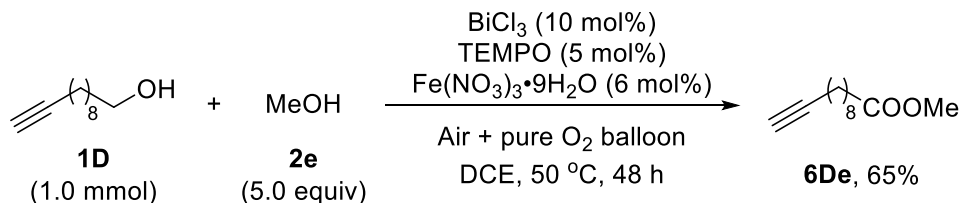
$J = 7.4$ Hz, 2 H, CH₂), 2.10-1.97 (m, 2 H, CH₂), 1.67-1.57 (m, 2 H, CH₂), 1.42-1.20 (m, 10 H, 5 x CH₂); ¹³C NMR (100 MHz, CDCl₃): $\delta = 174.2, 139.1, 114.1, 51.3, 34.0, 33.7, 29.2, 29.1, 29.05, 28.97, 28.8, 24.9$; IR (neat, cm⁻¹): $\nu = 3077, 2926, 2855, 1740, 1463, 1436, 1361, 1197, 1170$; MS (ESI) m/z : 199 (M+H)⁺.

(29) Preparation of methyl hept-6-ynoate (**6Ce**, yyb-7-074)



Following Typical Procedure II, the reaction of **1C** (126.9 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.5 mg, 0.06 mmol), TEMPO (8.2 mg, 0.05 mmol), BiCl₃ (31.8 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ce**¹⁵ (103.9 mg, 66%, 98% purity) [eluent: petroleum ether/ethyl acetate = 30/1 (~460 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): $\delta = 3.67$ (s, 3 H, OCH₃), 2.33 (t, $J = 7.6$ Hz, 2 H, CH₂), 2.20 (td, $J_1 = 7.0$ Hz, $J_2 = 2.7$ Hz, 2 H, CH₂), 1.95 (t, $J = 2.6$ Hz, 1 H, CH), 1.65 (quint, $J = 7.5$ Hz, 2 H, CH₂), 1.60-1.50 (m, 2 H, CH₂), 1.48-1.38 (m, 2 H, CH₂); ¹³C NMR (100 MHz, CDCl₃): $\delta = 174.0, 84.2, 68.3, 51.4, 33.8, 28.1, 28.0, 24.3, 18.1$; IR (neat, cm⁻¹): $\nu = 3295, 2940, 2863, 2116, 1735, 1460, 1436, 1364, 1202, 1172$; MS (ESI) m/z : 155 (M+H)⁺.

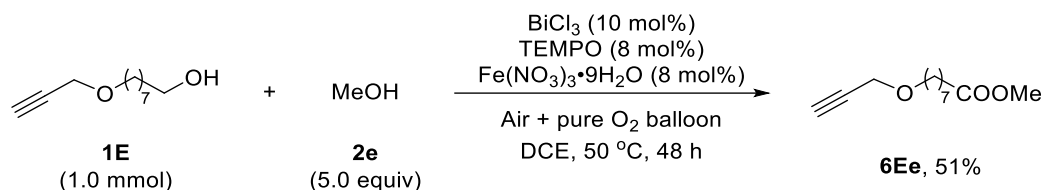
(30) Preparation of methyl undec-10-ynoate (**6De**, yyb-6-006)



Following Typical Procedure I, the reaction of **1D** (167.8 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (8.1 mg, 0.05 mmol), BiCl₃ (32.1 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6De**¹⁶ (127.6 mg, 65%) [eluent: petroleum ether/DCM = 10/1 (~220 mL) to 4/1 (~250 mL), then 2/1

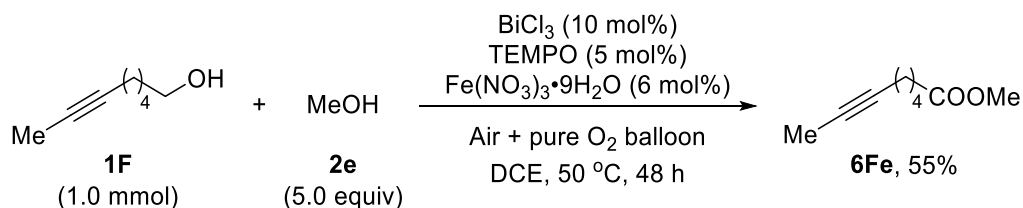
(~300 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.66 (s, 3 H, OCH₃), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.17 (t, *J* = 7.0 Hz, 2 H, CH₂), 1.94 (s, 1 H, CH), 1.68-1.57 (m, 2 H, CH₂), 1.52 (quint, *J* = 7.2 Hz, 2 H, CH₂), 1.45-1.24 (m, 8 H, 4 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.1, 84.5, 68.0, 51.3, 33.9, 29.0, 28.9, 28.8, 28.5, 28.3, 24.8, 18.2; **IR** (neat, cm⁻¹): ν = 3304, 2933, 2857, 2116, 1737, 1459, 1438, 1361, 1197, 1170; **MS** (ESI) *m/z*: 219 (M+Na)⁺, 197 (M+H)⁺.

(31) Preparation of methyl 8-(prop-2-ynoxy)octanoate (**6Ee**, yyb-7-081)



Following Typical Procedure II, the reaction of **1E** (182.7 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (31.9 mg, 0.08 mmol), TEMPO (12.7 mg, 0.08 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μL, 5 mmol) in DCE (3 mL) afforded **6Ee** (108.3 mg, 51%) [eluent: petroleum ether/ ethyl acetate = 15/1 (~320 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 4.13 (d, *J* = 2.4 Hz, 2 H, OCH₂), 3.66 (s, 3 H, OCH₃), 3.50 (t, *J* = 6.4 Hz, 2 H, OCH₂), 2.43 (t, *J* = 2.4 Hz, 1 H, CH), 2.30 (t, *J* = 7.4 Hz, 2 H, CH₂), 1.70-1.52 (m, 4 H, 2 x CH₂), 1.42-1.26 (m, 6 H, 3 x CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.1, 79.9, 74.0, 70.0, 57.9, 51.3, 33.9, 29.3, 28.92, 28.90, 25.8, 24.7; **IR** (neat, cm⁻¹): ν = 3274, 2932, 2857, 2115, 1735, 1437, 1357, 1249, 1171, 1098; **MS** (ESI) *m/z*: 235 (M+Na)⁺, 213 (M+H)⁺; **HRMS** (ESI) calcd *m/z* for C₁₂H₂₁O₃ [M+H]⁺: 213.1485, found 213.1484.

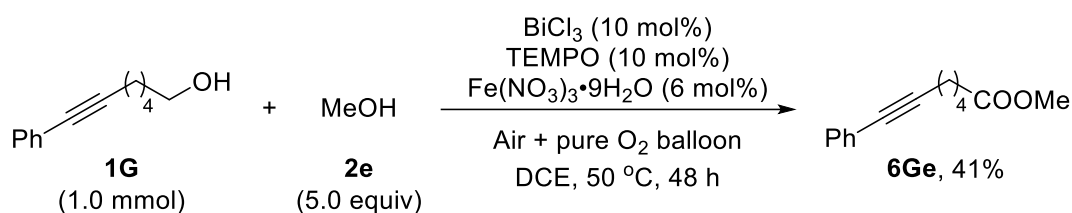
(32) Preparation of methyl 7-methylhept-6-ynoate (**6Fe**, zhangq-3-158)



Following Typical Procedure I, the reaction of **1F** (126.0 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.9 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.7 mg,

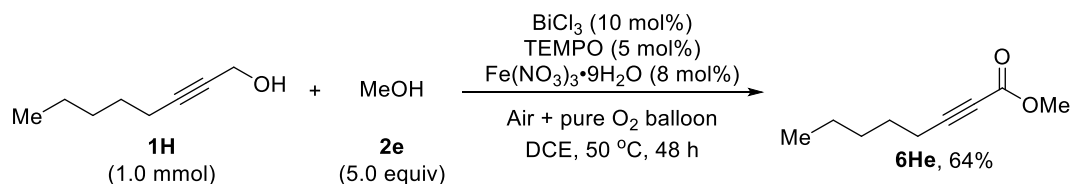
0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Fe**¹⁷ (85.3 mg, 55%) [eluent: petroleum ether/DCM = 10/1 (~440 mL) to 4/1 (~240 mL), then 2/1 (~150 mL)] as oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.33 (t, J = 7.6 Hz, 2 H, CH₂), 2.20-2.06 (m, 2 H, CH₂), 1.77 (s, 3 H, CH₃), 1.75-1.65 (m, 2 H, CH₂), 1.55-1.45 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 173.9, 78.5, 75.7, 51.3, 33.5, 28.3, 24.0, 18.3, 3.3; **IR** (neat, cm⁻¹): ν = 2946, 2924, 2220, 1736, 1437, 1358, 1201, 1172, 1152; **MS** (ESI) m/z : 177 (M+Na)⁺, 155 (M+H)⁺.

(33) Preparation of methyl 7-phenylhept-6-ynoate (**6Ge**, yyb-7-071)



Following Typical Procedure II, the reaction of **1G** (189.1 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.8 mg, 0.06 mmol), TEMPO (16.2 mg, 0.1 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ge**¹⁸ (90.7 mg, 41%, 98% purity) [eluent: petroleum ether/ethyl acetate = 50/1 (~510 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 7.45-7.35 (m, 2 H, Ar-H), 7.31-7.22 (m, 3 H, Ar-H), 3.67 (s, 3 H, OCH₃), 2.43 (t, J = 7.0 Hz, 2 H, CH₂), 2.37 (t, J = 7.4 Hz, 2 H, CH₂), 1.87-1.75 (m, 2 H, CH₂), 1.69-1.59 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 173.9, 131.5, 128.1, 127.5, 123.8, 89.5, 80.9, 51.5, 33.5, 28.1, 24.1, 19.1; **IR** (neat, cm⁻¹): ν = 2948, 2232, 1735, 1490, 1437, 1359, 1199, 1172, 1146; **MS** (70 eV, EI) m/z (%): 216 (M⁺, 28.56), 115 (100).

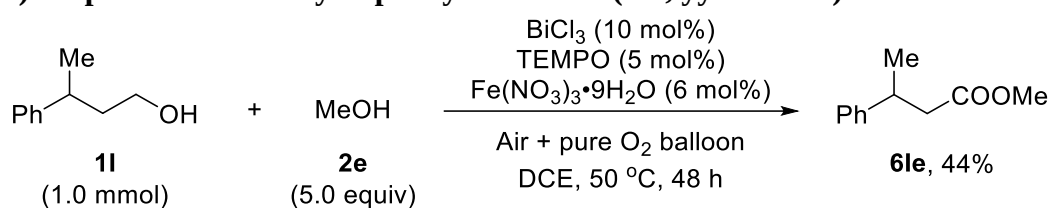
(34) Preparation of methyl oct-2-ynoate (**6He**, yyb-11-173A)



Following Typical Procedure II, the reaction of **1H** (127.7 mg, 99% purity, 1.0 mmol), Fe(NO₃)₃·9H₂O (32.6 mg, 0.08 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃

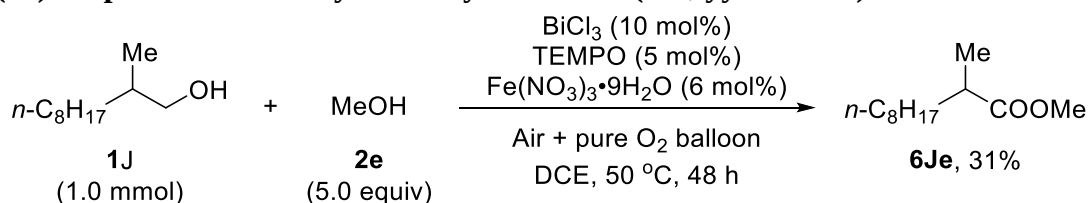
(31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6He**¹⁹ (99.3 mg, 64%) [eluent: petroleum ether/DCM = 10/1 (~270 mL) to 5/1 (~240 mL), then 3/1 (~200 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 3.76 (s, 3 H, OCH₃), 2.33 (t, J = 7.0 Hz, 2 H, CH₂), 1.59 (quint, J = 7.2 Hz, 2 H, CH₂), 1.44-1.24 (m, 4 H, 2 x CH₂), 0.91 (t, J = 7.0 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 154.2, 89.8, 72.7, 52.4, 30.8, 27.1, 22.0, 18.5, 13.7; IR (neat, cm⁻¹): ν = 2956, 2863, 2236, 1713, 1458, 1434, 1247, 1075; MS (ESI) m/z : 155 (M+H)⁺.

(35) Preparation of methyl 3-phenylbutanoate (**6Ie**, yyb-12-158)



Following Typical Procedure II, the reaction of **1I** (150.0 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.3 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.3 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Ie**²⁰ (77.6 mg, 44%) [eluent: petroleum ether/ethyl acetate = 60/1 (~300 mL) to 40/1 (~200 mL)] as light yellow oil. ¹H NMR (400 MHz, CDCl₃): δ = 7.34-7.26 (m, 2 H, Ar-H), 7.25-7.15 (m, 3 H, Ar-H), 3.62 (s, 3 H, OCH₃), 3.36-3.20 (m, 1 H, CH), 2.63 (dd, J_1 = 15.2 Hz, J_2 = 6.8 Hz, 1 H, one proton of CH₂), 2.54 (dd, J_1 = 15.0 Hz, J_2 = 8.2 Hz, 1 H, one proton of CH₂), 1.30 (d, J = 6.8 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 172.8, 145.7, 128.5, 126.7, 126.4, 51.4, 42.7, 36.4, 21.7; IR (neat, cm⁻¹): ν = 3029, 2964, 2930, 1735, 1495, 1436, 1378, 1290, 1164, 1085, 1022; MS (ESI) m/z : 179 (M+H)⁺.

(36) Preparation of methyl 2-methyldecanoate (**6Je**, yyb-12-157)



Following Typical Procedure II, the reaction of **1J** (172.8 mg, 1.0 mmol), Fe(NO₃)₃·9H₂O (24.7 mg, 0.06 mmol), TEMPO (7.9 mg, 0.05 mmol), BiCl₃ (31.5 mg, 0.1 mmol), and MeOH (202 μ L, 5 mmol) in DCE (3 mL) afforded **6Je**²¹ (62.3 mg, 31%)

[eluent: petroleum ether/ethyl acetate = 50/1 (~410 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.50-2.37 (m, 1 H, CH), 1.71-1.57 (m, 1 H, one proton of CH₂), 1.46-1.35 (m, 1 H, one proton of CH₂), 1.35-1.17 (m, 12 H, 6 x CH₂), 1.14 (d, *J* = 6.8 Hz, 3 H, CH₃), 0.88 (t, *J* = 6.6 Hz, 3 H, CH₃); **¹³C NMR** (100 MHz, CDCl₃): δ = 177.4, 51.4, 39.4, 33.8, 31.8, 29.5, 29.4, 29.2, 27.2, 22.6, 17.0, 14.0; **IR** (neat, cm⁻¹): ν = 2953, 2925, 2855, 1738, 1463, 1434, 1378, 1195, 1164; **MS** (ESI) *m/z*: 201 (M+H)⁺.

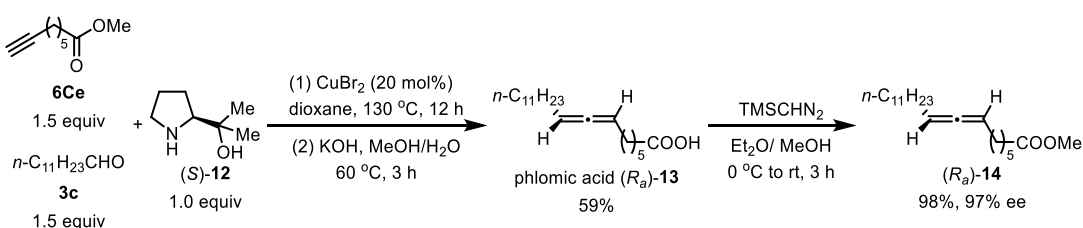
Total synthesis of phlomic acid:

(1) Preparation of methyl hept-6-ynoate (**6Ce**, yyb-7-156A)



Following Typical Procedure II, the reaction of **1C** (631.8 mg, 5.0 mmol), Fe(NO₃)₃·9H₂O (121.6 mg, 0.3 mmol), TEMPO (40.0 mg, 0.25 mmol), BiCl₃ (157.1 mg, 0.5 mmol), MgSO₄ (249.6 mg, 2.1 mmol), and MeOH (0.6 mL, 15 mmol) in DCE (15 mL) afforded **6Ce**¹⁵ (516.2 mg, 65%, 97% purity) [eluent: petroleum ether/ethyl acetate = 49/1 (~600 mL) to 24/1 (~300 mL)] as light yellow oil. **¹H NMR** (400 MHz, CDCl₃): δ = 3.67 (s, 3 H, OCH₃), 2.33 (t, *J* = 7.4 Hz, 2 H, CH₂), 2.25-2.13 (m, 2 H, CH₂), 1.99-1.91 (m, 1 H, CH), 1.65 (quint, *J* = 7.3 Hz, 2 H, CH₂), 1.55 (quint, *J* = 7.0 Hz, 2 H, CH₂), 1.50-1.38 (m, 2 H, CH₂); **¹³C NMR** (100 MHz, CDCl₃): δ = 174.0, 84.2, 68.3, 51.4, 33.8, 28.1, 28.0, 24.3, 18.1.

(2) Preparation of (*R*)-7,8-Eicosadienoic acid (phlomic acid) ((*R_a*)-**13**, yyb-7-158, yyb-7-160, yyb-7-163)



To an oven dried screw cap reaction tube were added CuBr₂ (90.6 mg, 0.4 mmol),

(*S*)-**12** (271.6 mg, 95% purity, 2.0 mmol), **6Ce** (464.3 mg, 3.0 mmol)/dioxane (3 mL), and **3c** (573.4 mg, 97% purity, 3.0 mmol)/dioxane (3 mL) sequentially. The resulting mixture was stirred at 130 °C for 12 h as monitored by TLC, diluted with 60 mL of ethyl ether, washed with an aqueous solution of HCl (3 M, 40 mL), extracted with ethyl ether (20 mL x 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (*R_a*)-**14** (397.4 mg) [eluent: petroleum ether/ethyl acetate = 99/1 (~1200 mL)] as light yellow oil.

To a round bottom flask were added KOH (198.3 mg, 3.0 mmol, 85% purity), (*R_a*)-**14** (397.4 mg), and MeOH/H₂O (12 mL, 4:1/v:v) sequentially. The resulting mixture was stirred at 60 °C for 3 h as monitored by TLC. HCl (3 M, 1.2 mL) was added dropwise at 0 °C with an ice-water bath. The resulting mixture was evaporated to remove MeOH, treated with DCM (30 mL) and H₂O (25 mL), extracted with DCM (20 mL x 3), washed with brine, and dried over anhydrous Na₂SO₄. After filtration and evaporation, the residue was purified by chromatography on silica gel to afford (*R_a*)-**13**¹⁵ (361.8 mg, 59%) [eluent: petroleum ether/ethyl acetate = 10/1 (~440 mL) to 4/1 (~370 mL)] as light yellow solid: m.p. 30.4-31.5 °C (We were not able to recrystallize from the solvents tested, the m.p. of (*R_a*)-**13** was determined by using the solid right after evaporation of the eluent). $[\alpha]_D^{27} = -41.3$ (c = 0.94, CHCl₃) (reported:¹⁵ $[\alpha]_D^{30.5} = -40.7$ (c = 1.03, CHCl₃)); ¹H NMR (400 MHz, CDCl₃) δ = 5.12-5.01 (m, 2 H, CH=C=CH), 2.35 (t, *J* = 7.6 Hz, 2 H, CH₂), 2.04-1.92 (m, 4 H, 2 x CH₂), 1.65 (quint, *J* = 7.3 Hz, 2 H, CH₂), 1.48-1.18 (m, 22 H, 11 x CH₂), 0.88 (t, *J* = 6.6 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 203.9, 180.5, 91.1, 90.5, 34.1, 31.9, 29.6, 29.5, 29.4, 29.2, 29.1, 29.0, 28.71, 28.68, 28.5, 24.5, 22.7, 14.1.

The ee of (*R_a*)-**13** was determined by HPLC analysis after being converted to (*R_a*)-**14**: To the flask containing (*R_a*)-**13** (62.0 mg, 0.2 mmol) was added ethyl ether/MeOH (5.0 mL, 4:1/v:v). TMSCHN₂ (0.15 mL, 2.0 M in hexane, 0.3 mmol) was added dropwise at 0 °C. The resulting mixture was warmed up to room temperature and then stirred for 3 h as monitored by TLC. The solvent was removed under reduced pressure

and the residue was purified by chromatography on silica gel to afford (*R_a*)-**14**¹⁵ (63.5 mg, 98%, 97% ee) [eluent: petroleum ether/ethyl acetate = 100/1 (~200 mL) to 20/1 (~210 mL)] as colorless oil; HPLC conditions: LC column, hexane, 1.5 mL/min, λ = 214 nm, t_R (major) = 8.1 min, t_R (minor) = 9.8 min; $[\alpha]_D^{27}$ = -39.3 (c = 1.075, CHCl₃) (reported:¹⁵ $[\alpha]_D^{30.5}$ = -39.9 (c = 0.99, CHCl₃)); ¹H NMR (400 MHz, CDCl₃) δ = 5.16-4.98 (m, 2 H, CH=C=CH), 3.66 (s, 3 H, OCH₃), 2.31 (t, J = 7.4 Hz, 2 H, CH₂), 2.05-1.90 (m, 4 H, 2 x CH₂), 1.63 (quint, J = 7.5 Hz, 2 H, CH₂), 1.48-1.16 (m, 22 H, 11 x CH₂), 0.88 (t, J = 6.6 Hz, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 203.8, 174.2, 91.1, 90.5, 51.4, 34.0, 31.9, 29.65, 29.63, 29.62, 29.5, 29.3, 29.2, 29.1, 29.0, 28.73, 28.72, 28.6, 24.8, 22.7, 14.1.

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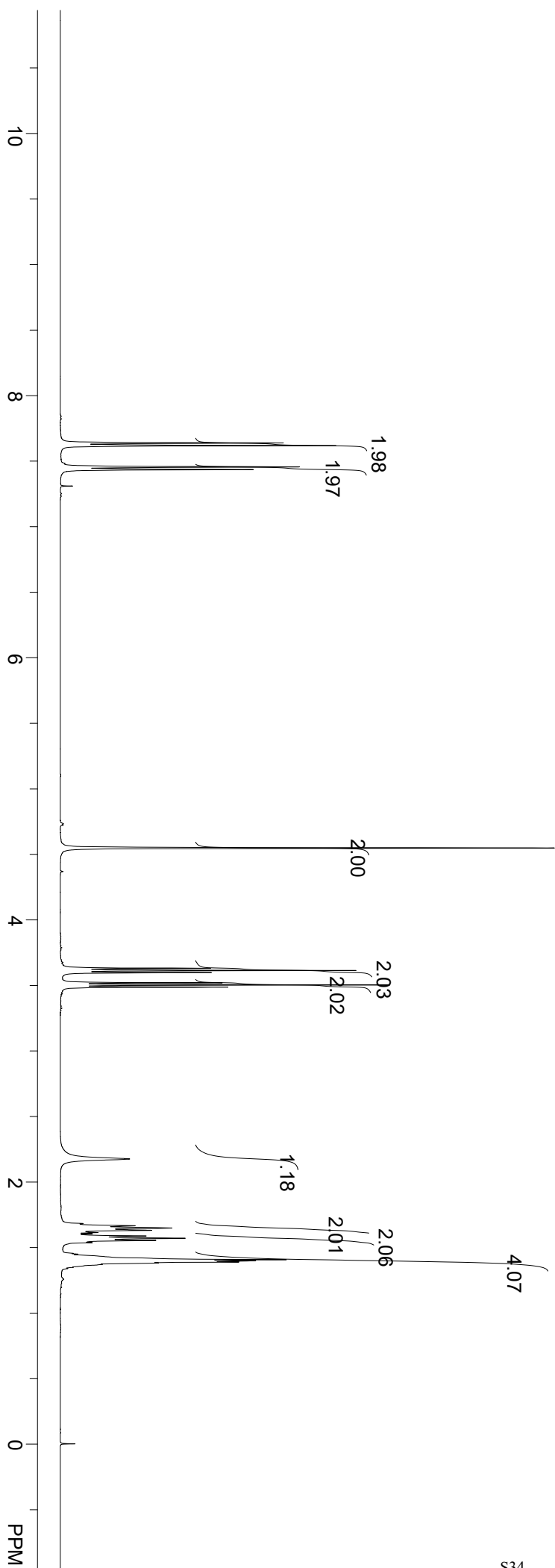
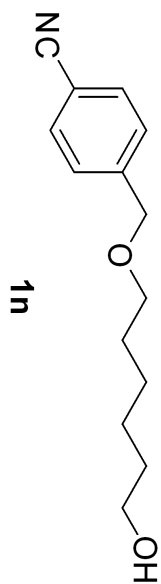
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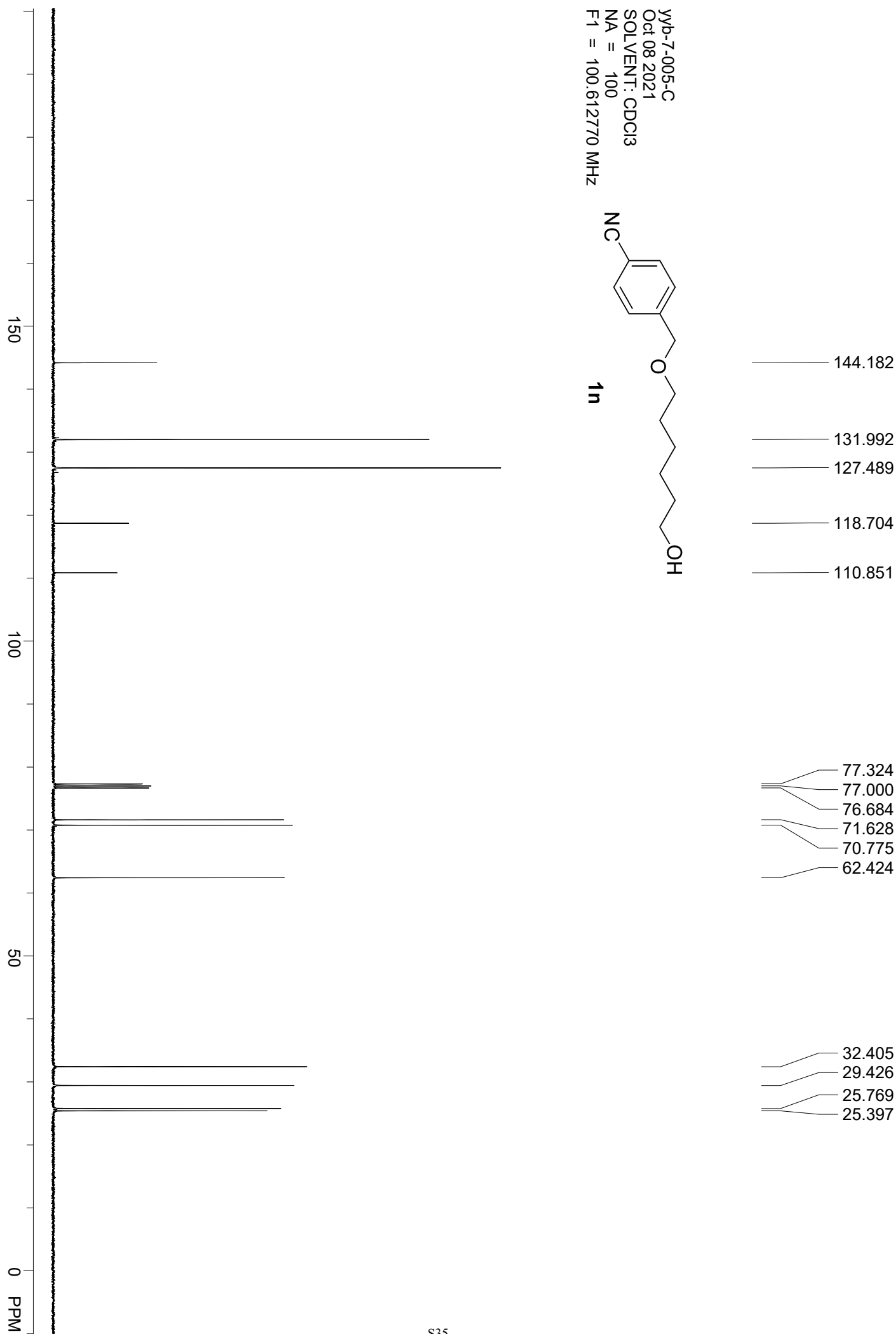
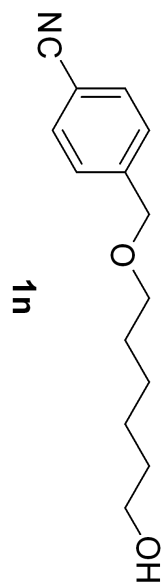
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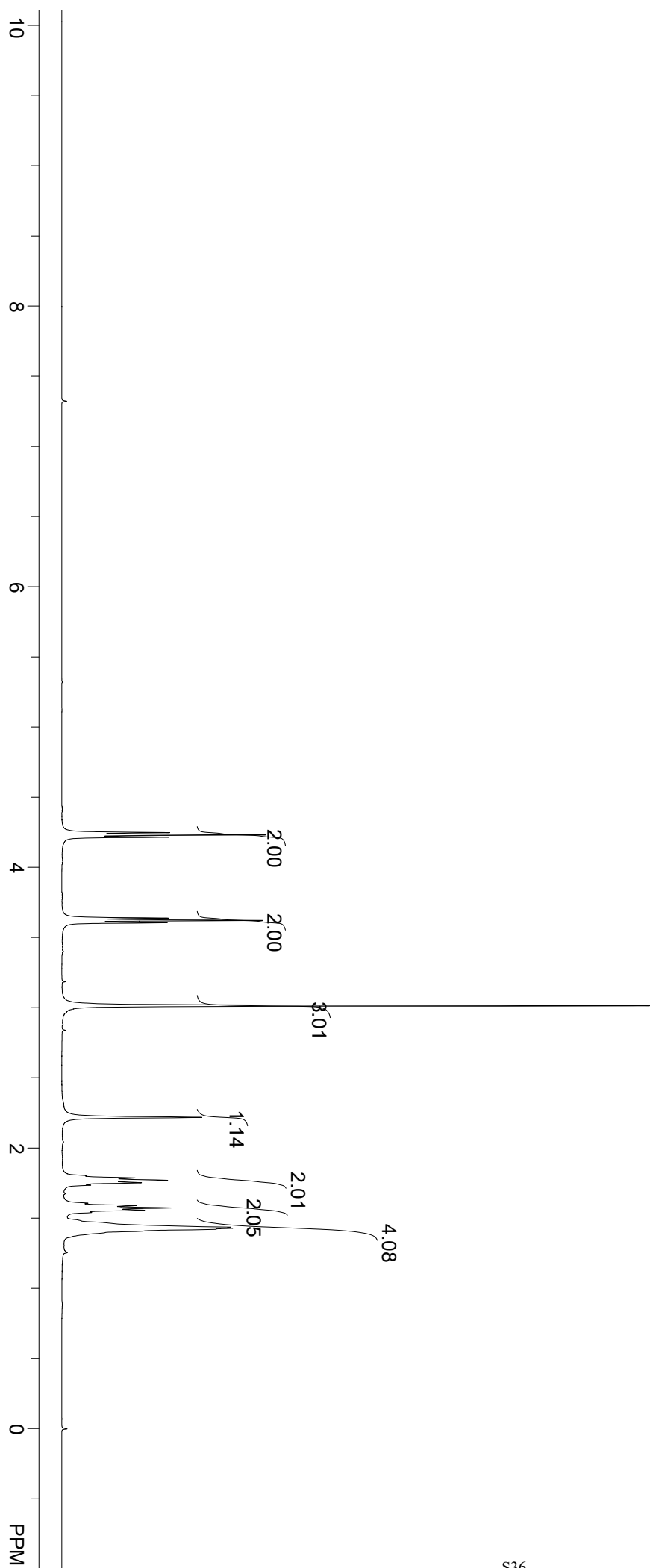
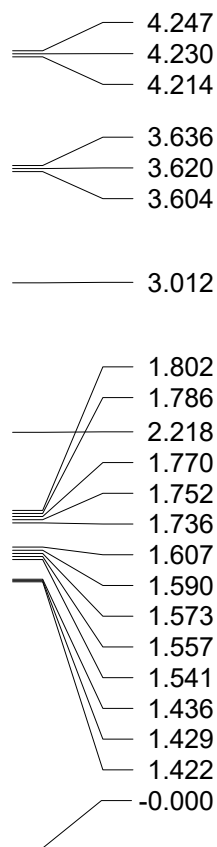
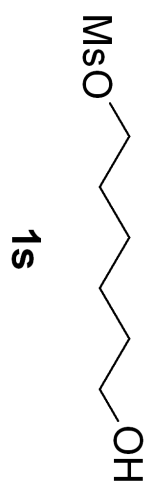
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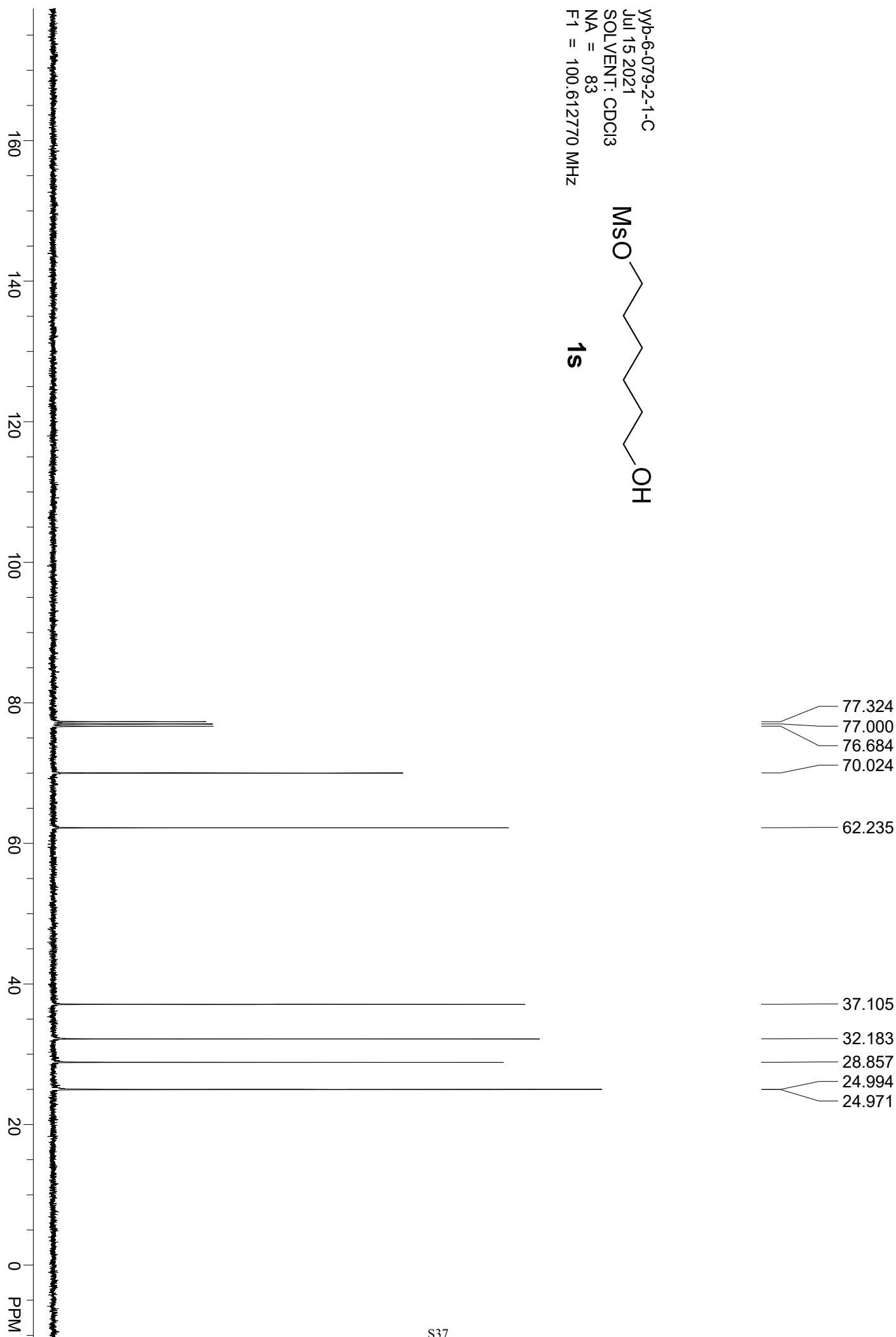
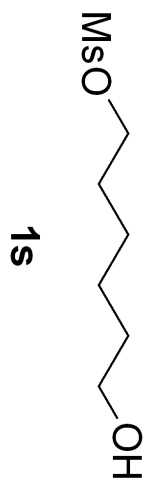
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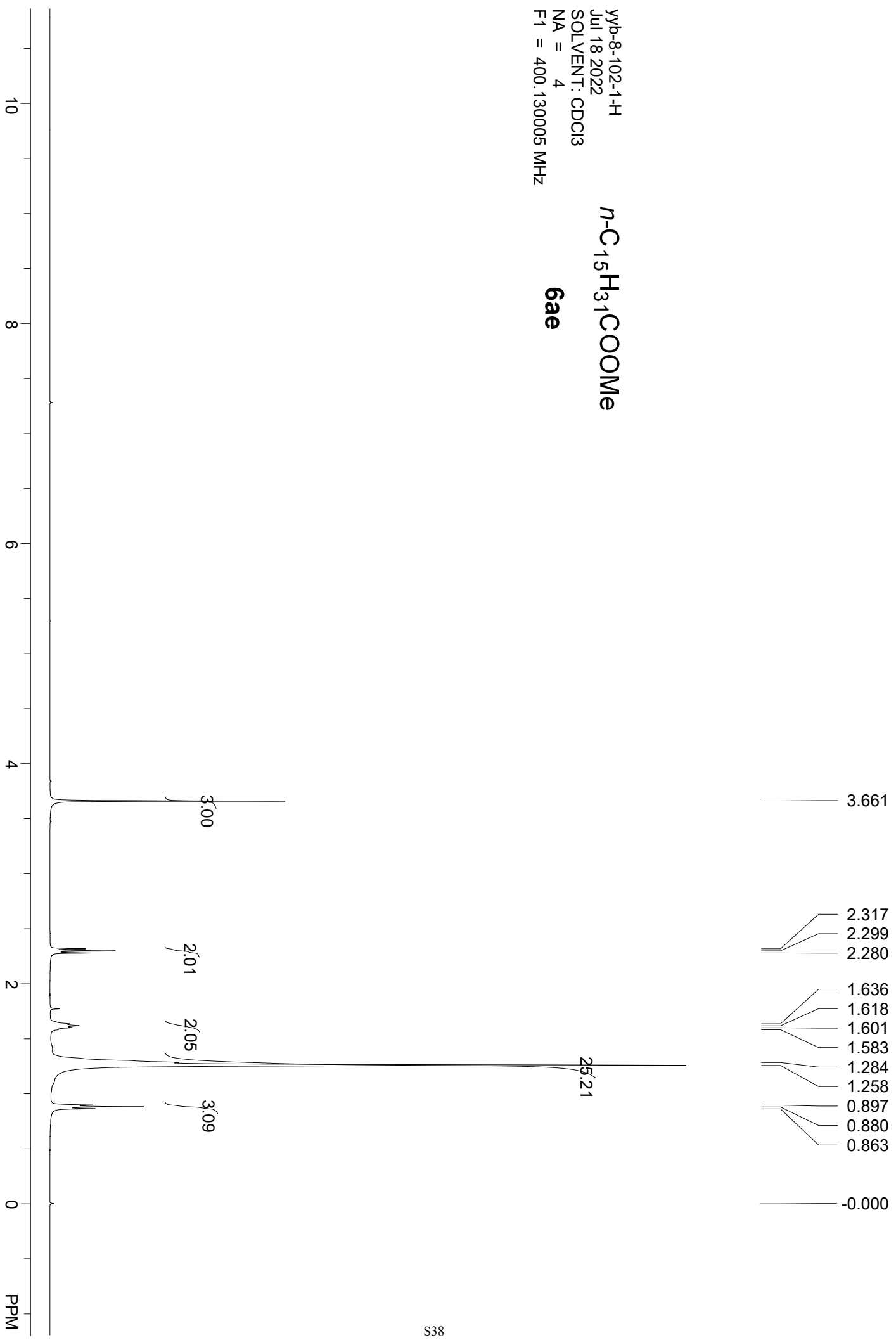
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NA = 4
F1 = 400.130005 MHz

n-C₁₅H₃₁COOMe

6ae

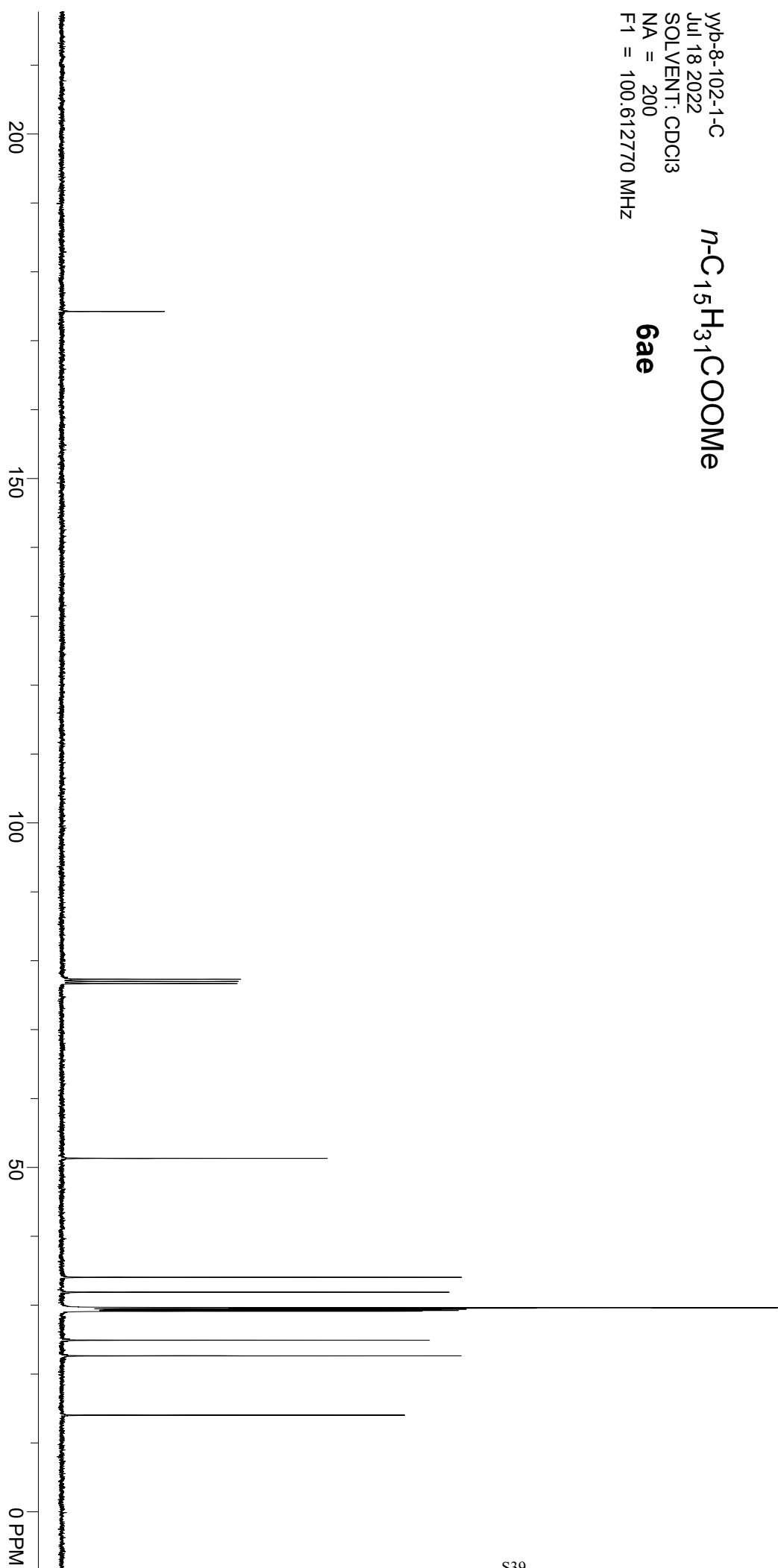


174.241

77.316
77.000
76.676

51.309
34.048
31.883
29.647
29.632
29.616
29.553
29.410
29.324
29.221
29.118
24.915
22.640
14.029

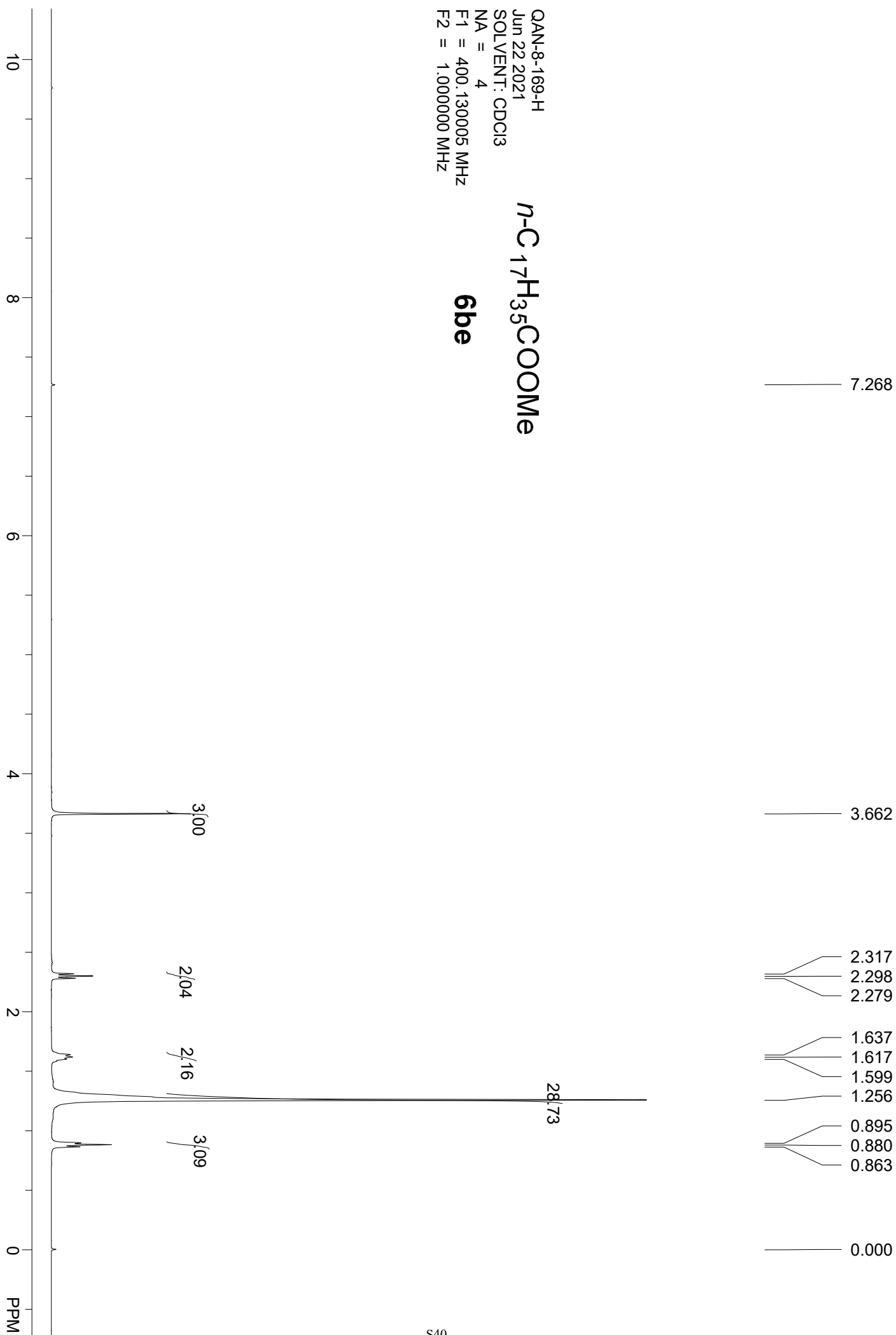
yyb-8-102-1-C
Jul 18 2022
SOLVENT: CDCl₃
NA = 200
F1 = 100.612770 MHz
***n*-C₁₅H₃₁COOMe**
6ae



QAN-8-169-H
Jun 22 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz
F2 = 1.000000 MHz

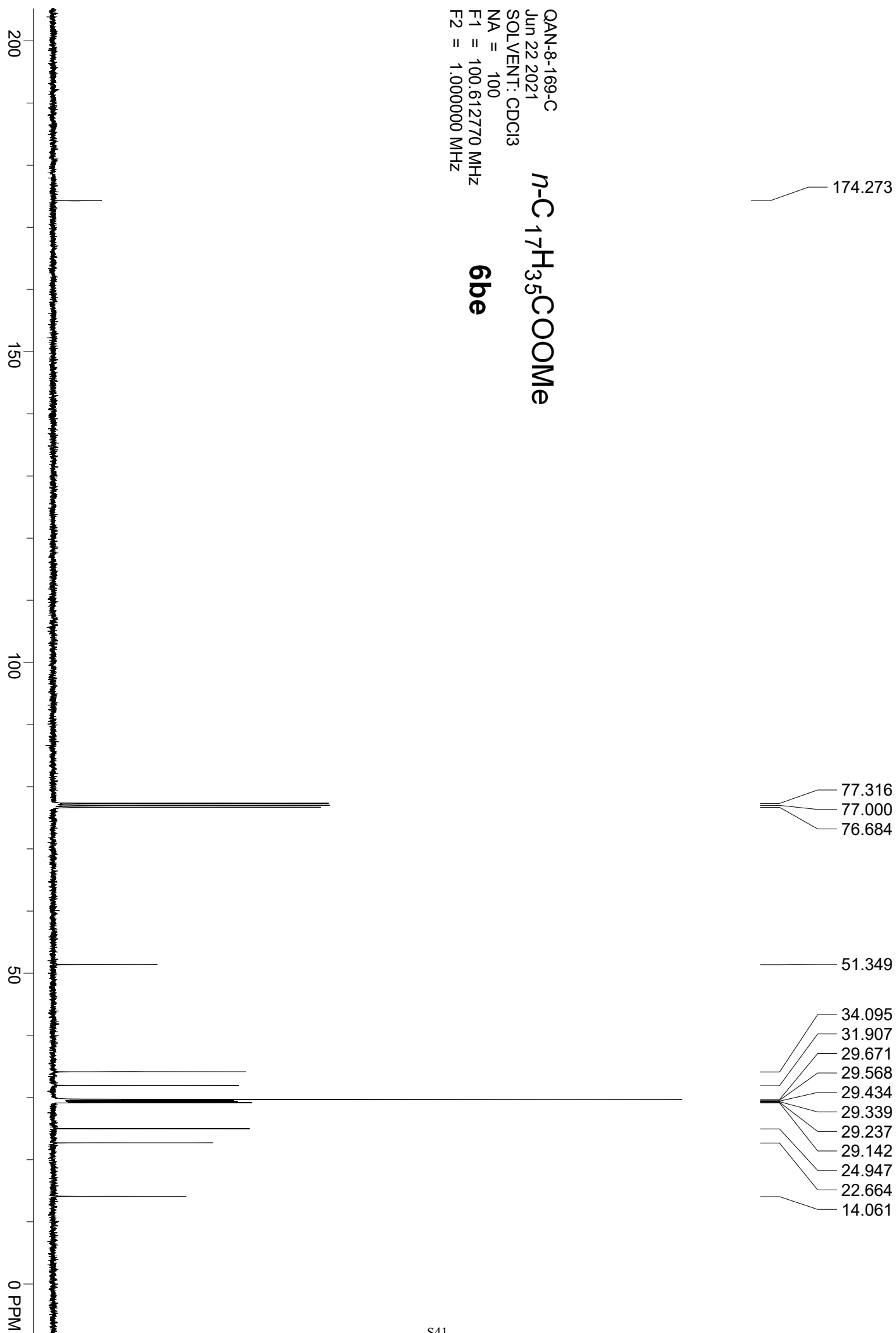
n-C₁₇H₃₅COOMe

6be



QAN-8-169-C
Jun 22 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz
F2 = 1.000000 MHz

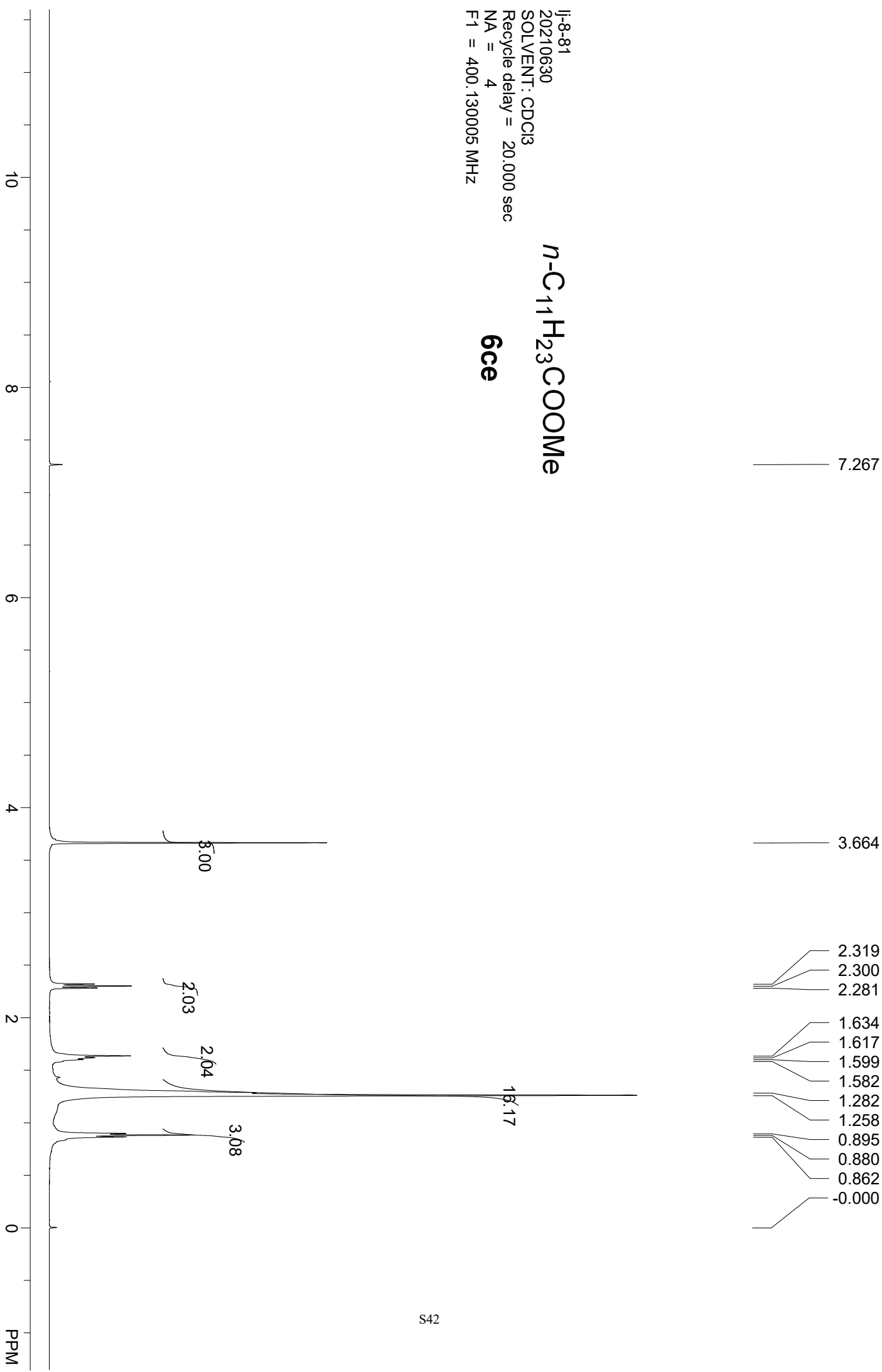
n-C₁₇H₃₅COOMe
6be



lj-8-81
20210630
SOLVENT: CDCl3
Recycle delay = 20.000 sec
NA = 4
F1 = 400.130005 MHz

n-C₁₁H₂₃COOMe

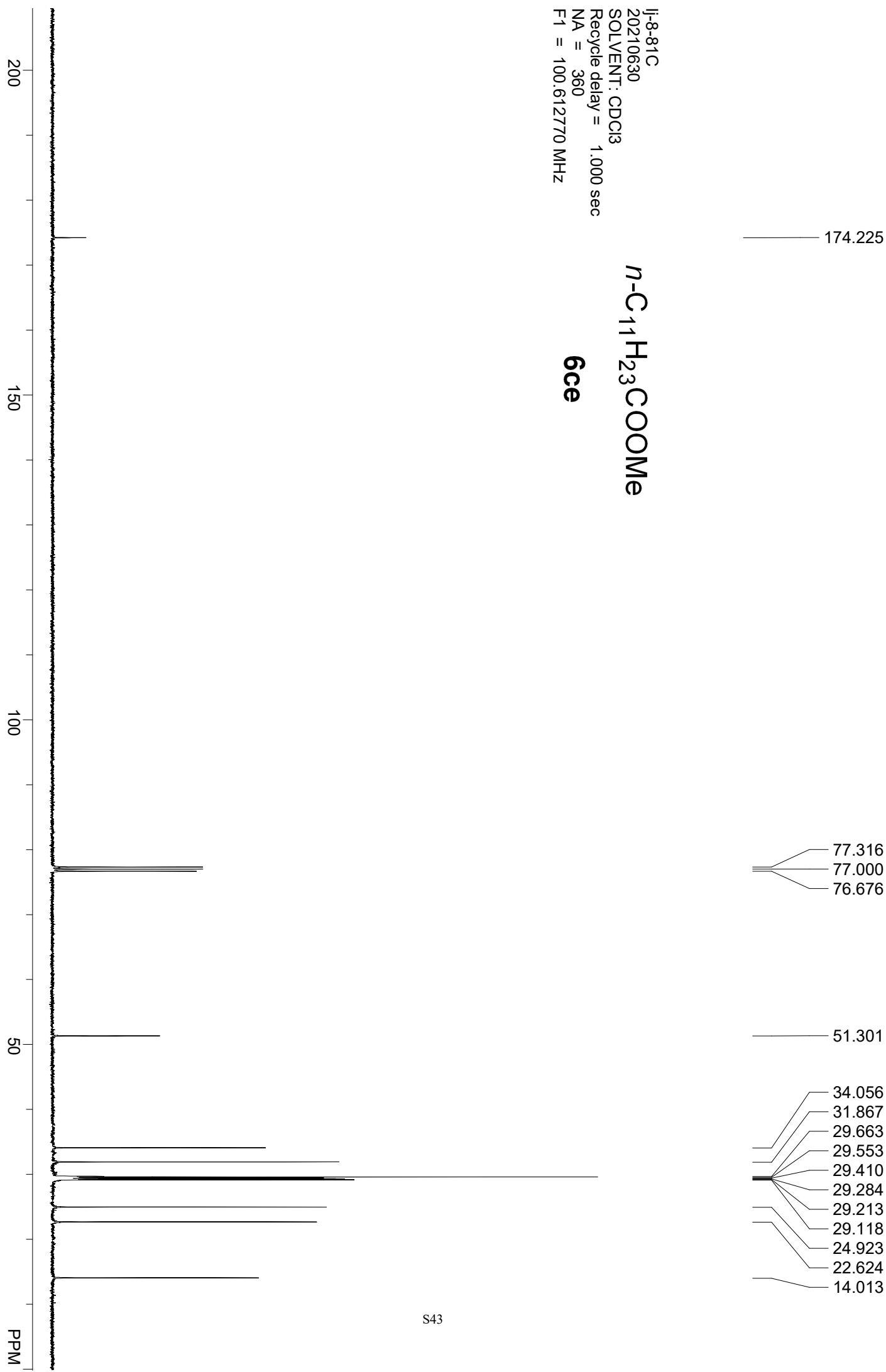
6ce



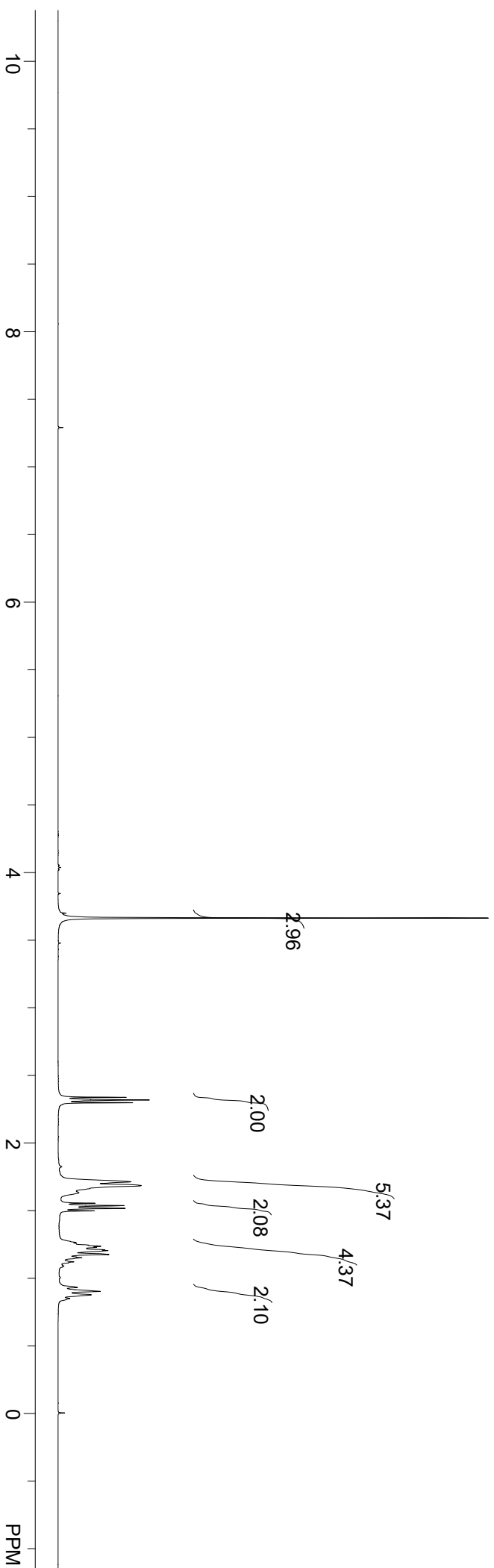
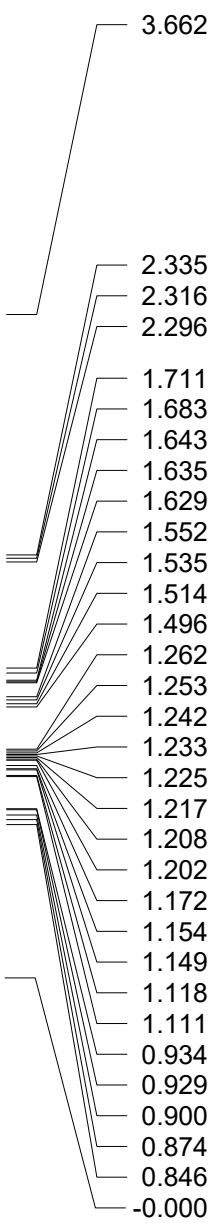
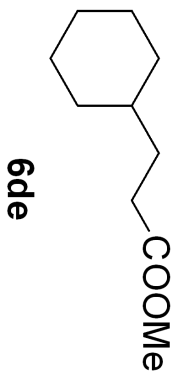
lj-8-81C
20210630
SOLVENT: CDCl₃
Recycle delay = 1.000 sec
NA = 360
F1 = 100.612770 MHz



6ce



YYB-7-092-H
Nov 30 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz



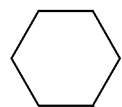
174.518

77.316
77.000
76.676

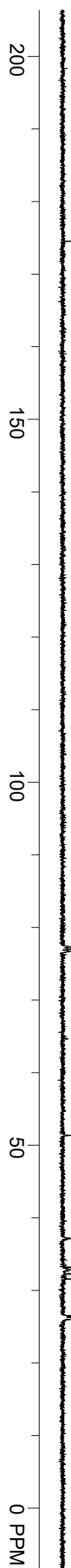
51.349

37.129
32.871
32.254
31.575
26.440
26.132

YYB-7-092-C
Nov 30 2021
SOLVENT: CDCl₃
NA = 64
F1 = 100.612770 MHz



6de

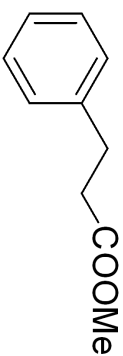


7.299
7.281
7.262
7.242
7.201
7.183

3.662

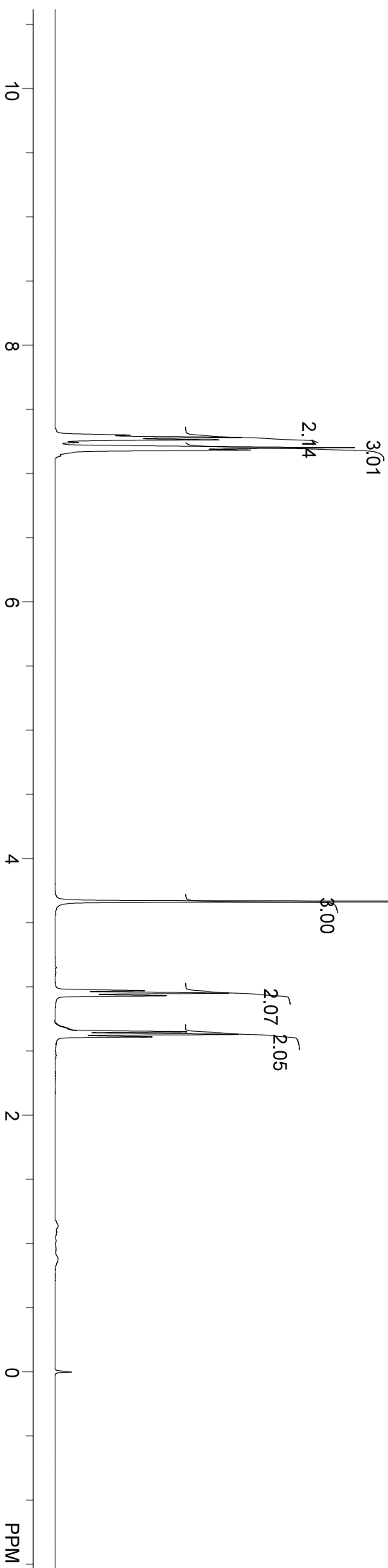
2.969
2.949
2.929
2.648
2.628
2.609

-0.000

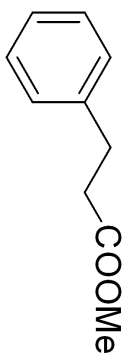


lj-8-82
20210624
SOLVENT: CDCl₃
Recycle delay = 20.000 sec
NA = 4
F1 = 400.130005 MHz

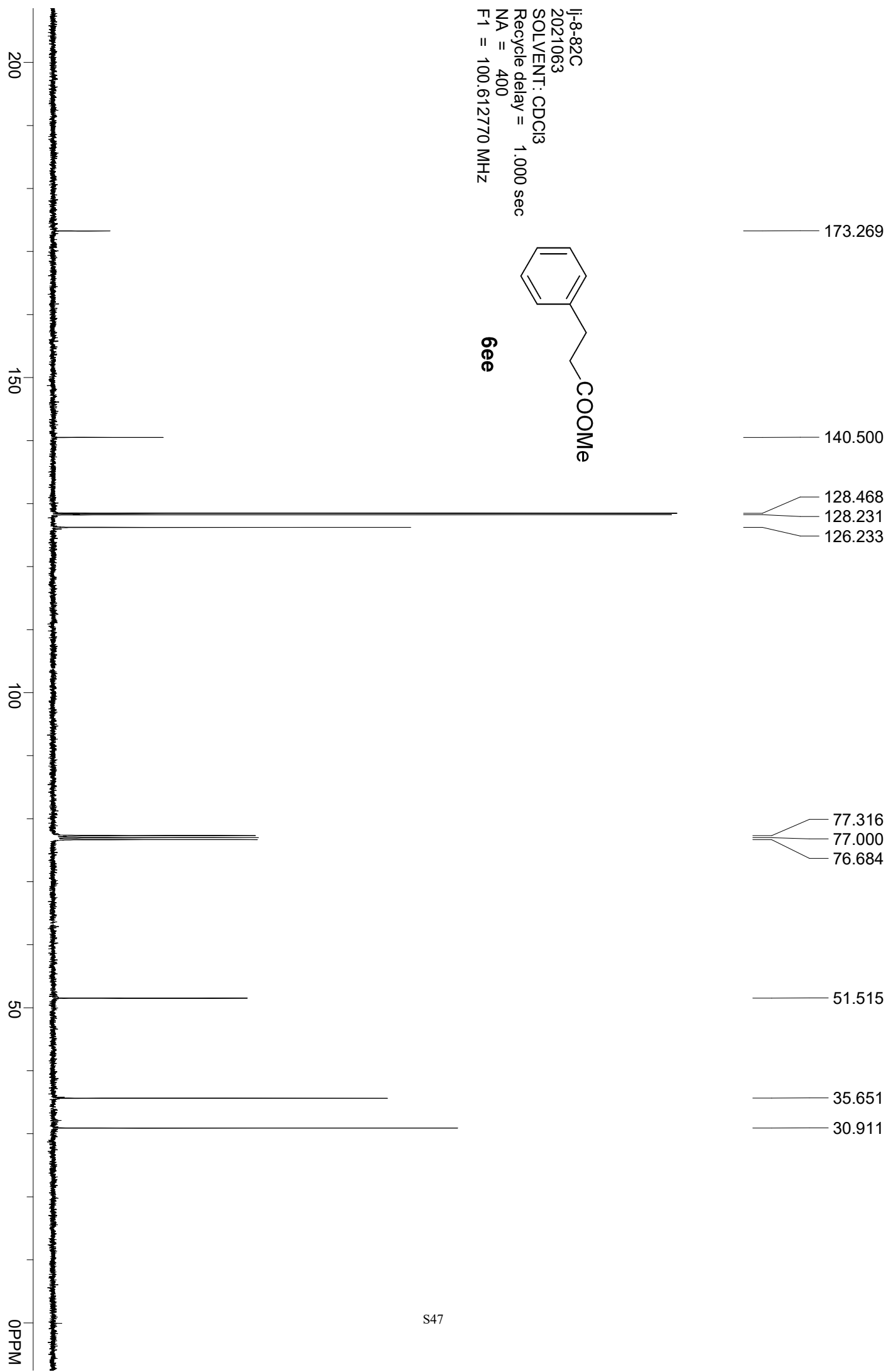
6ee



lj-8-82C
2021063
SOLVENT: CDCl₃
Recycle delay = 1.000 sec
NA = 400
F1 = 100.612770 MHz



6ee



7.284
7.265
7.247
7.189
7.171
7.152

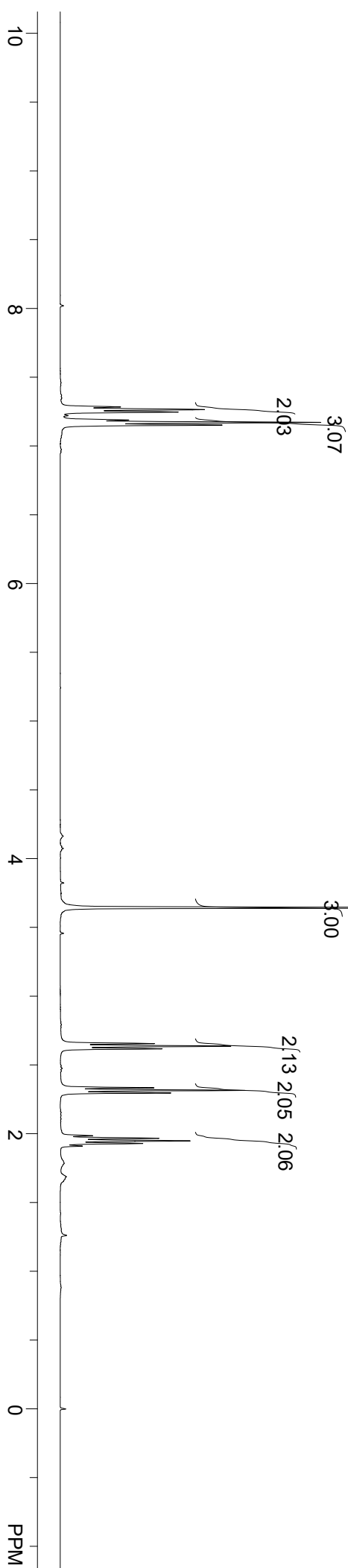
3.641

2.654
2.636
2.616
2.333
2.314
2.295
1.985
1.966
1.947
1.928
1.910

0.000



yyb-6-060-H
Jul 02 2021
SOLVENT : CDCl₃
NA = 4
F1 = 400.130035 MHz



7.296
7.277
7.259
7.201
7.182
7.164

4.918

3.658

2.666
2.648
2.629
2.347
2.328
2.310
1.995
1.976
1.957
1.939
1.920

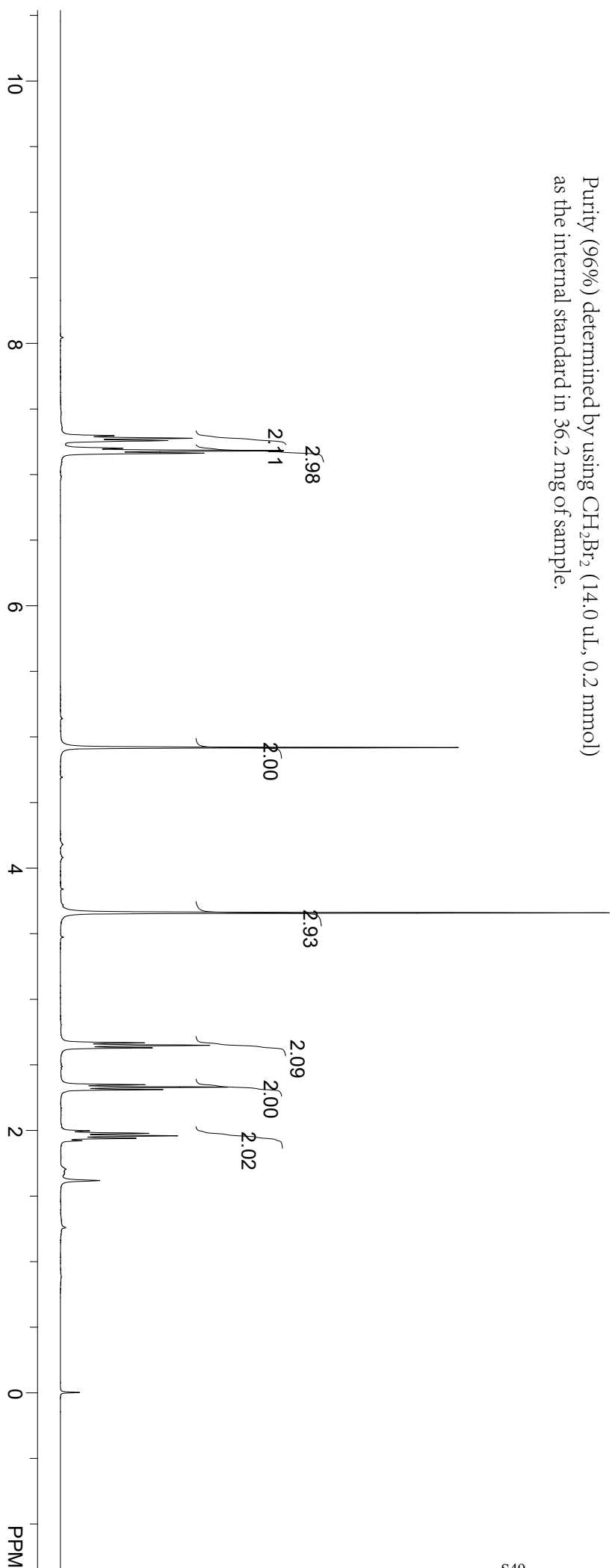
0.000

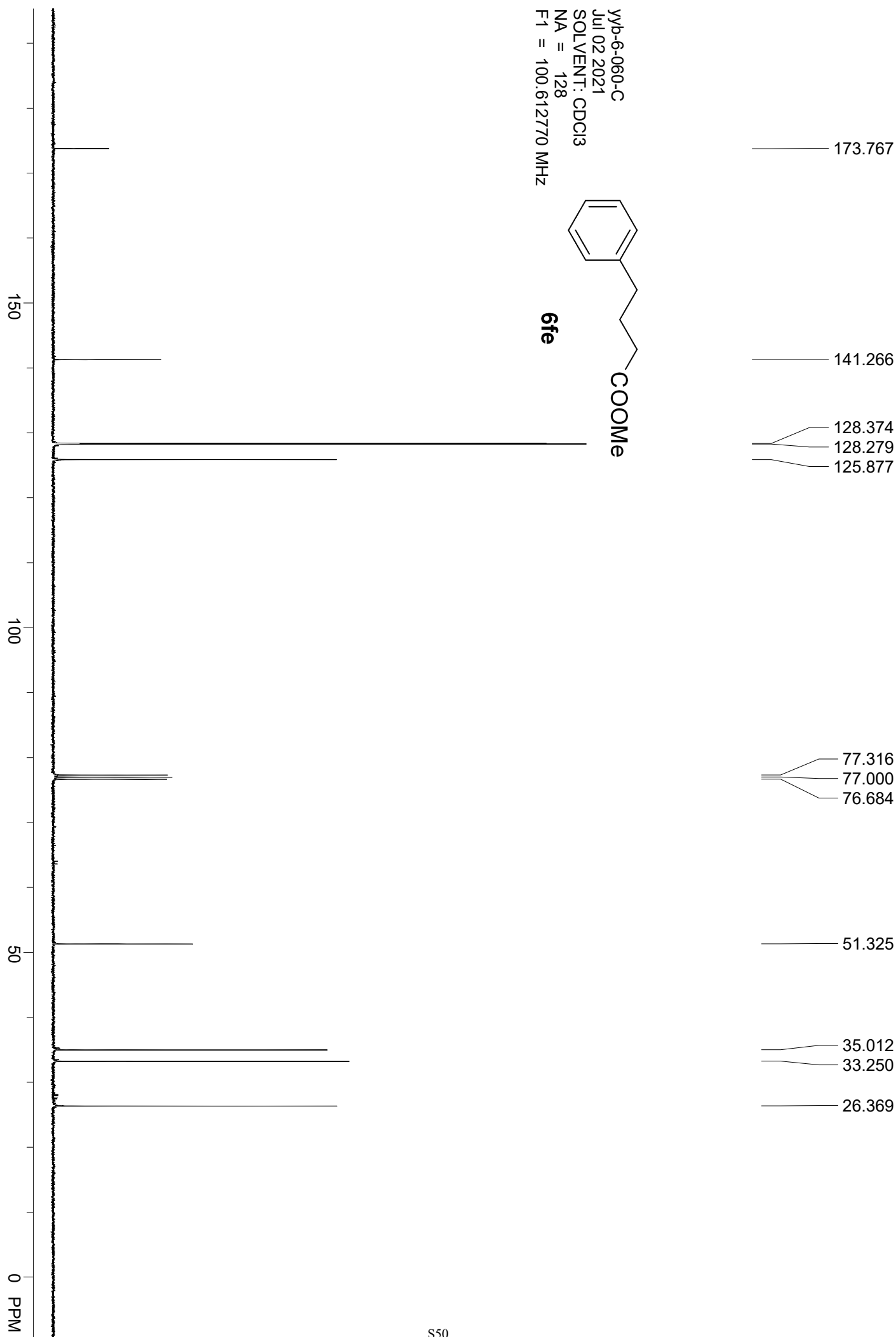


yyb-6-060-p
Jul 03 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz

6fe

Purity (96%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
as the internal standard in 36.2 mg of sample.



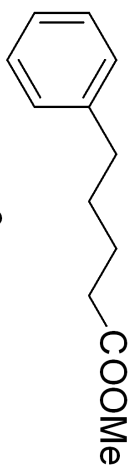


7.281
7.262
7.244
7.168
7.150

3.647

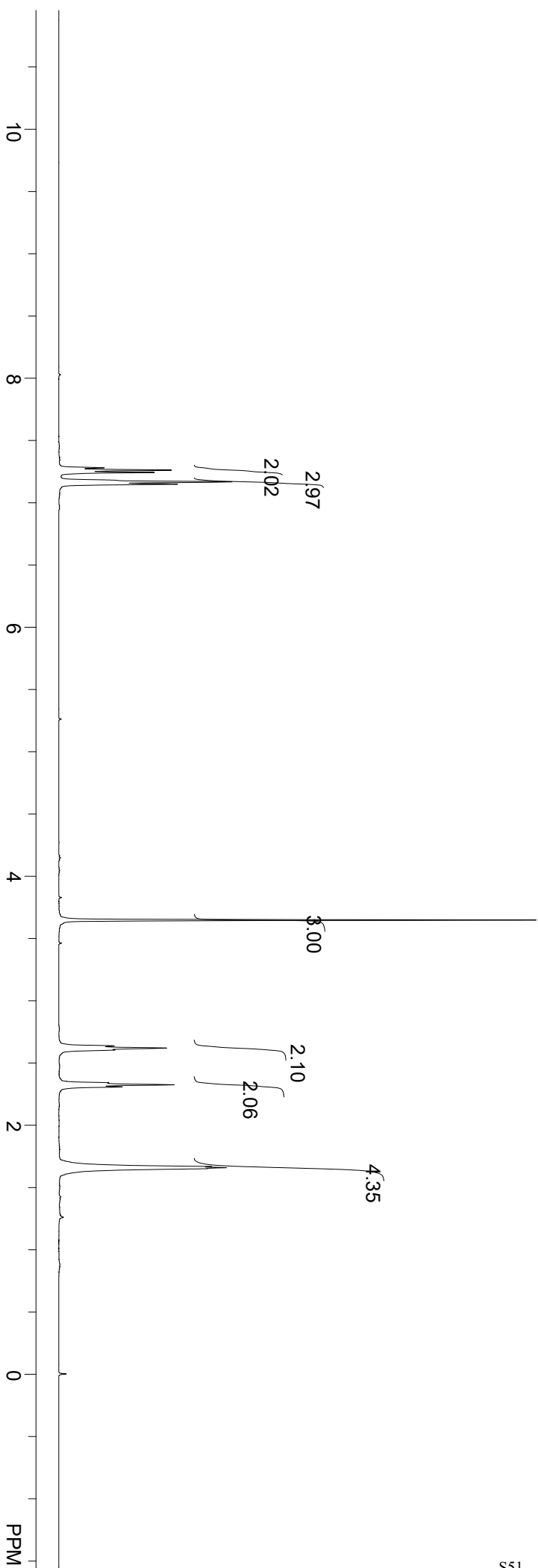
2.636
2.618
2.603
2.339
2.323
2.306
1.666
1.657
1.649

0.000

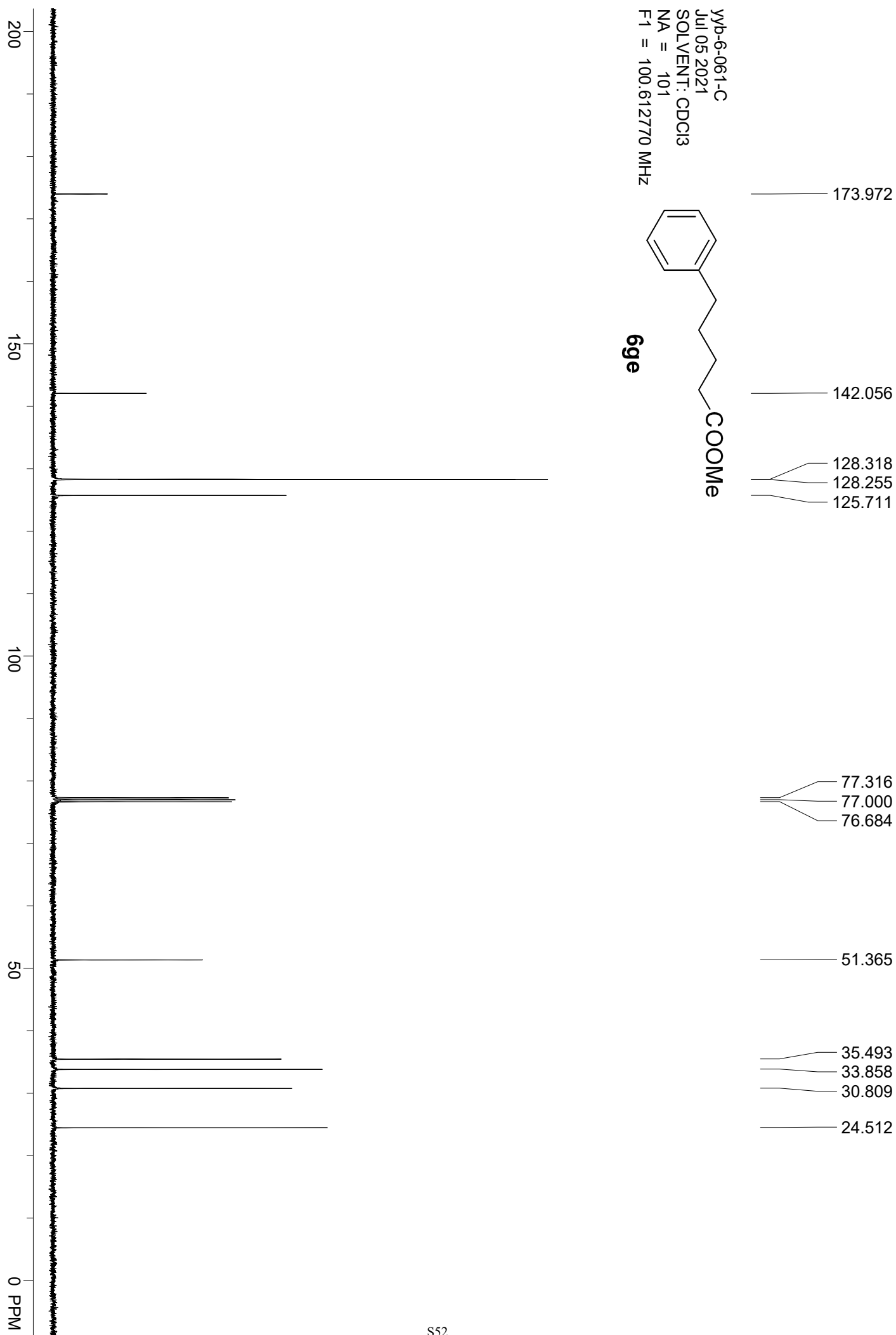
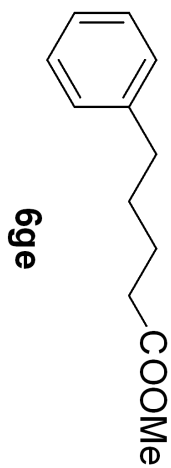


6ge

yyb-6-061-H
Jul 05 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130035 MHz



yyb-6-061-C
Jul 05 2021
SOLVENT: CDCl3
NA = 101
F1 = 100.612770 MHz



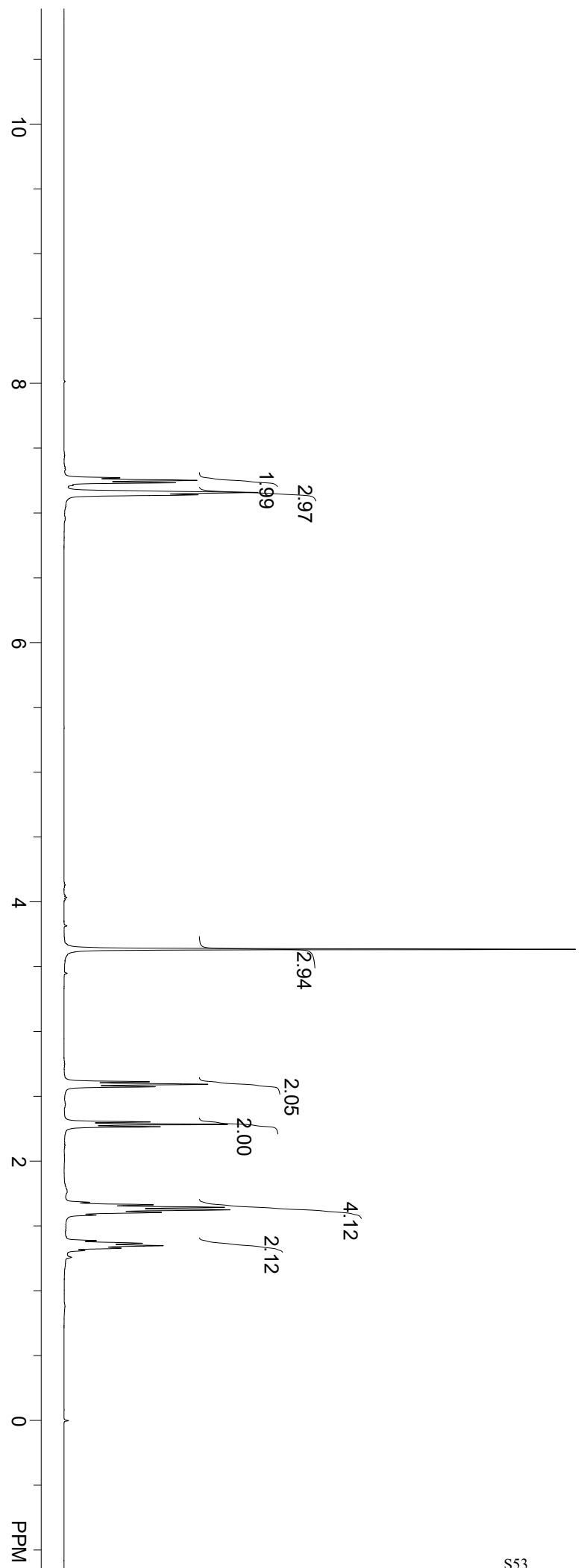
7.270
7.252
7.234
7.159
7.140

3.635
2.613
2.594
2.575
2.304
2.285
2.266
1.684
1.665
1.646
1.626
1.606
1.585
1.388
1.367
1.349
1.330
1.312
-0.000



6he

yyb-6-096-H
Jul 22 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130035 MHz



174.012

142.372

128.271
128.160
125.561

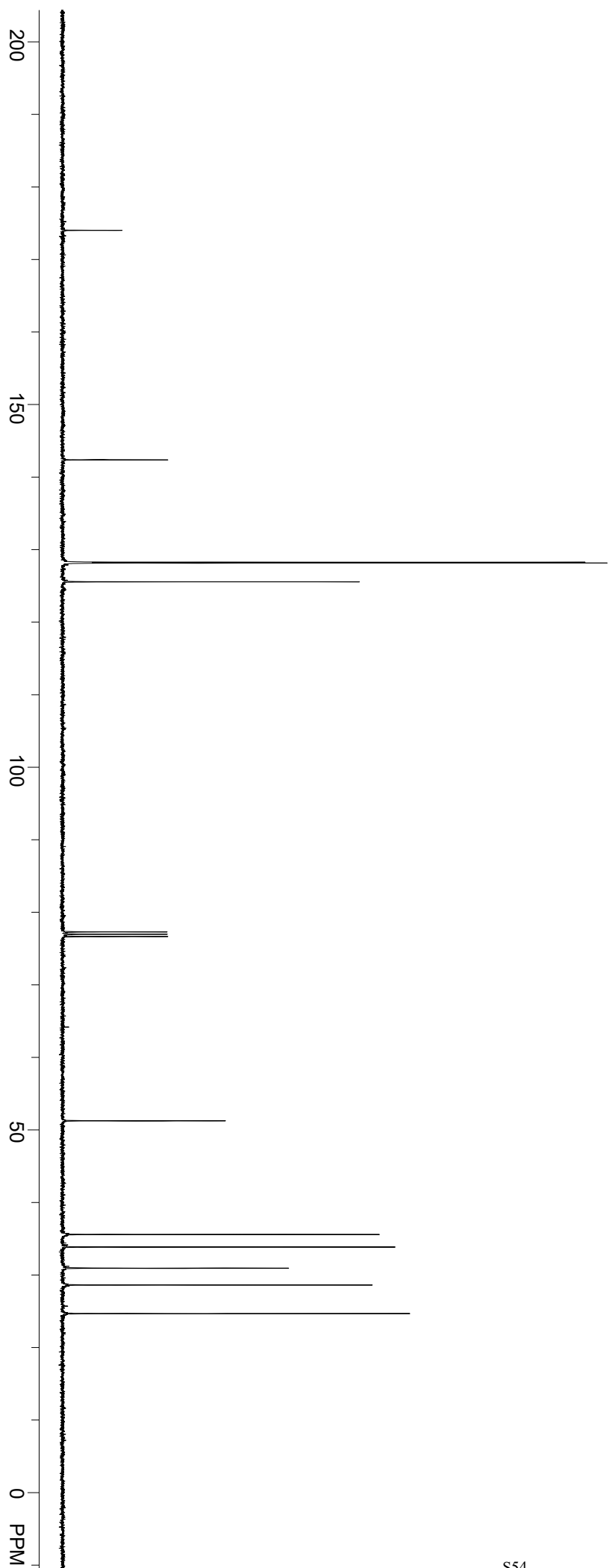
77.324
77.000
76.684

51.270

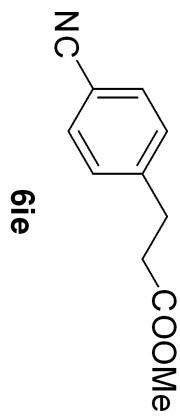
35.596
33.866
30.943
28.628
24.678



yyb-6-096-C
Jul 22 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz



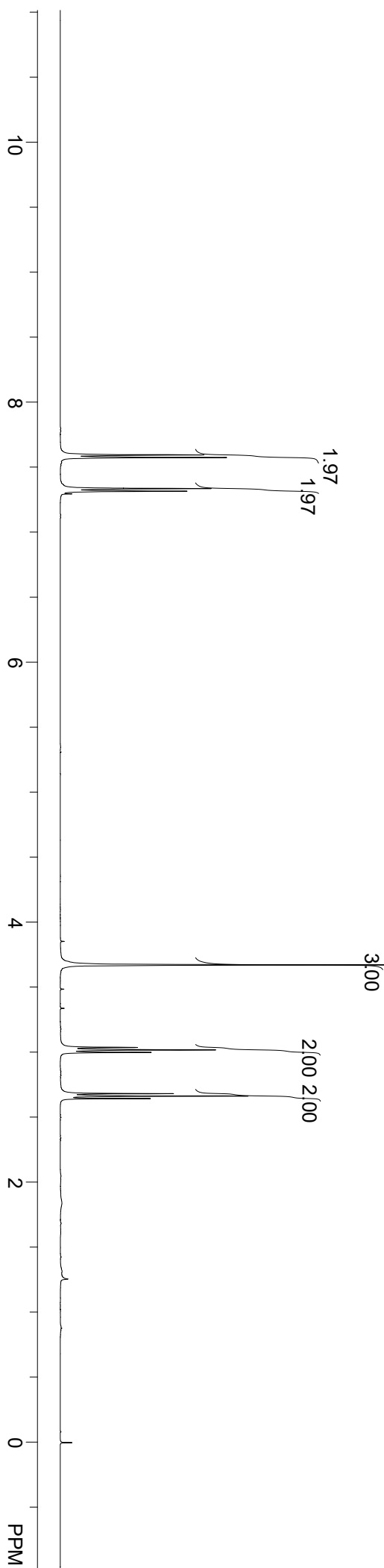
yyb-7-024-H
Oct 21 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



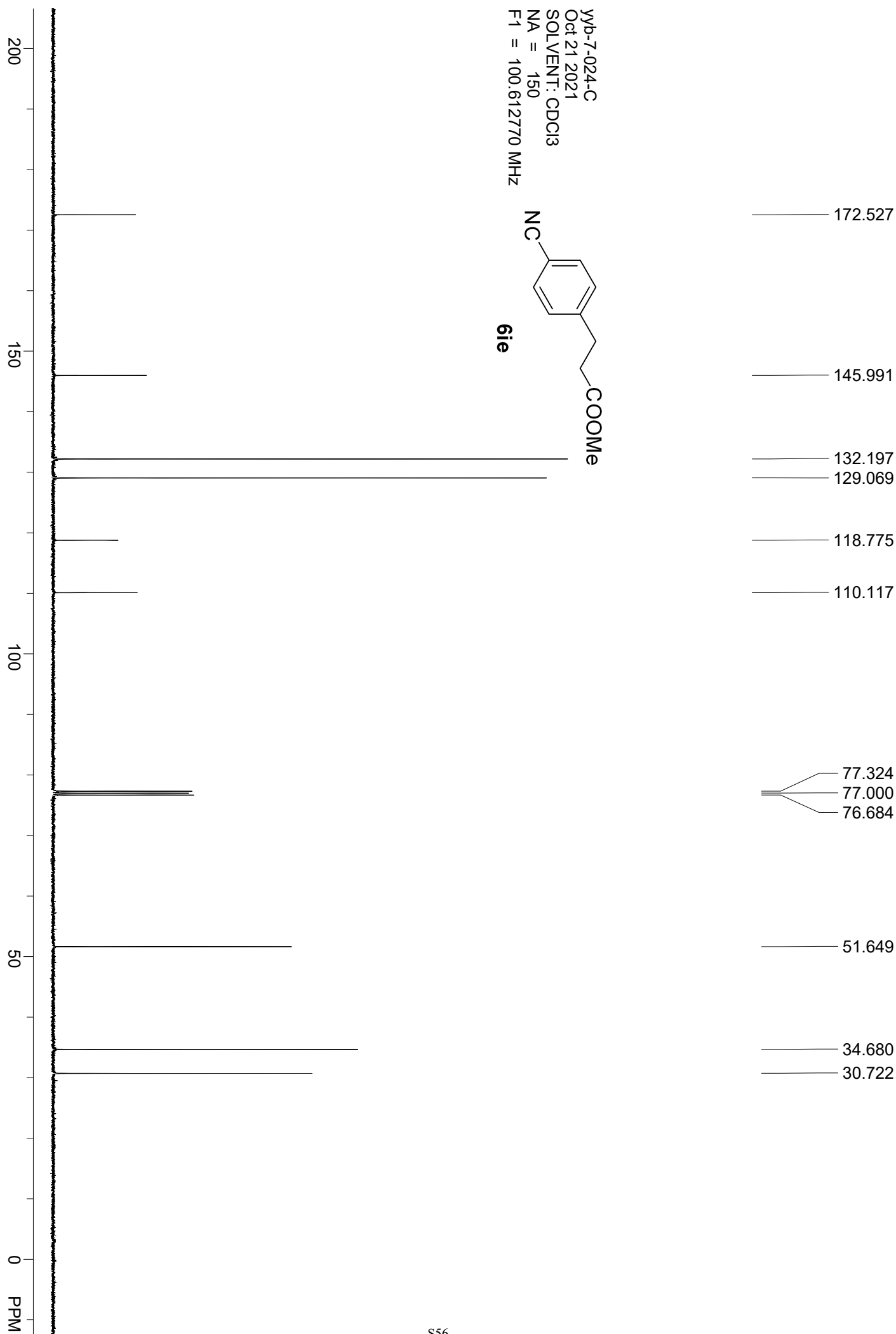
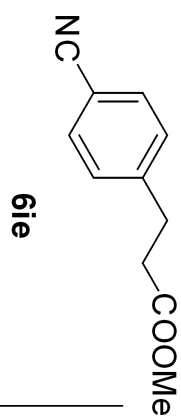
7.594
7.573
7.335
7.314

3.670
3.035
3.016
2.997
2.681
2.662
2.643

-0.000



yyb-7-024-C
Oct 21 2021
SOLVENT: CDCl₃
NA = 150
F1 = 100.612770 MHz



8.160
8.139

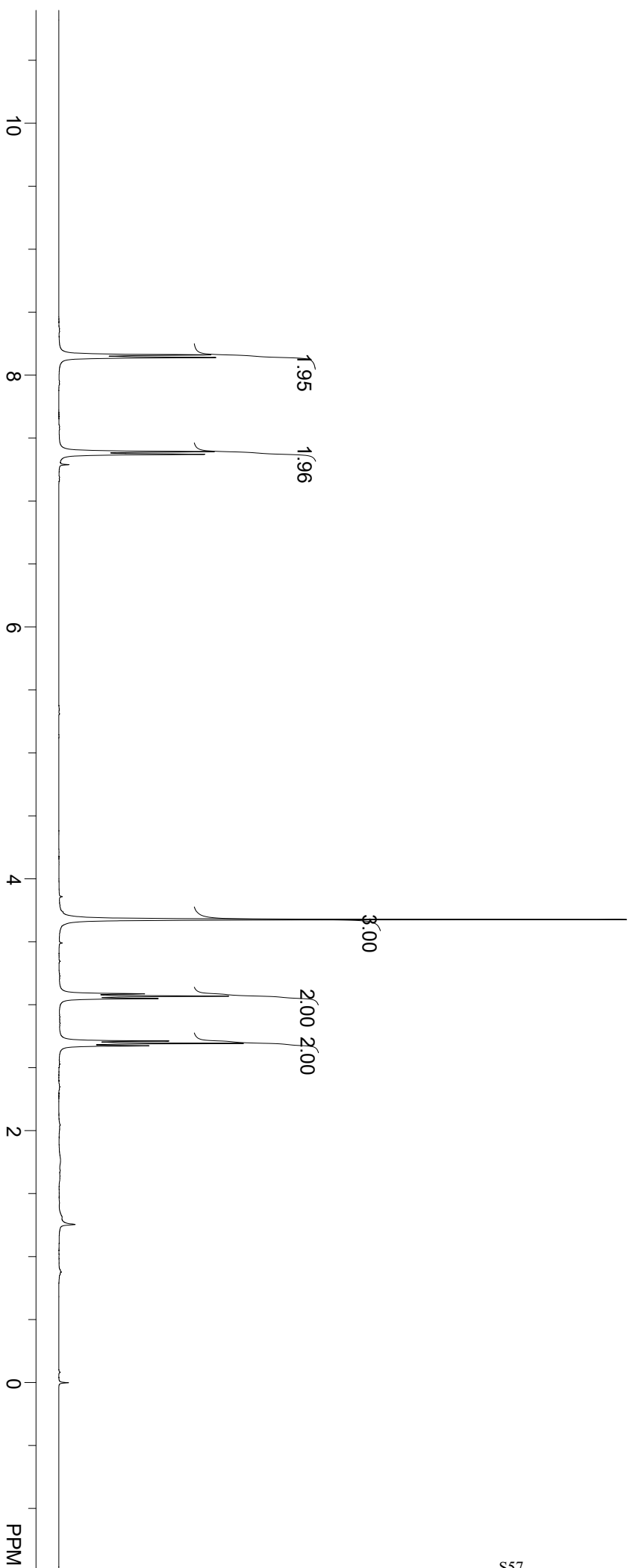
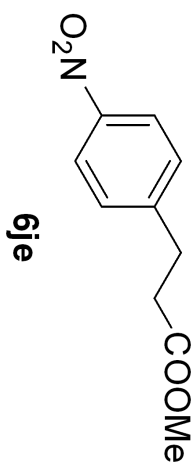
7.391
7.370

3.677

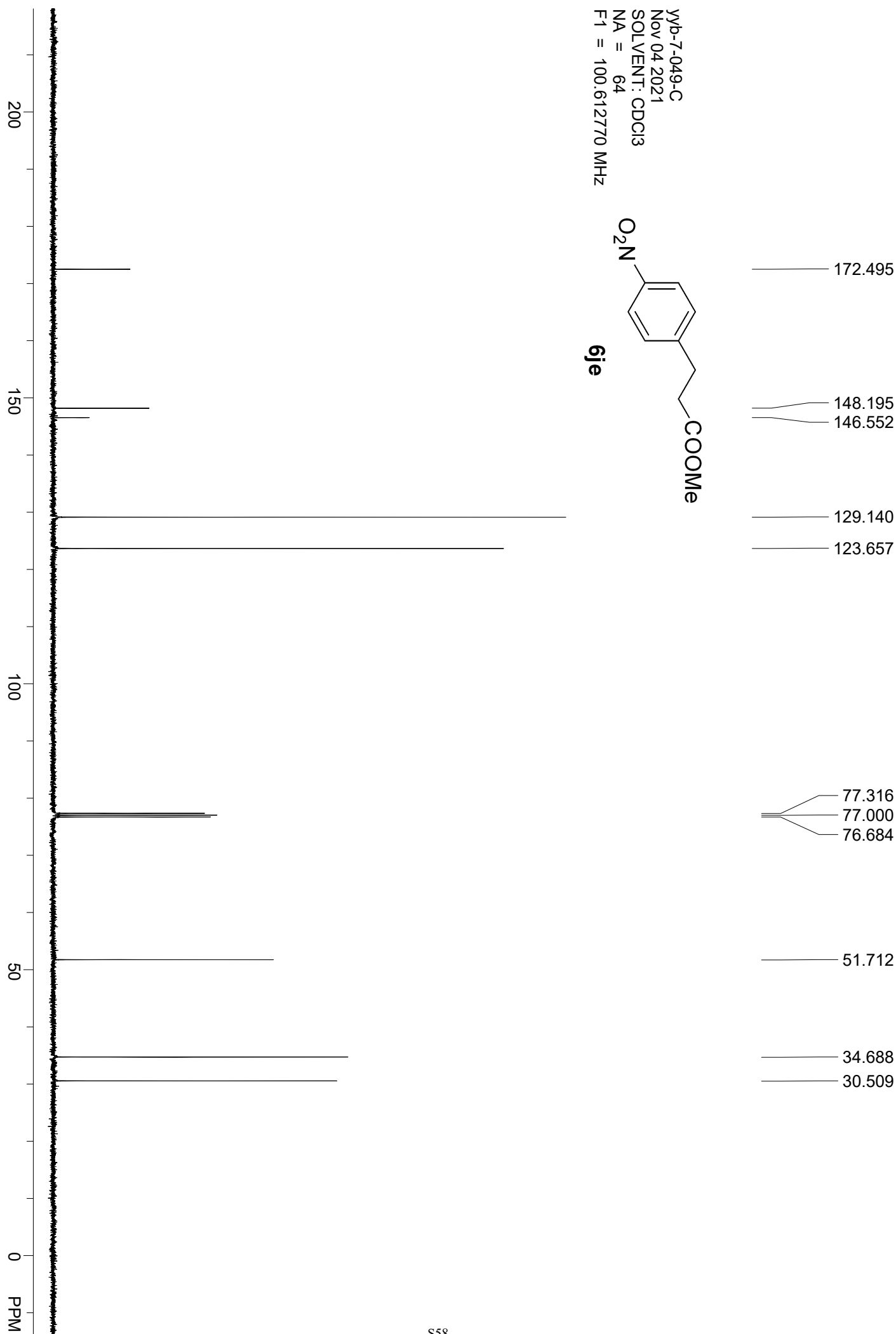
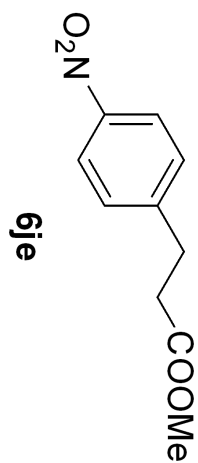
3.087
3.069
3.050
2.712
2.694
2.675

-0.000

yyb-7-049-H
Nov 04 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



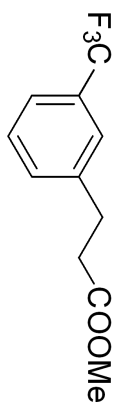
ywb-7-049-C
Nov 04 2021
SOLVENT: CDCl₃
NA = 64
F1 = 100.612770 MHz



7.471
7.457
7.406
7.402
7.393
7.390

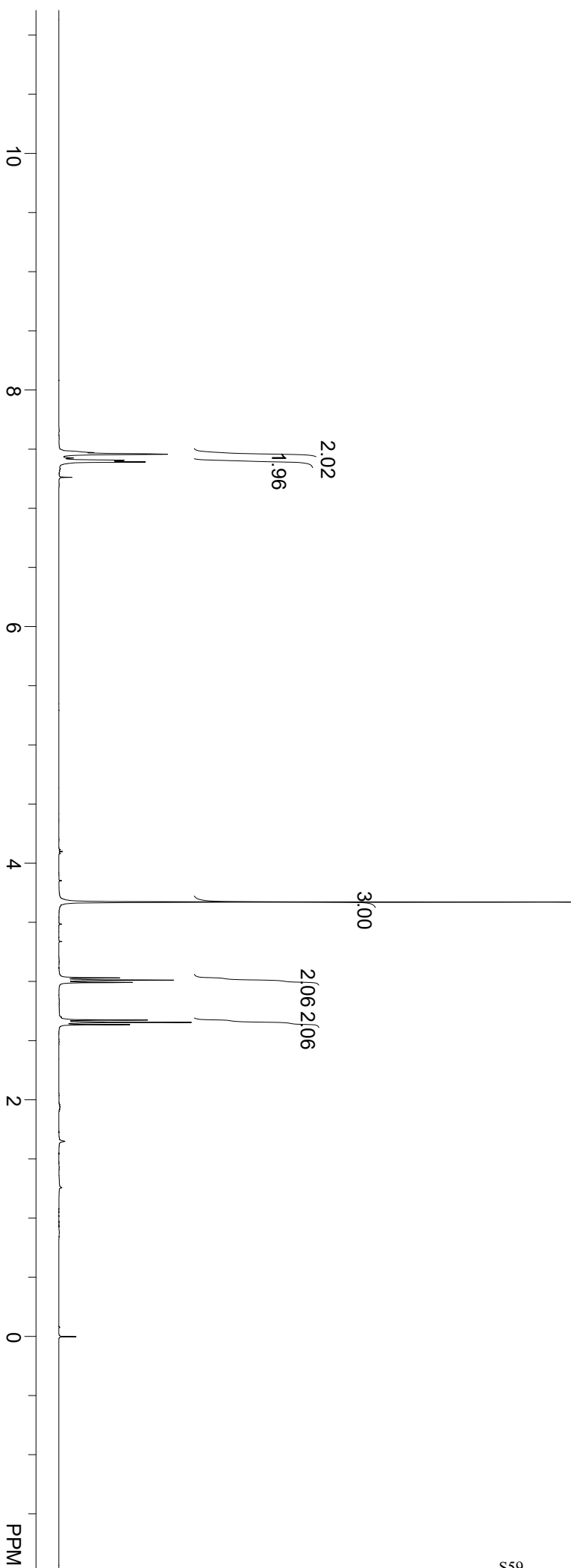
3.672
3.032
3.013
2.993
2.675
2.656
2.637

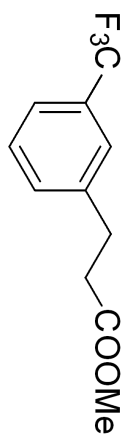
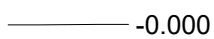
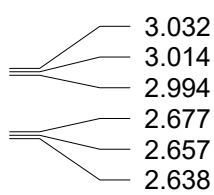
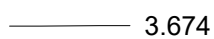
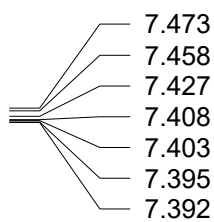
-0.000



6ke

ywb-7-072-H
Nov 16 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

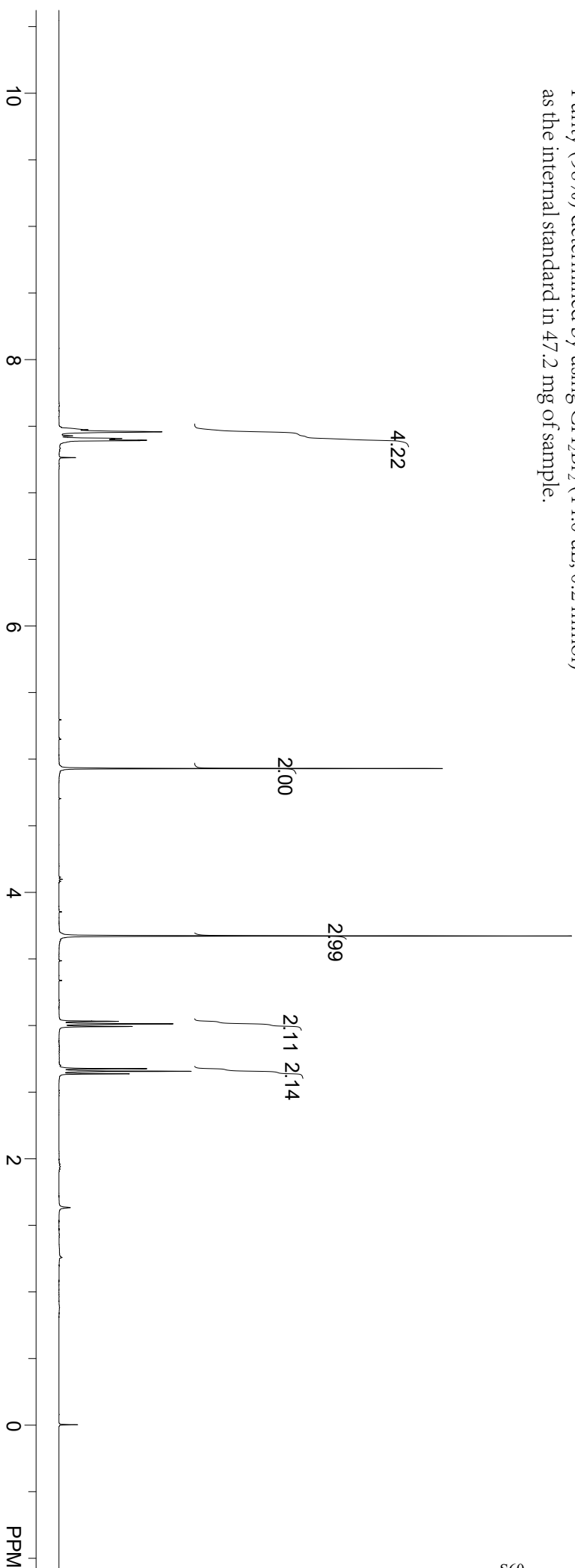




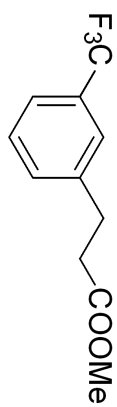
6ke

yyb-7-072-p
Nov 16 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz

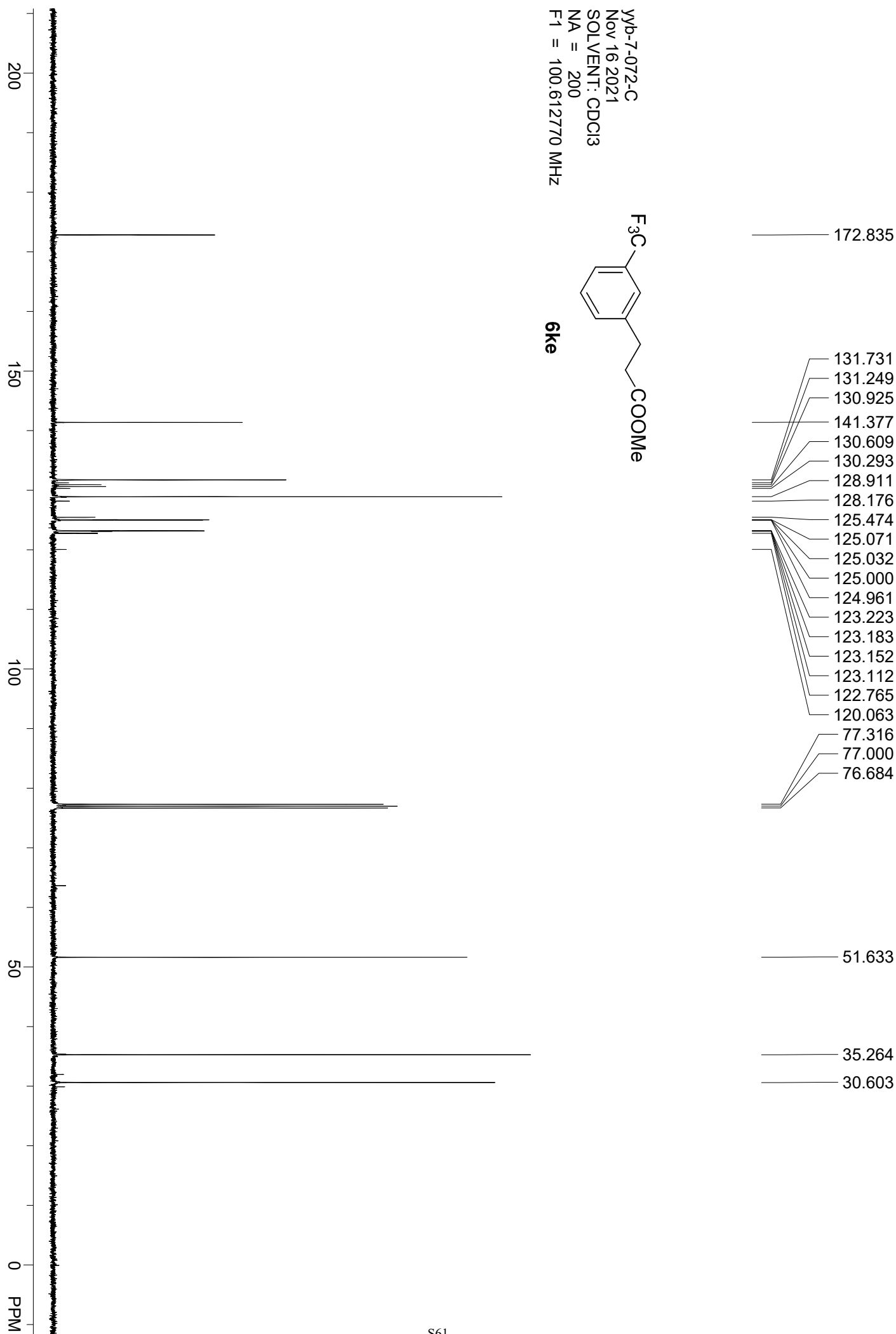
Purity (98%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
as the internal standard in 47.2 mg of sample.



yyb-7-072-C
Nov 16 2021
SOLVENT: CDCl3
NA = 200
F1 = 100.612770 MHz



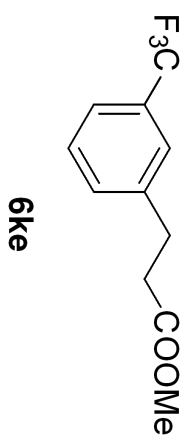
6ke



-0.000

-63.116

yyb-7-072-F
Nov 16 2021
SOLVENT: CDCl₃
NA = 16
F1 = 376.460724 MHz



0

-20

-40

-60

-80

PPM

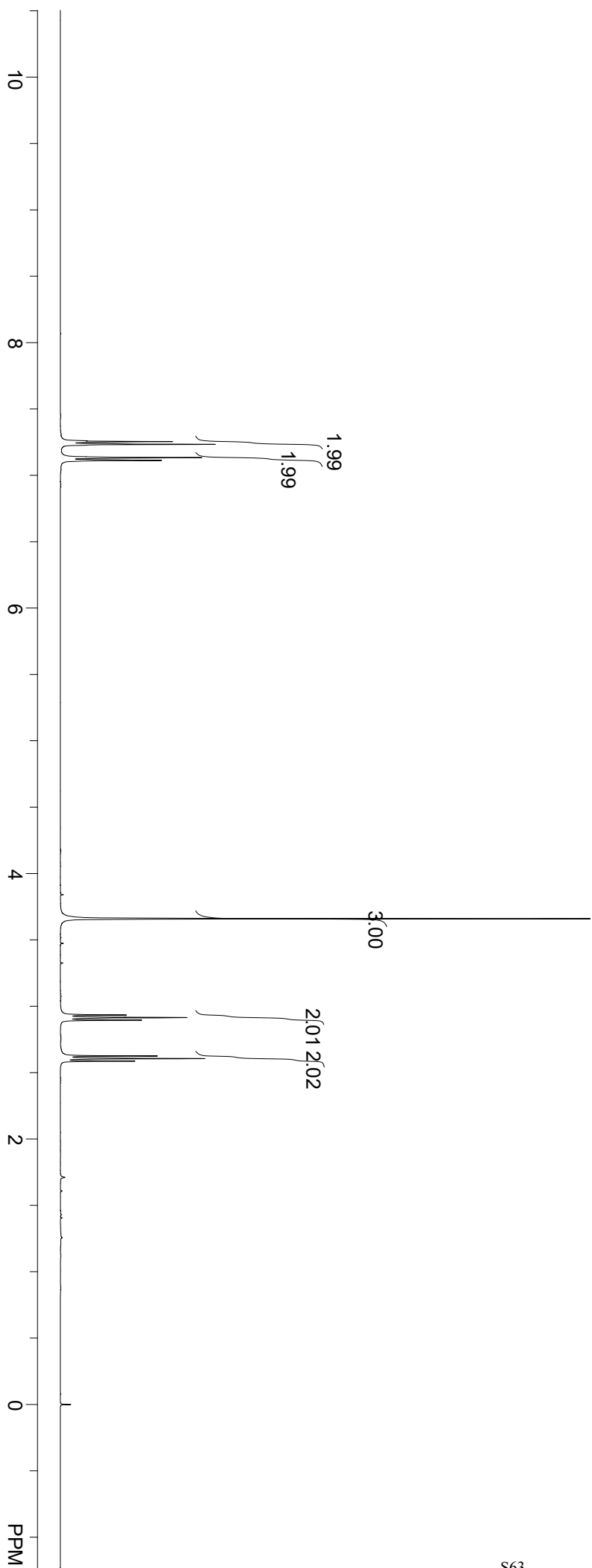
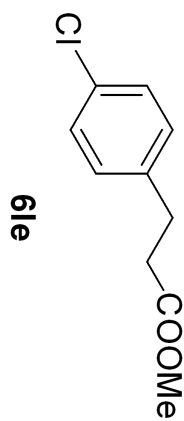
7.260
7.254
7.234
7.134
7.113

3.658

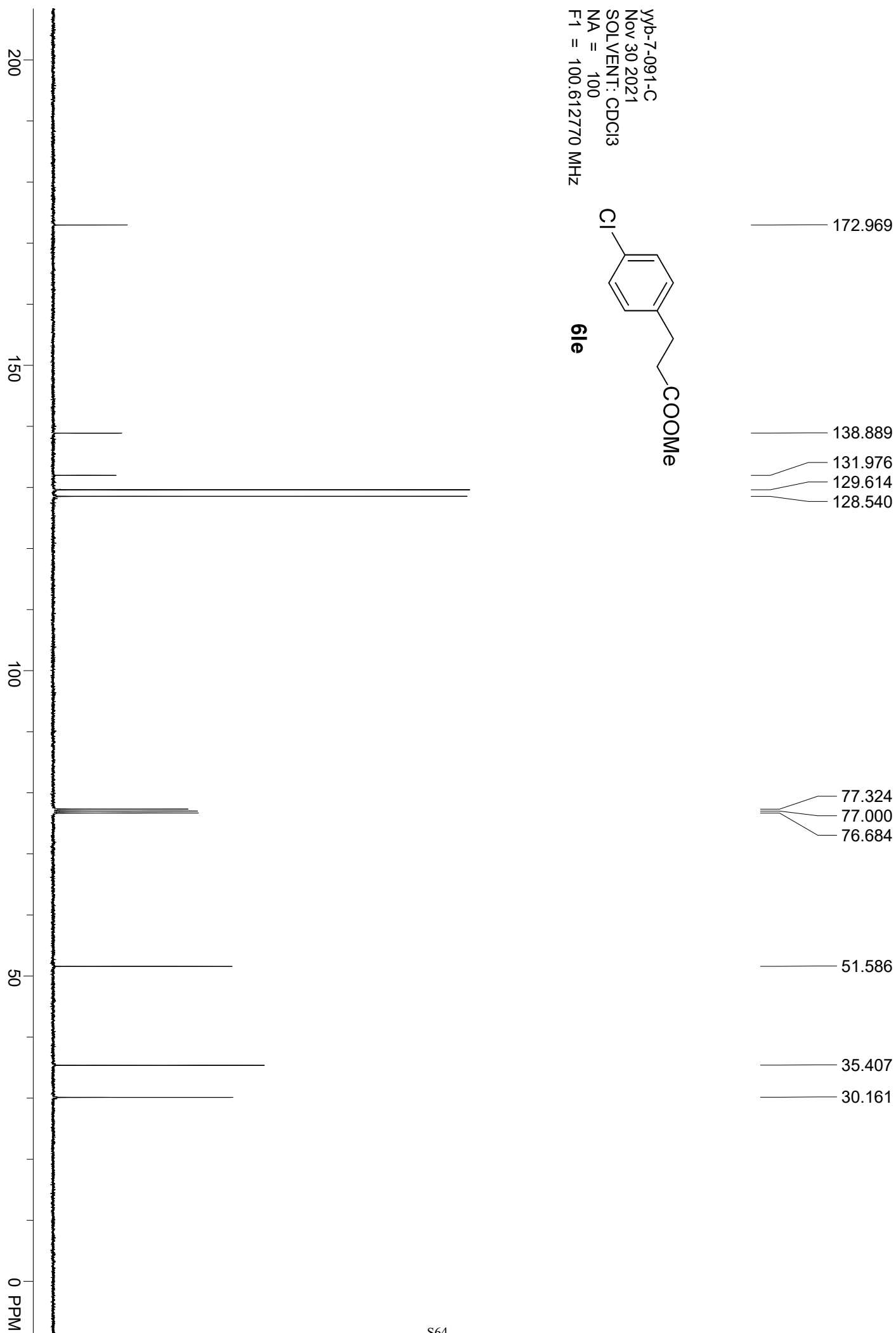
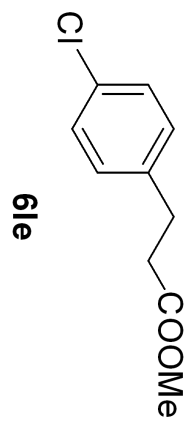
2.932
2.913
2.894
2.624
2.604
2.585

-0.000

yyb-7-091-H
Nov 30 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



yyb-7-091-C
Nov 30 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz

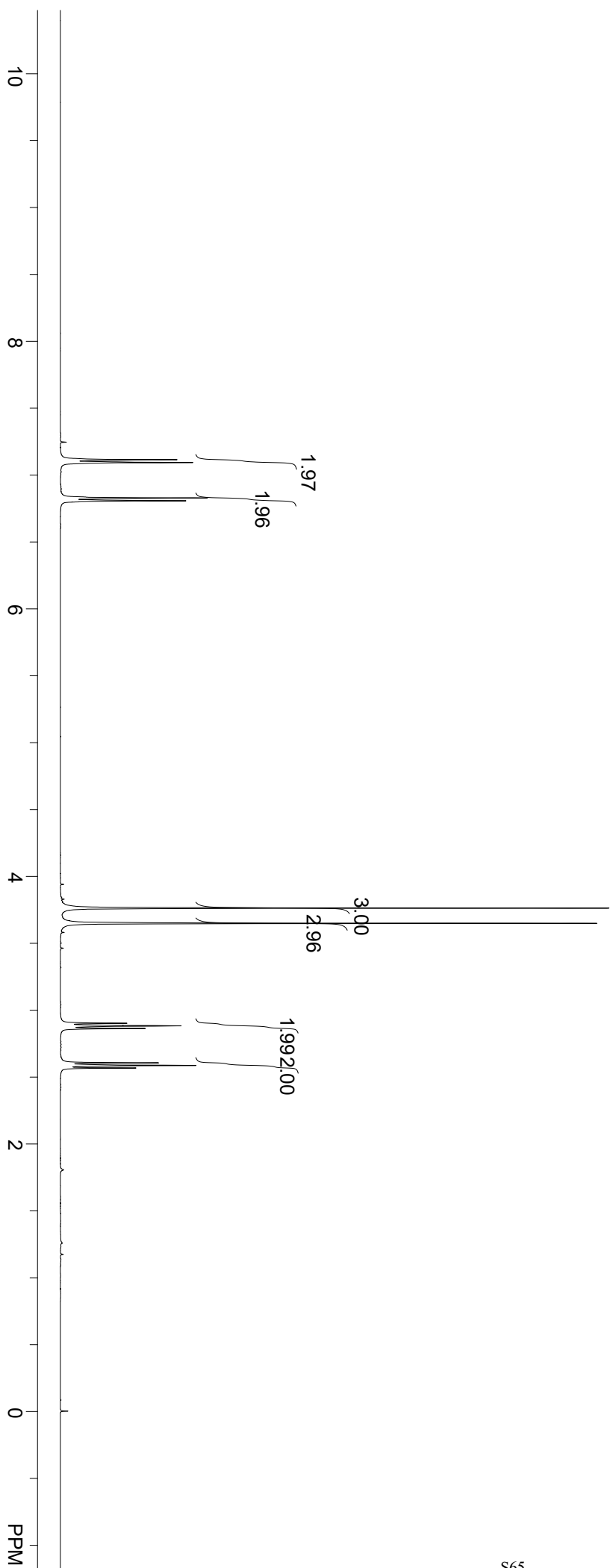
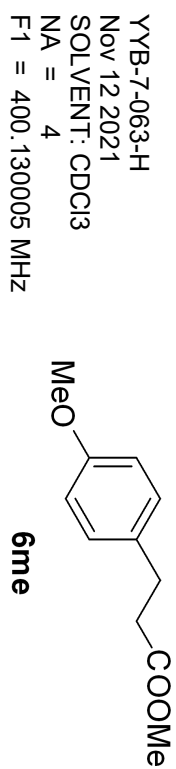


7.116
7.094
6.829
6.808

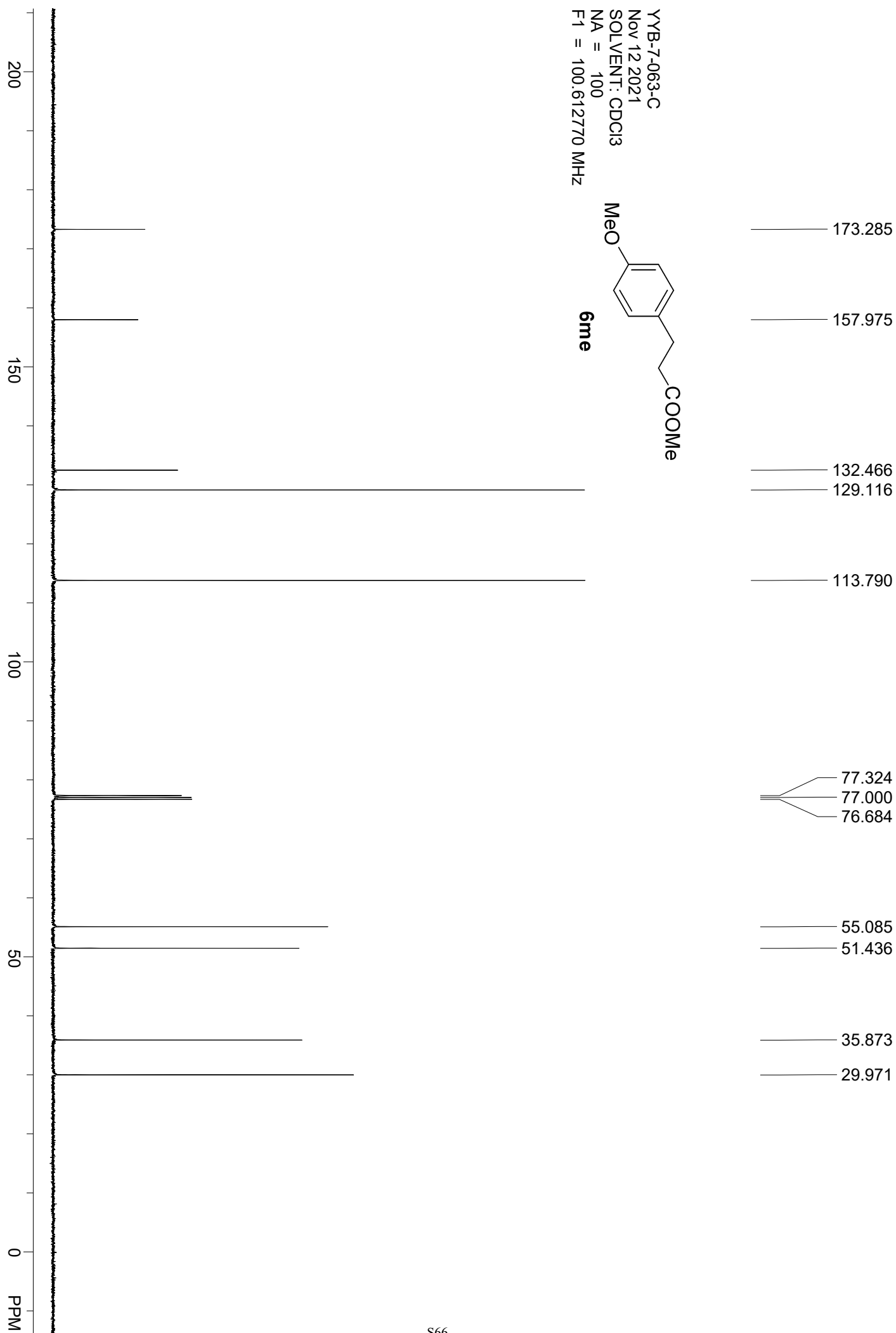
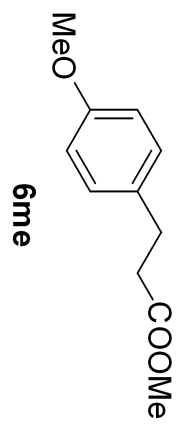
3.764
3.650

2.903
2.884
2.864
2.607
2.587
2.568

-0.000



YYB-7-063-C
Nov 12 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz

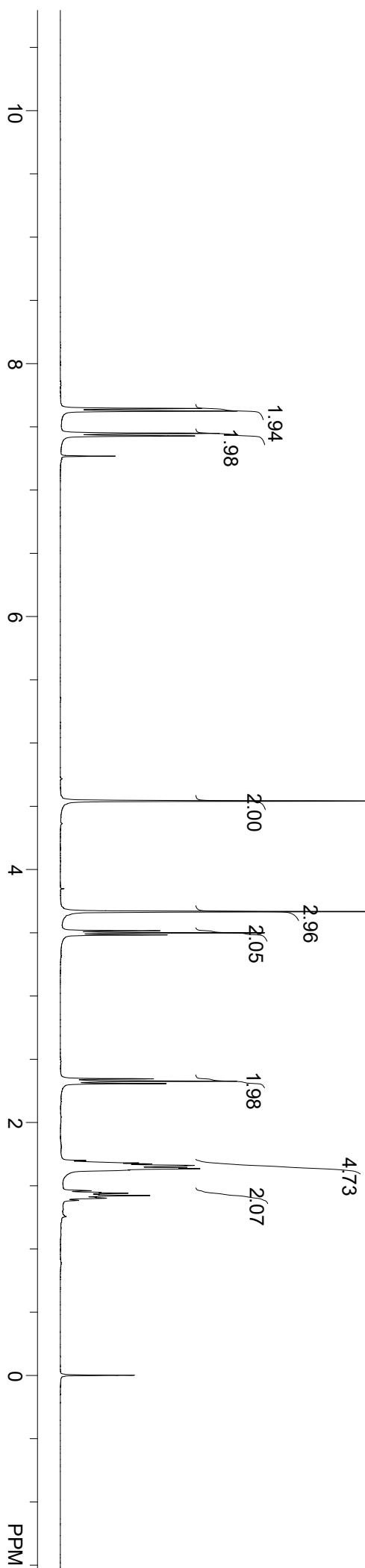
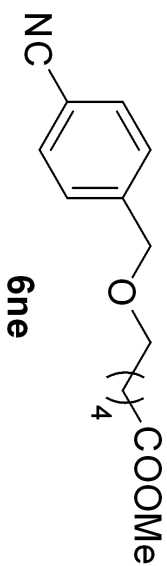


7.644
7.624
7.448
7.428

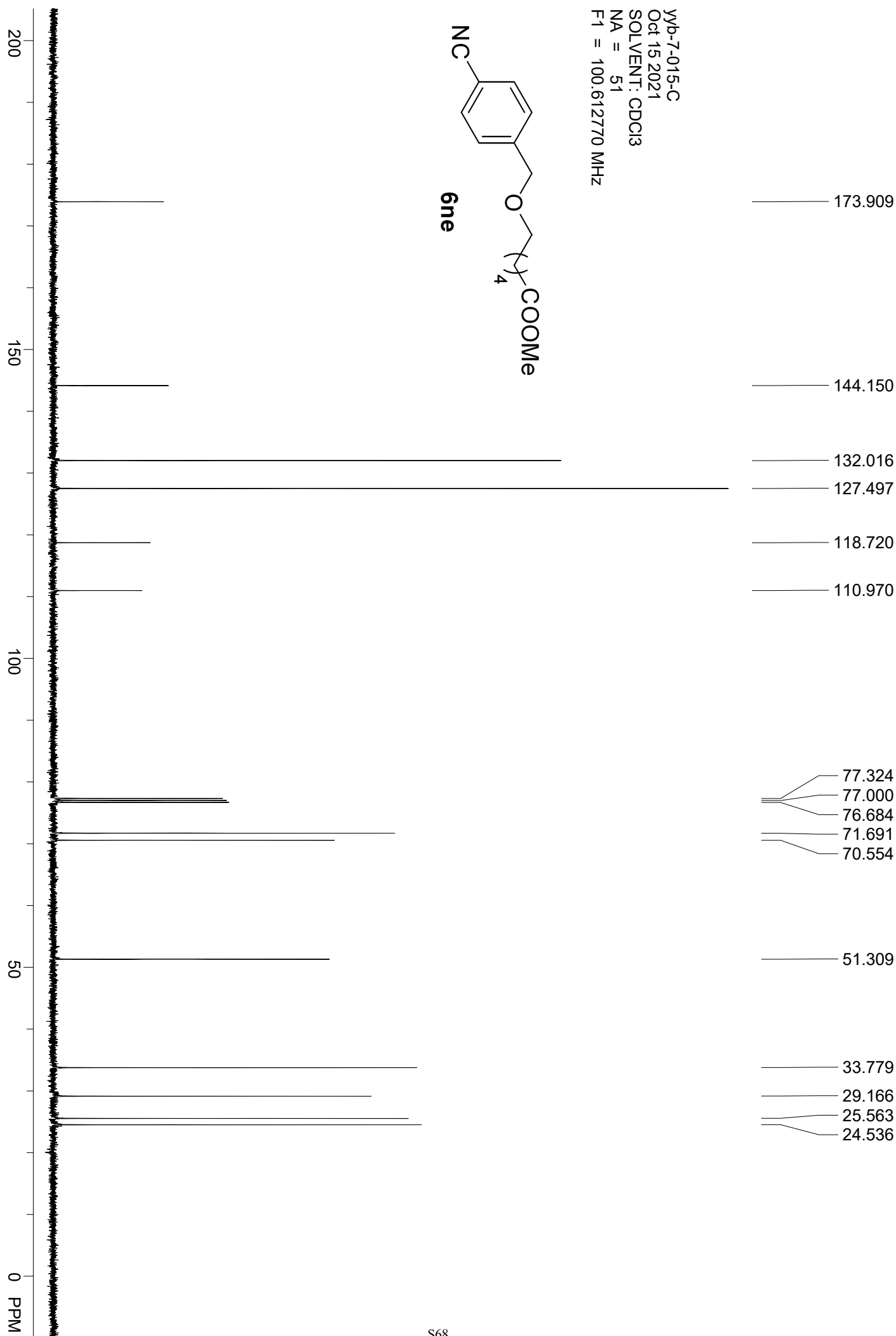
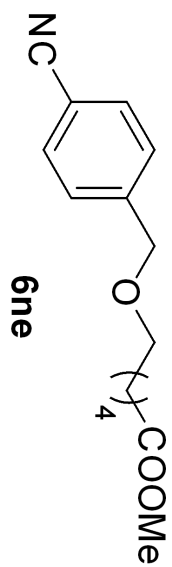
4.543
3.668
3.517
3.501
3.484

2.346
2.327
2.308
1.680
1.672
1.660
1.655
1.641
1.635
1.624
1.460
1.440
1.422
1.401
1.384
-0.000

yyb-7-015-H
Jul 06 2022
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz



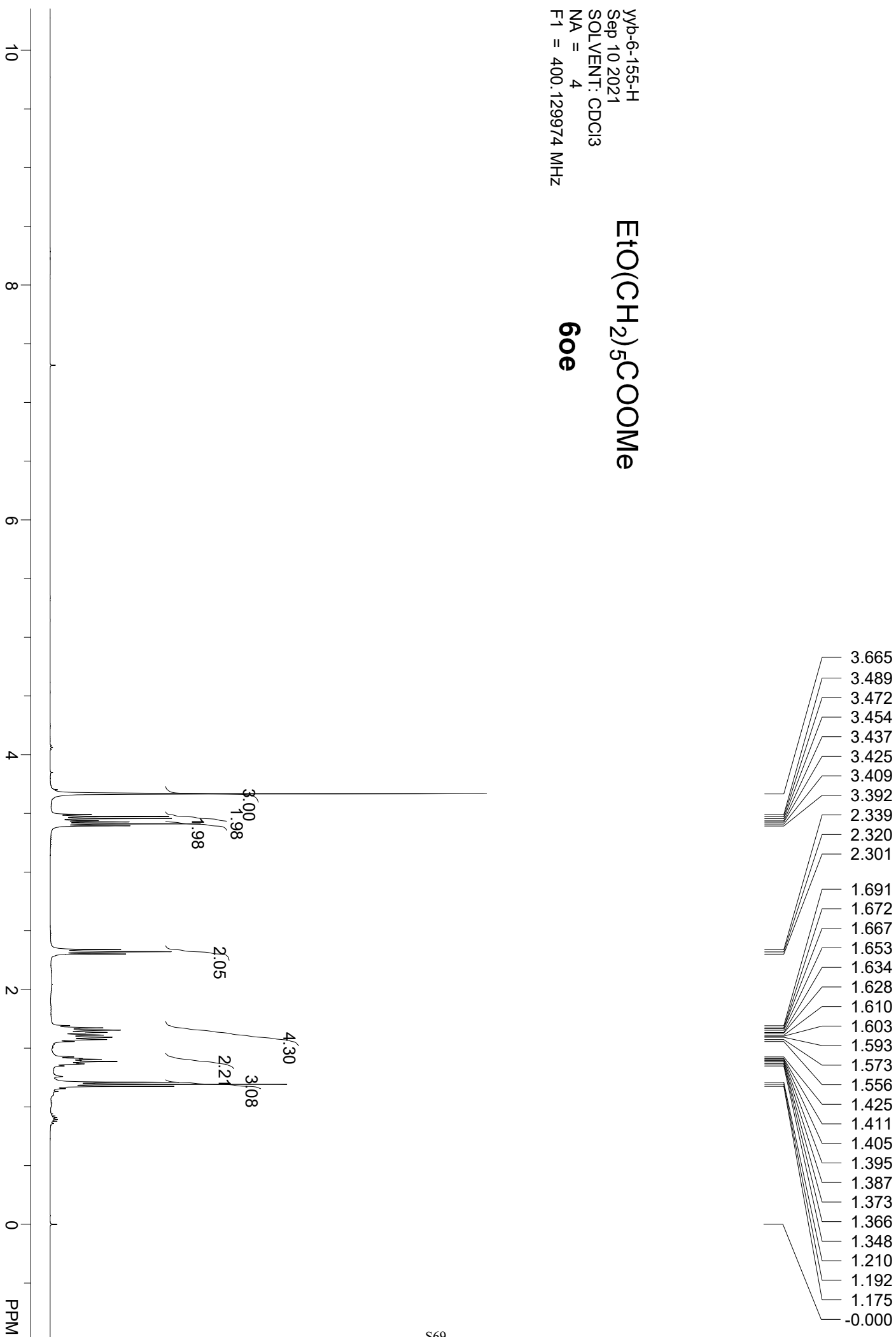
yyb-7-015-C
Oct 15 2021
SOLVENT: CDCl3
NA = 51
F1 = 100.612770 MHz



yyb-6-155-H
Sep 10 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.129974 MHz

Eto(CH₂)₅COOMe

60e

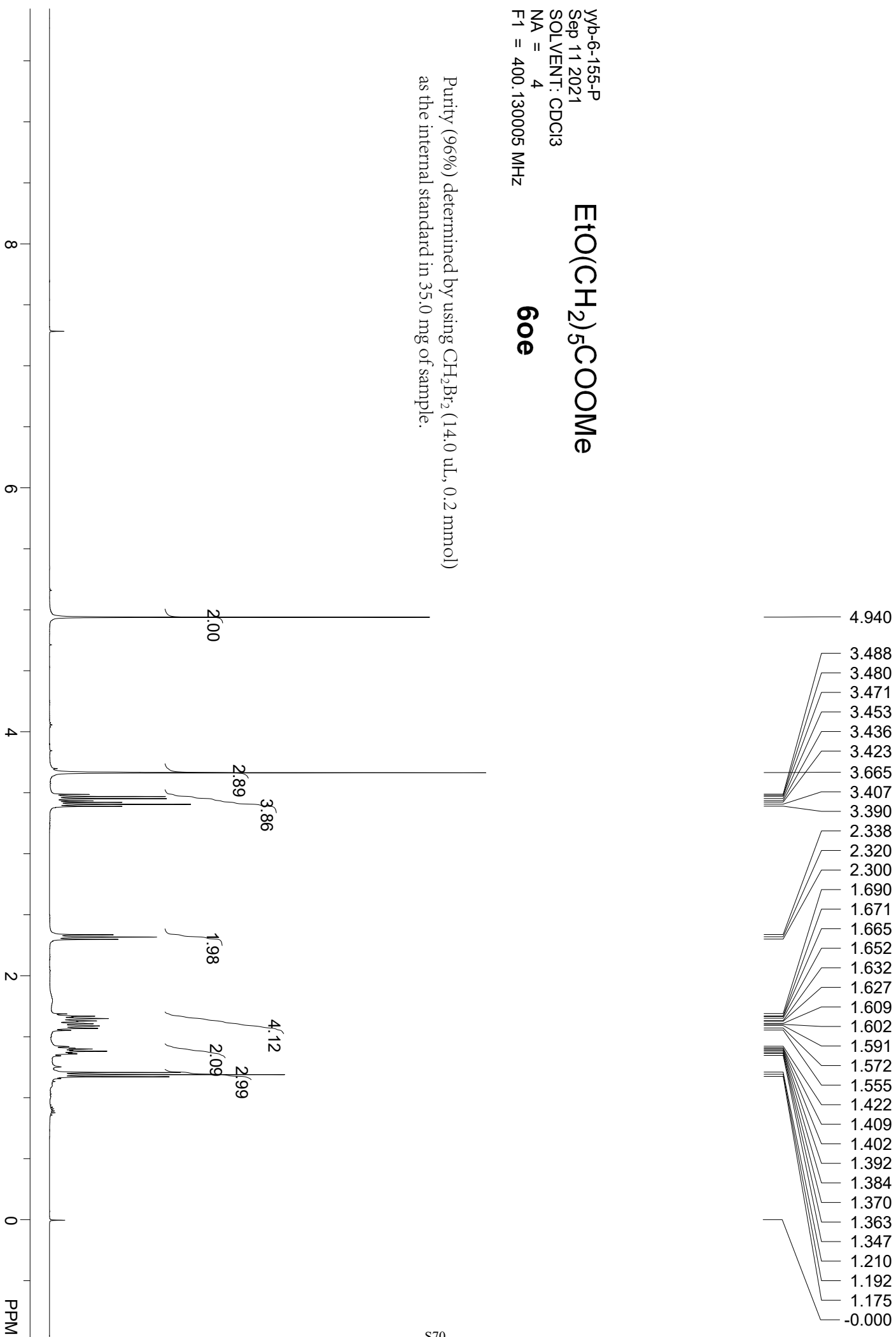


Wb-6-155-P
Sep 11 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz

EtO(CH₂)₅COOMe

6oe

Purity (96%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
as the internal standard in 35.0 mg of sample.



174.028

77.324
77.000
76.684
70.238
65.956

51.286

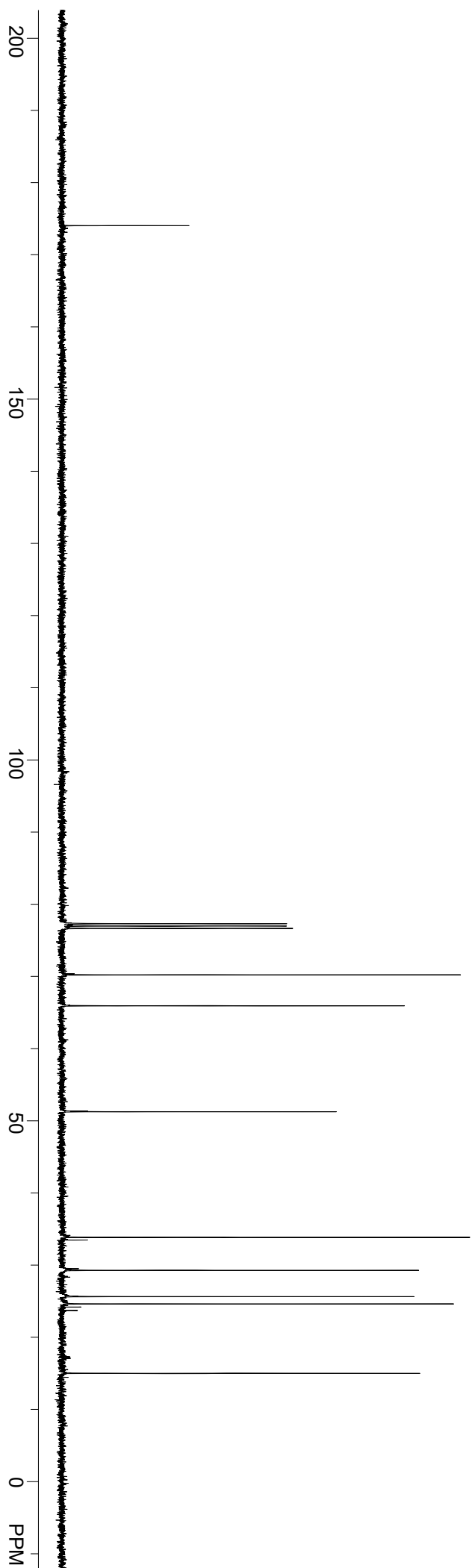
33.882
29.316
25.674
24.663

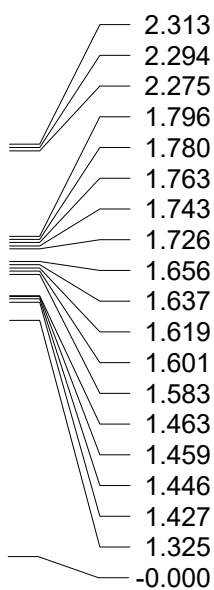
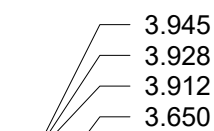
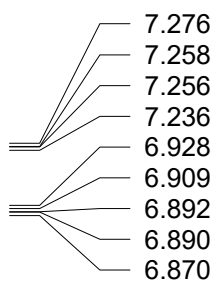
15.048

EtO(CH₂)₅COOMe

yyb-6-155-C
Sep 10 2021
SOLVENT: CDCl₃
NA = 58
F1 = 100.612770 MHz

60e

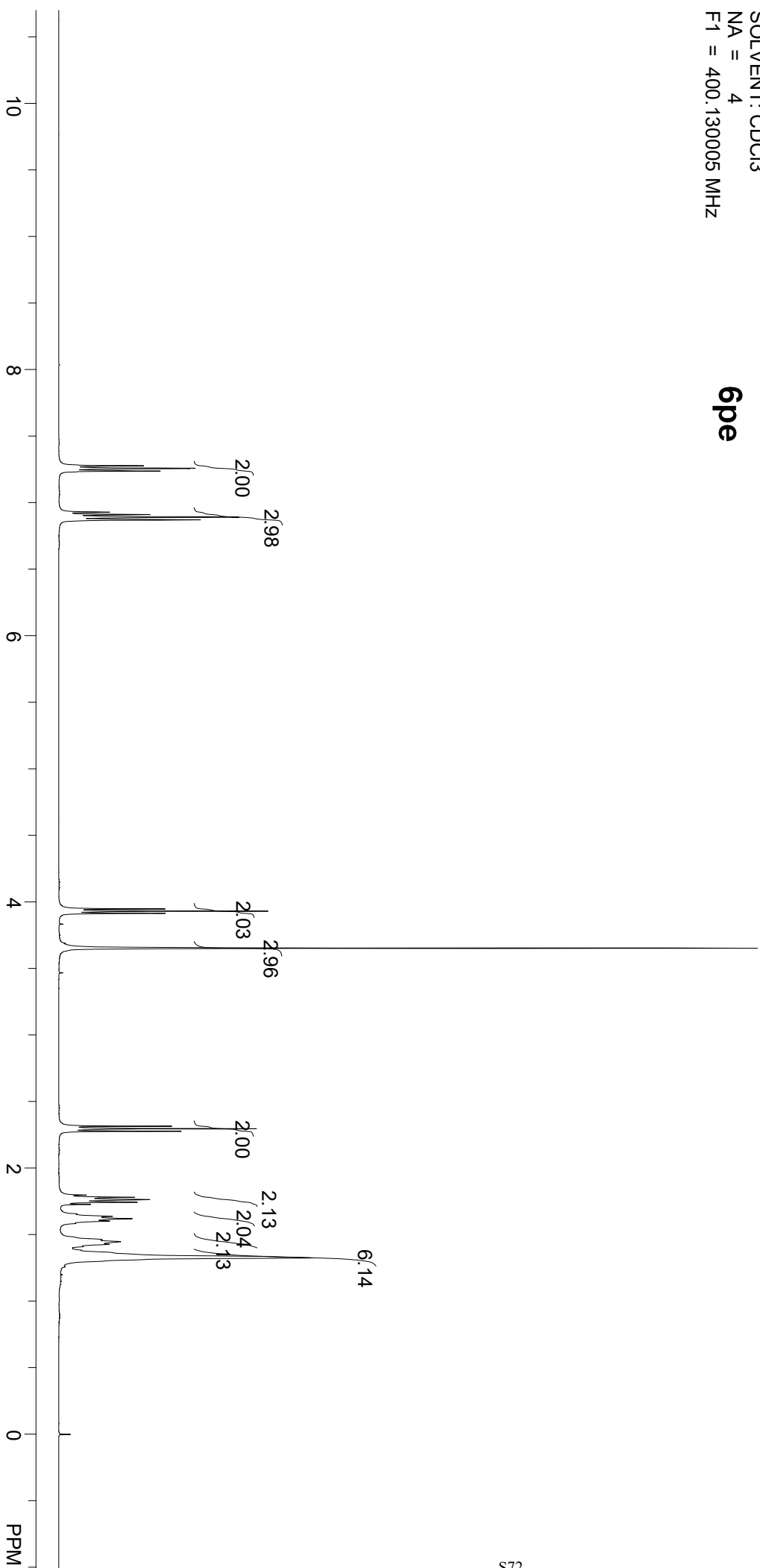




YYB-6-172-H
Sep 20 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

Pho(CH₂)₈COOMe

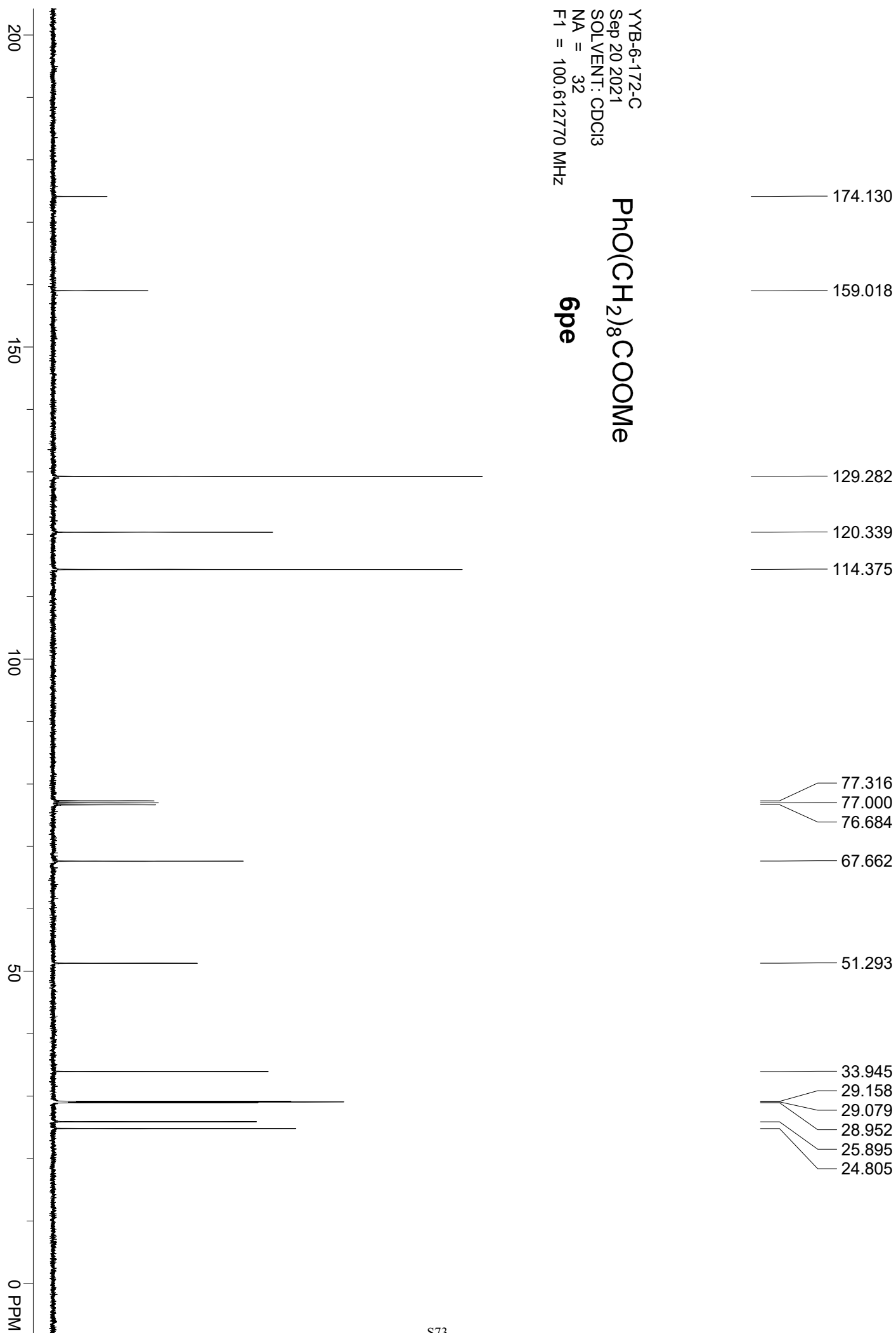
6pe



YYB-6-172-C
Sep 20 2021
SOLVENT: CDCl3
NA = 32
F1 = 100.612770 MHz

PhO(CH₂)₈COOMe

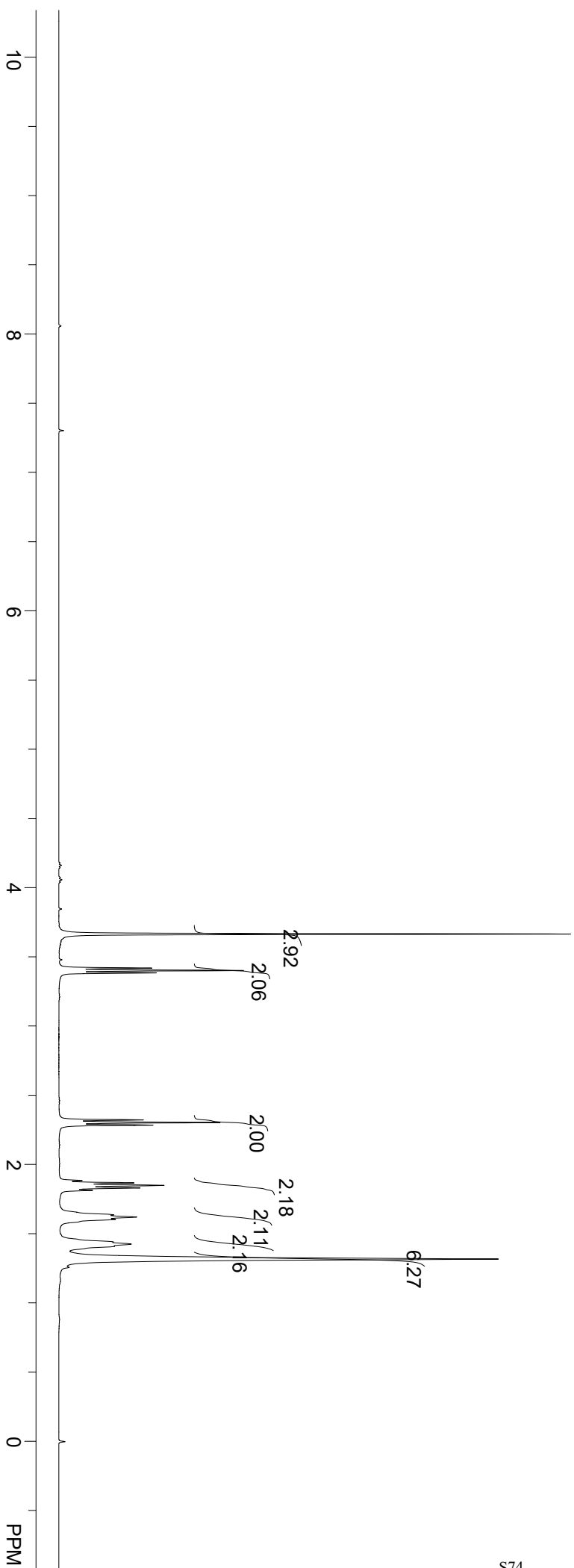
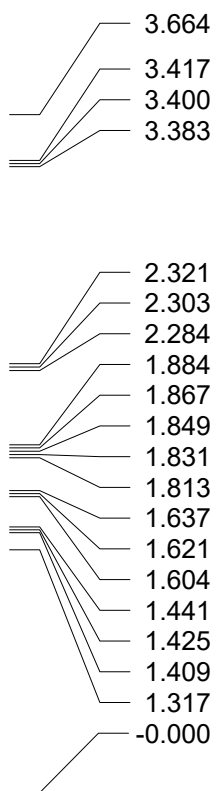
6pe



yyb-6-085-H
Jul 15 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

Br(CH₂)₈COOMe

6qe

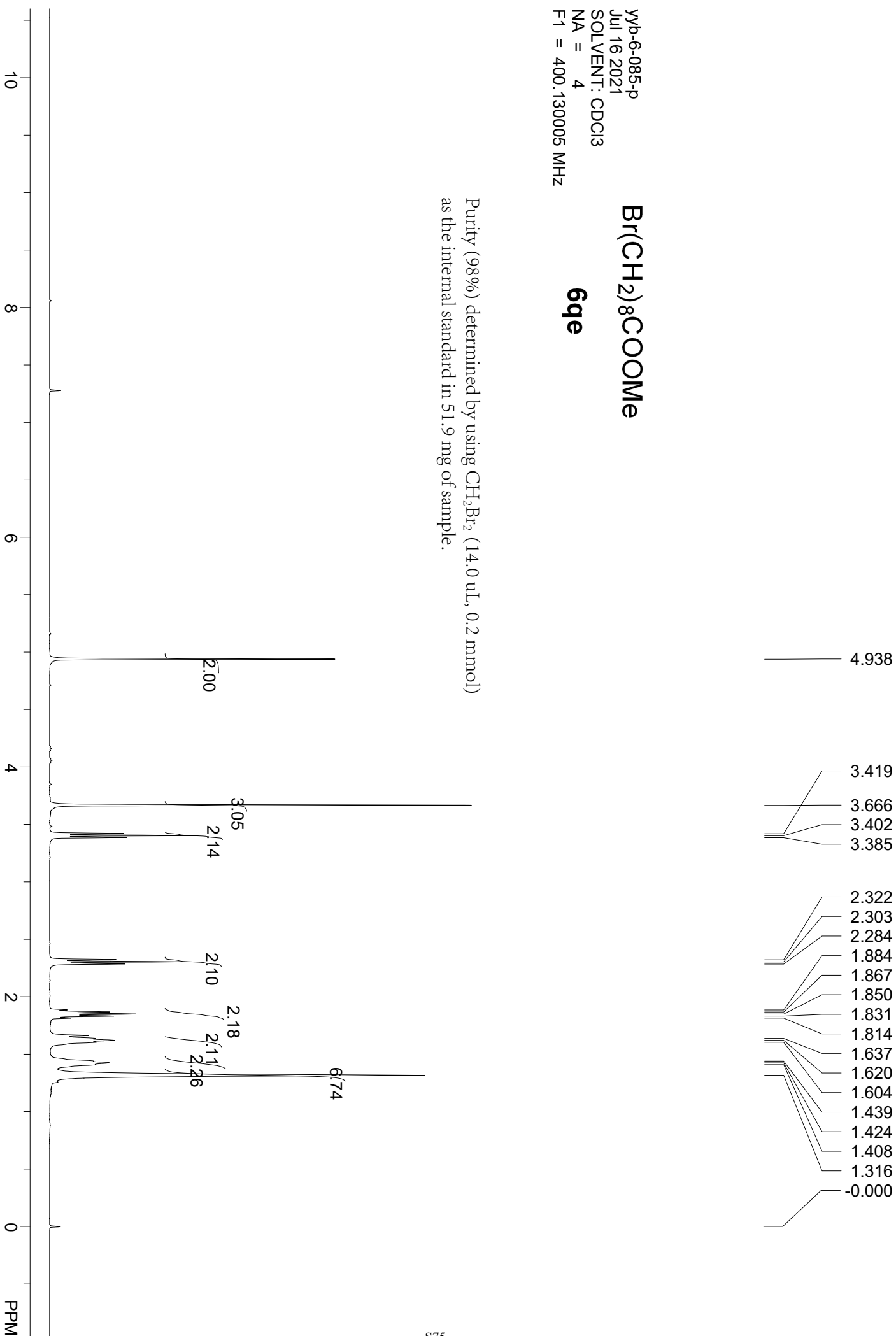


y/b-6-085-p
Jul 16 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

Br(CH₂)₈COOMe

6qe

Purity (98%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
as the internal standard in 51.9 mg of sample.



174.044

77.316
77.000
76.676

51.278

33.906
33.732
32.657
28.913
28.873
28.423
27.957
24.749

yyb-6-085-C
Jul 16 2021

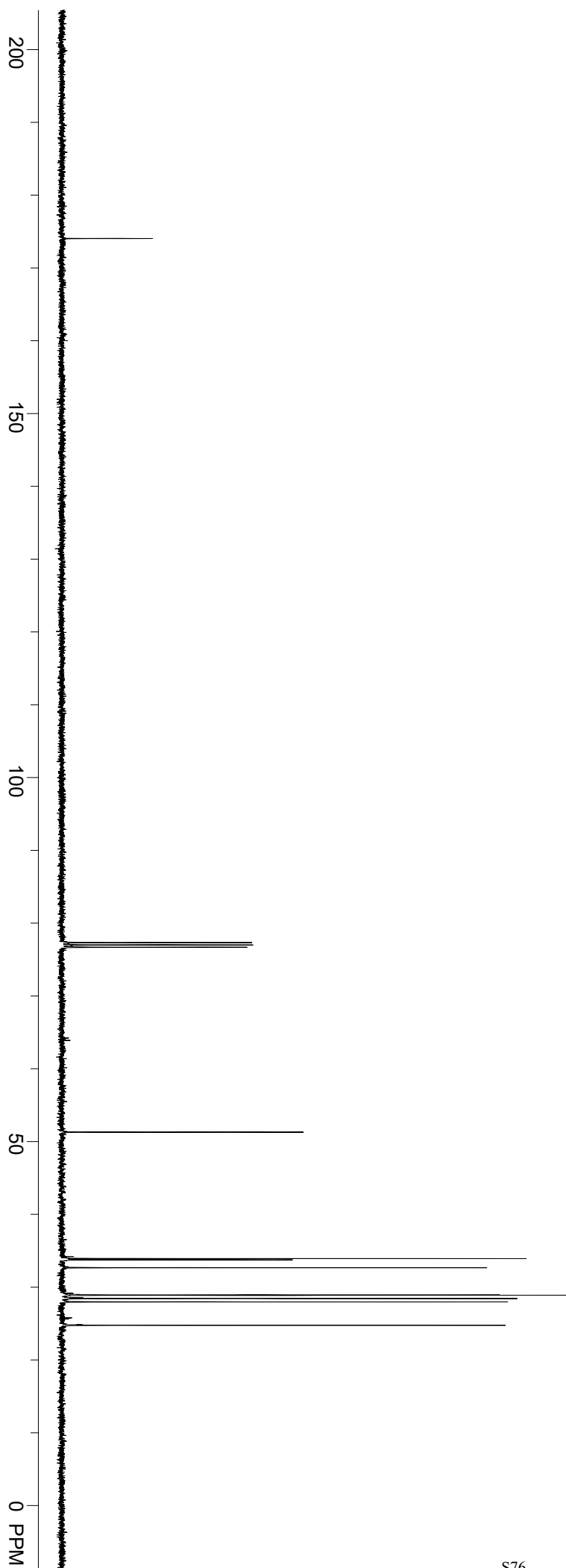
SOLVENT: CDCl₃

NA = 100

F1 = 100.612770 MHz

Br(CH₂)₈COOMe

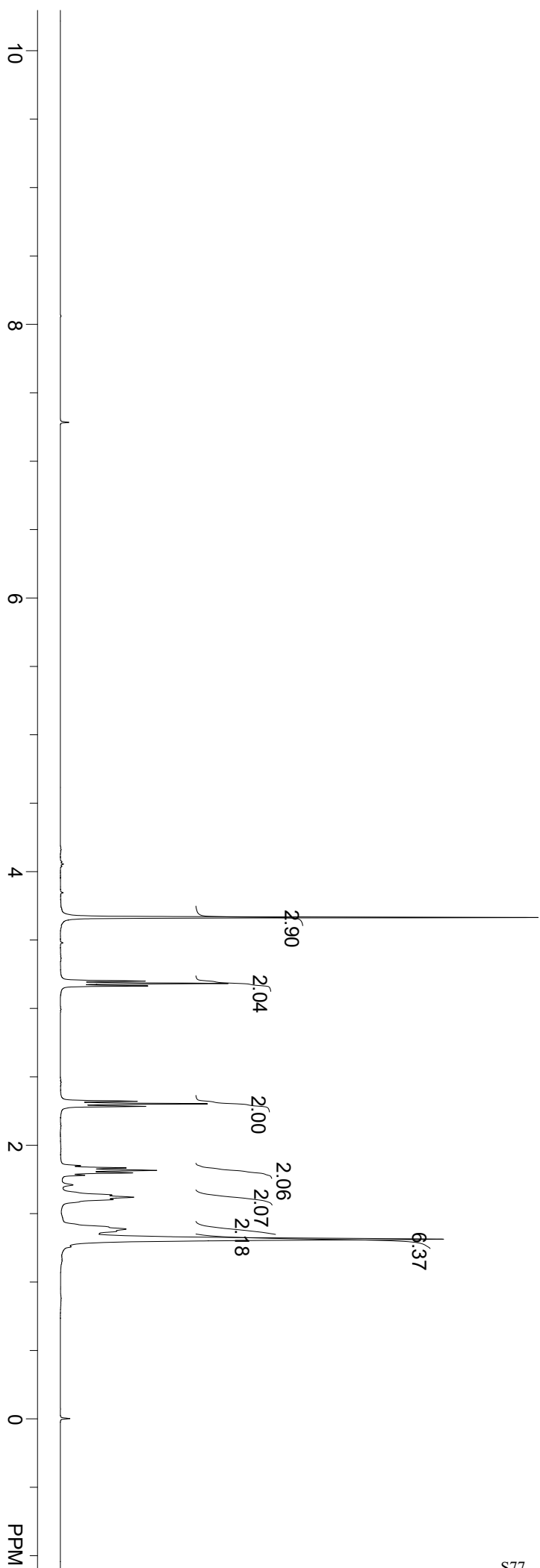
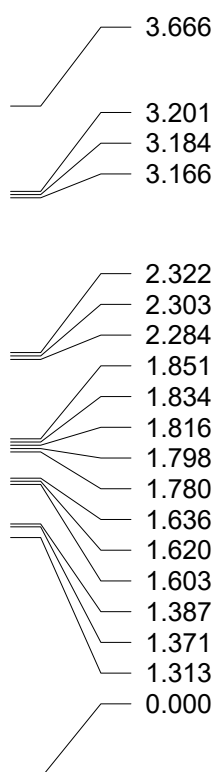
6qe



yyb-6-088-2-H
Jul 19 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

I(CH₂)₈COOMe

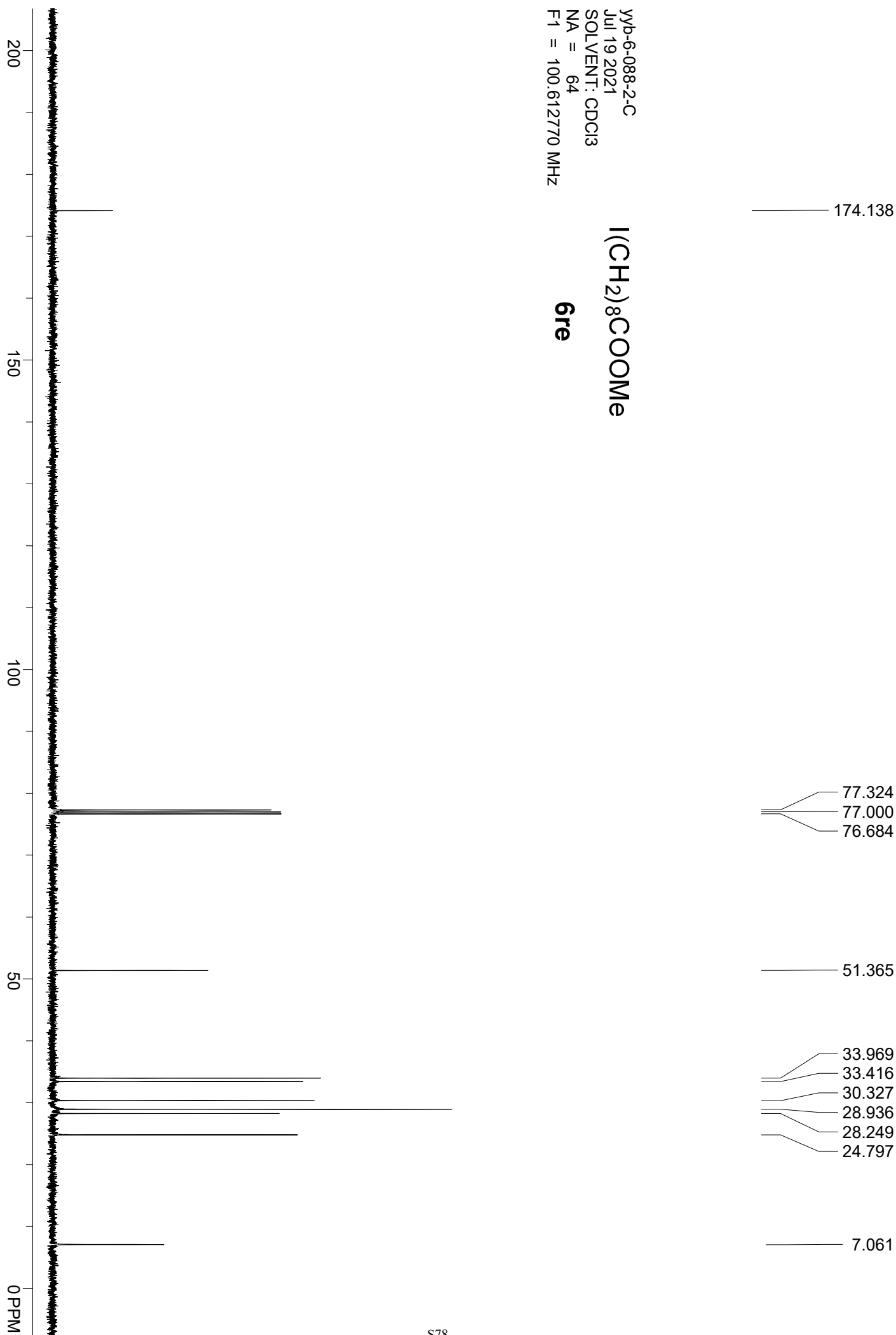
6re



yyb-6-088-2-C
Jul 19 2021
SOLVENT: CDCl3
NA = 64
F1 = 100.612770 MHz

I(CH₂)₈COOMe

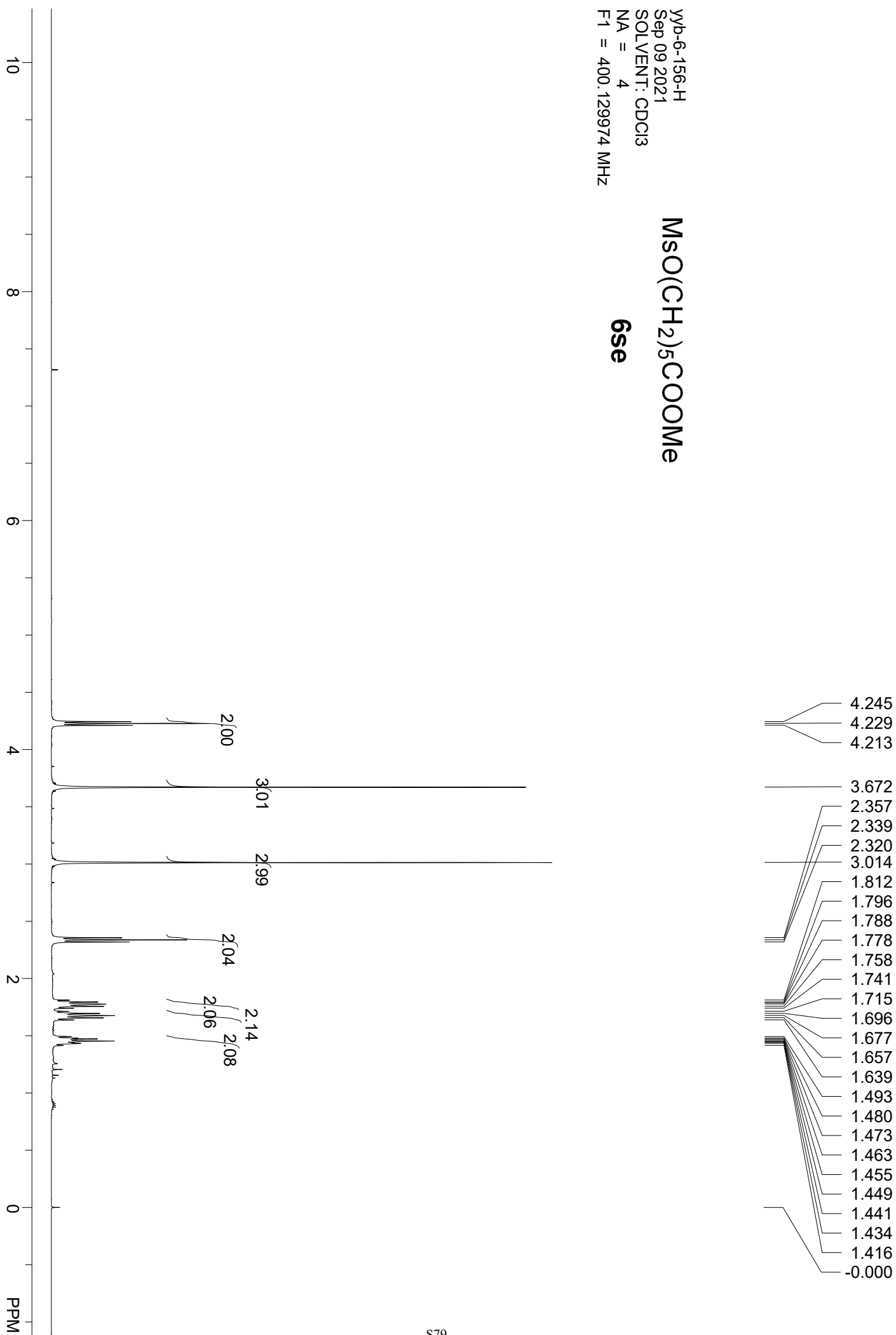
6re



yyb-6-156-H
Sep 09 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.129974 MHz

MSO(CH₂)₅COOMe

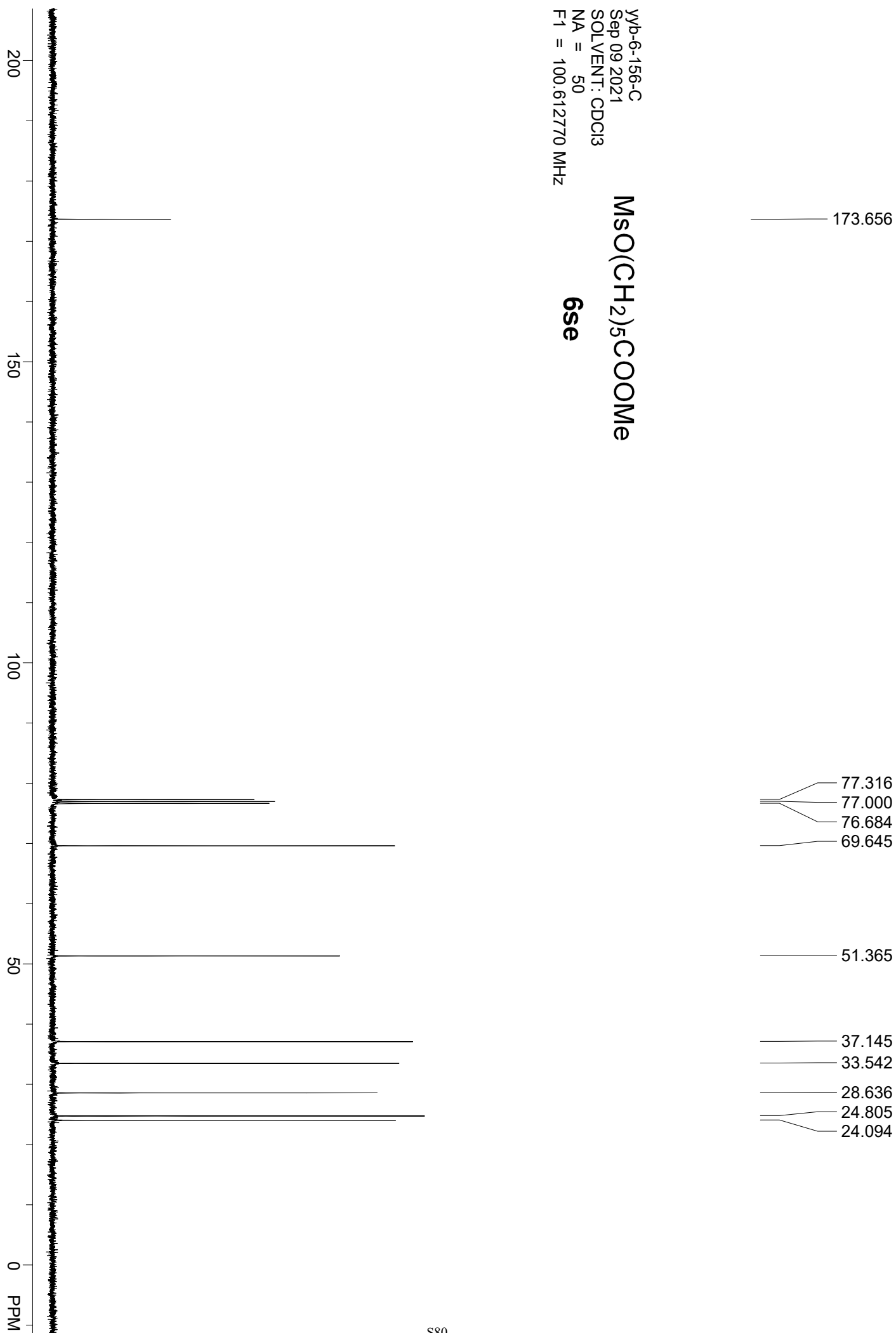
6se



W/b-6-156-C
Sep 09 2021
SOLVENT: CDCl₃
NA = 50
F1 = 100.612770 MHz

MSO(CH₂)₅COOMe

6se



7.796
7.776
7.358
7.339
7.281

4.030
4.014
3.998
3.661

2.451
2.296
2.277
2.258

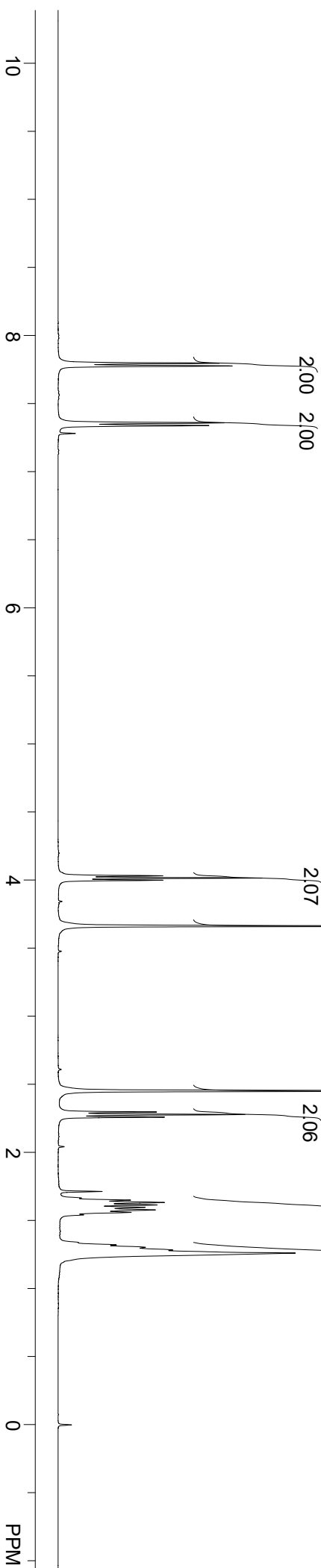
1.649
1.632
1.614
1.596
1.578
1.560
1.321
1.303
1.261

0.000

TsO(CH₂)₇COOMe

6te

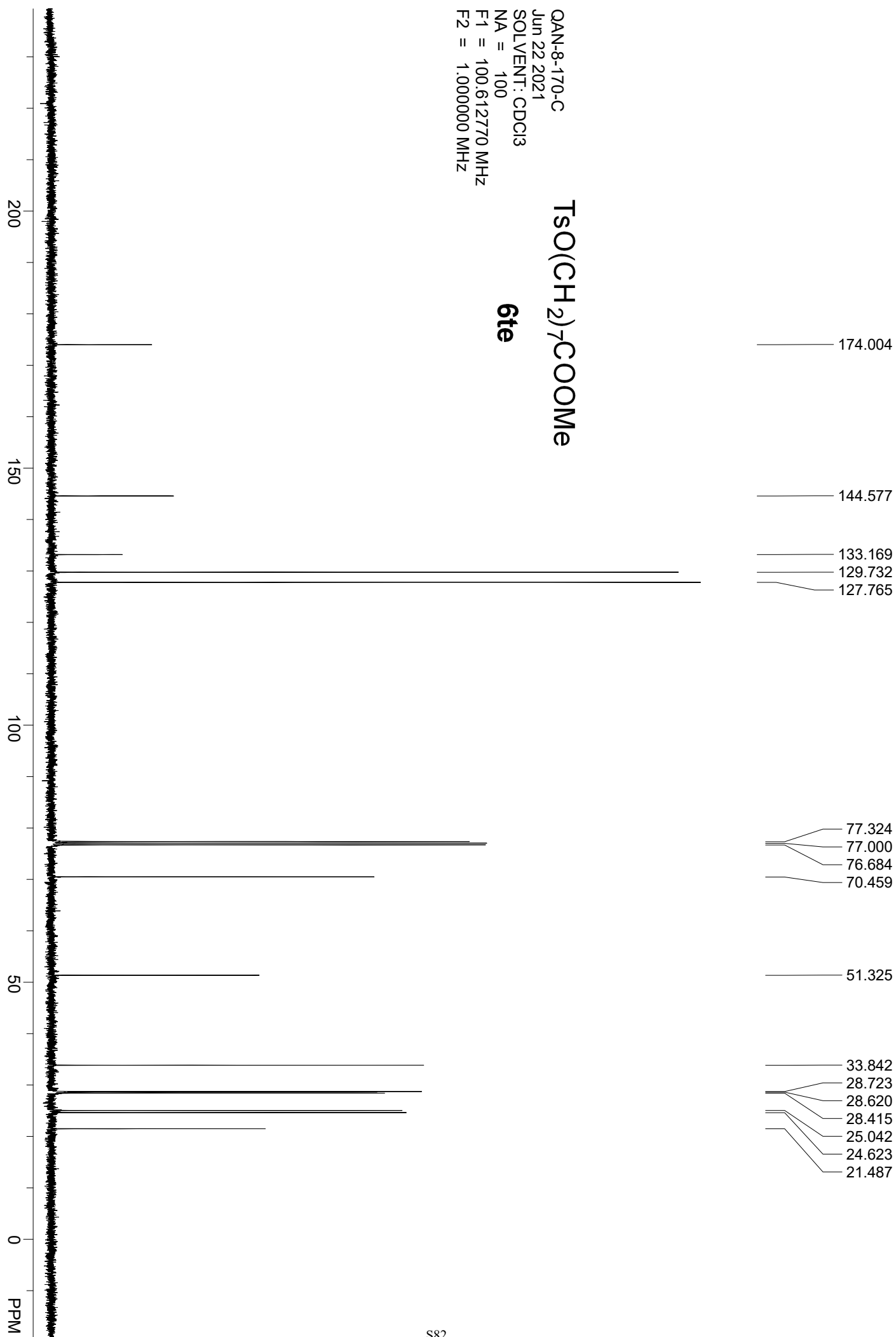
QAN-8-170-H
Jun 22 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz
F2 = 1.000000 MHz



QAN-8-170-C
Jun 22 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz
F2 = 1.000000 MHz

TsO(CH₂)₇COOMe

6te



yyb-7-076-H
Nov 19 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.129974 MHz

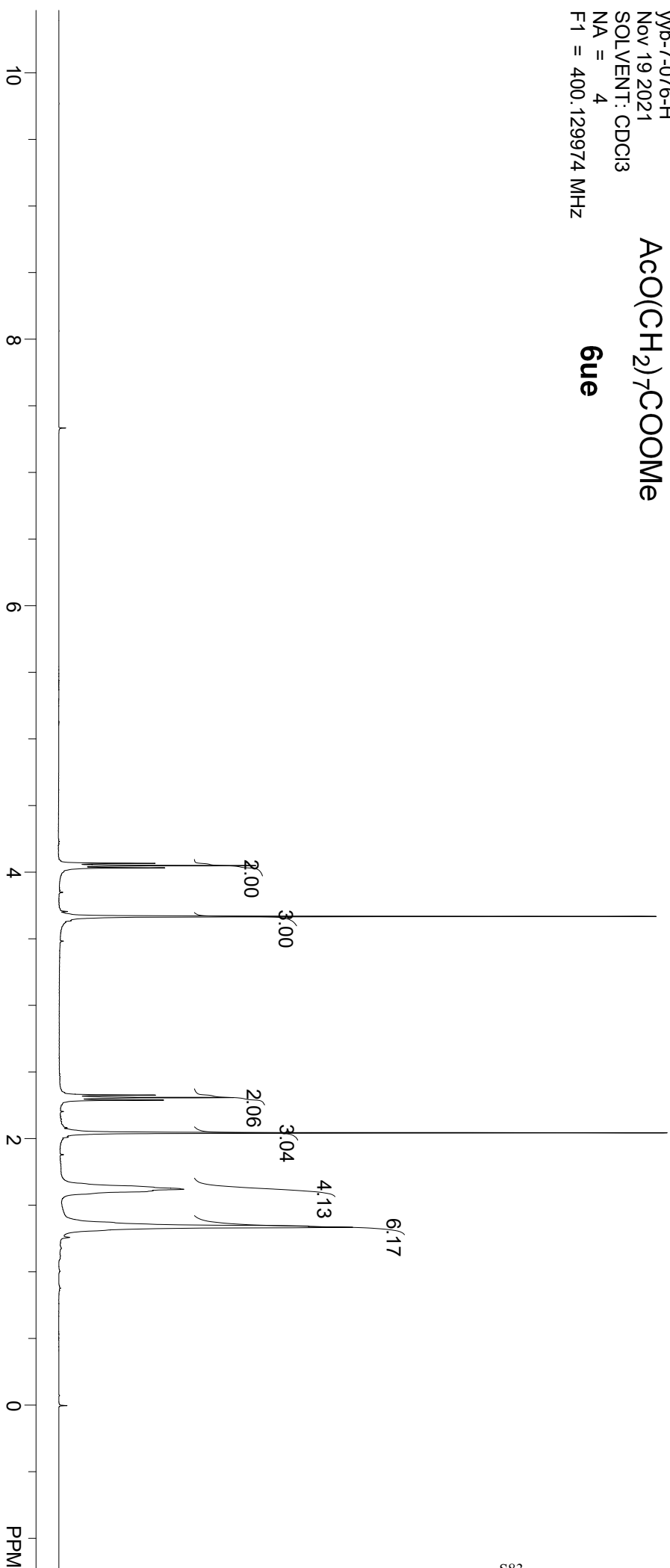
AcO(CH₂)₇COOMe

6ue

4.066
4.049
4.032
3.667

2.327
2.309
2.290
2.044
1.621
1.610
1.345
1.336

-0.000



174.020
171.010

77.316
77.000
76.684

64.344

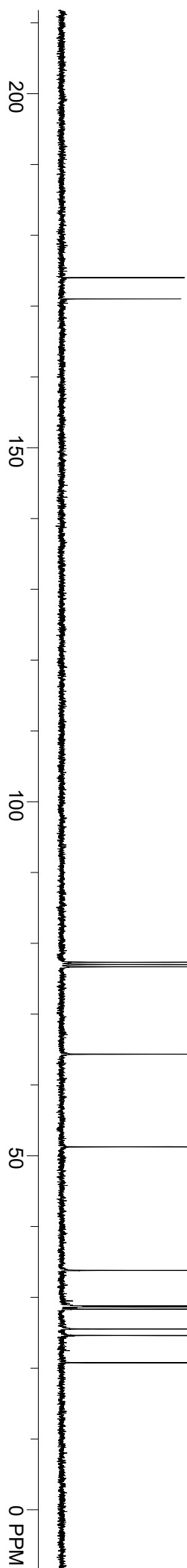
51.270

33.834
28.818
28.707
28.368
25.563
24.655
20.807

AcO(CH₂)₇COOMe

6ue

Yyb-7-076-C
Nov 19 2021
SOLVENT: CDCl₃
NA = 64
F1 = 100.612770 MHz



8.048
8.030
7.563
7.544
7.526
7.449
7.429
7.411

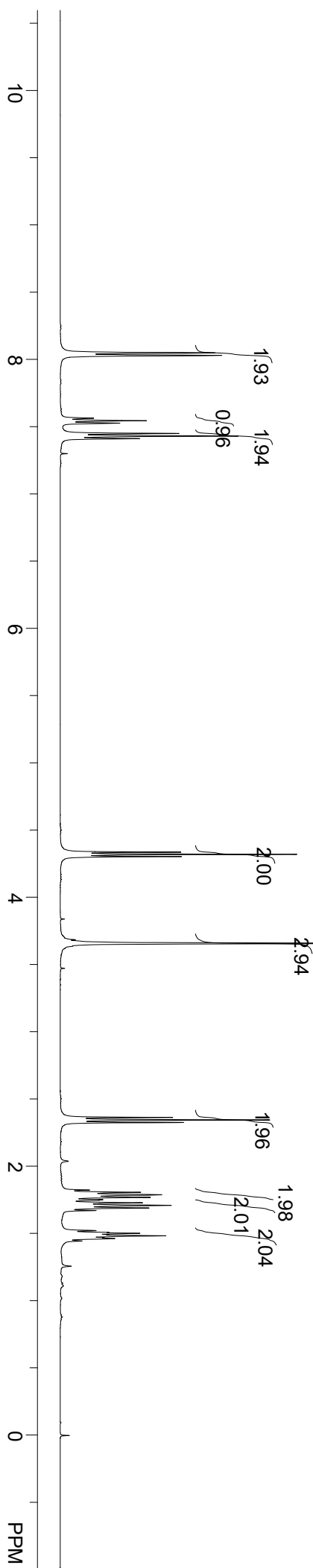
4.335
4.318
4.302

3.656
2.362
2.344
2.325
1.823
1.806
1.788
1.769
1.752
1.748
1.728
1.709
1.690
1.671
1.522
1.508
1.502
1.491
1.483
1.469
1.463
1.445
0.000

BZO(CH₂)₅COOMe

yyb-7-077-H
Nov 19 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz

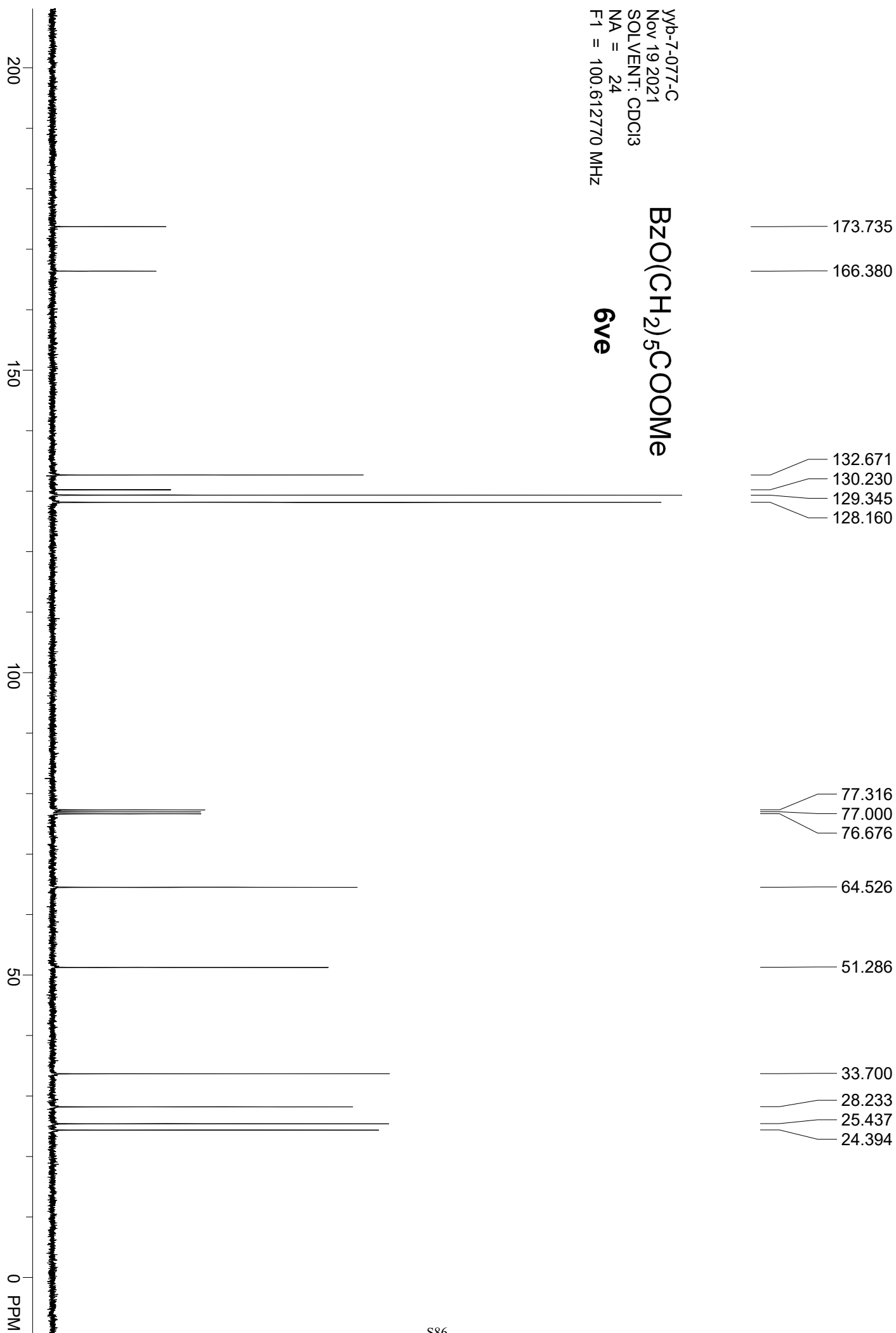
6ve



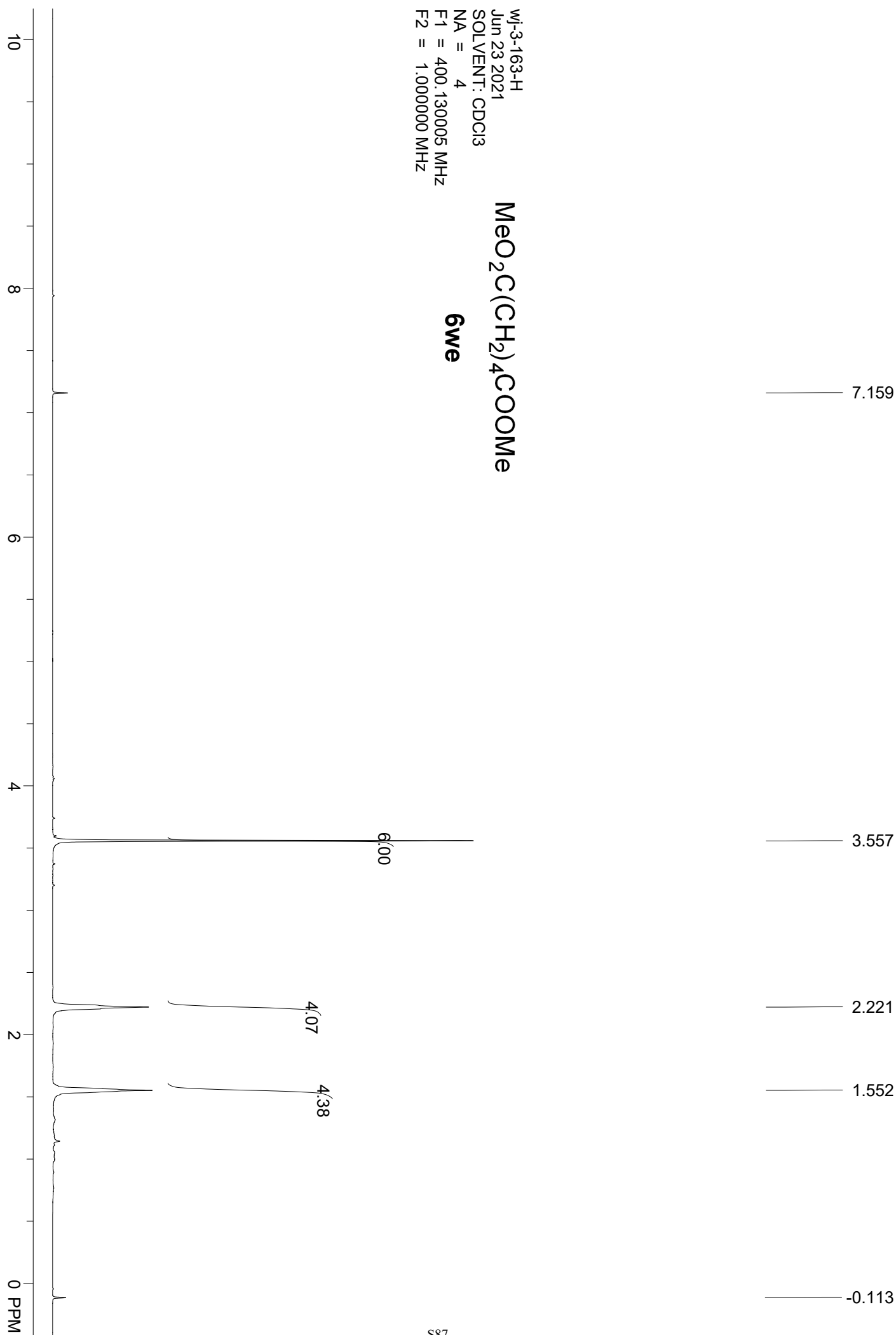
yyb-7-077-C
Nov 19 2021
SOLVENT: CDCl3
NA = 24
F1 = 100.612770 MHz

BzO(CH₂)₅COOMe

6ve



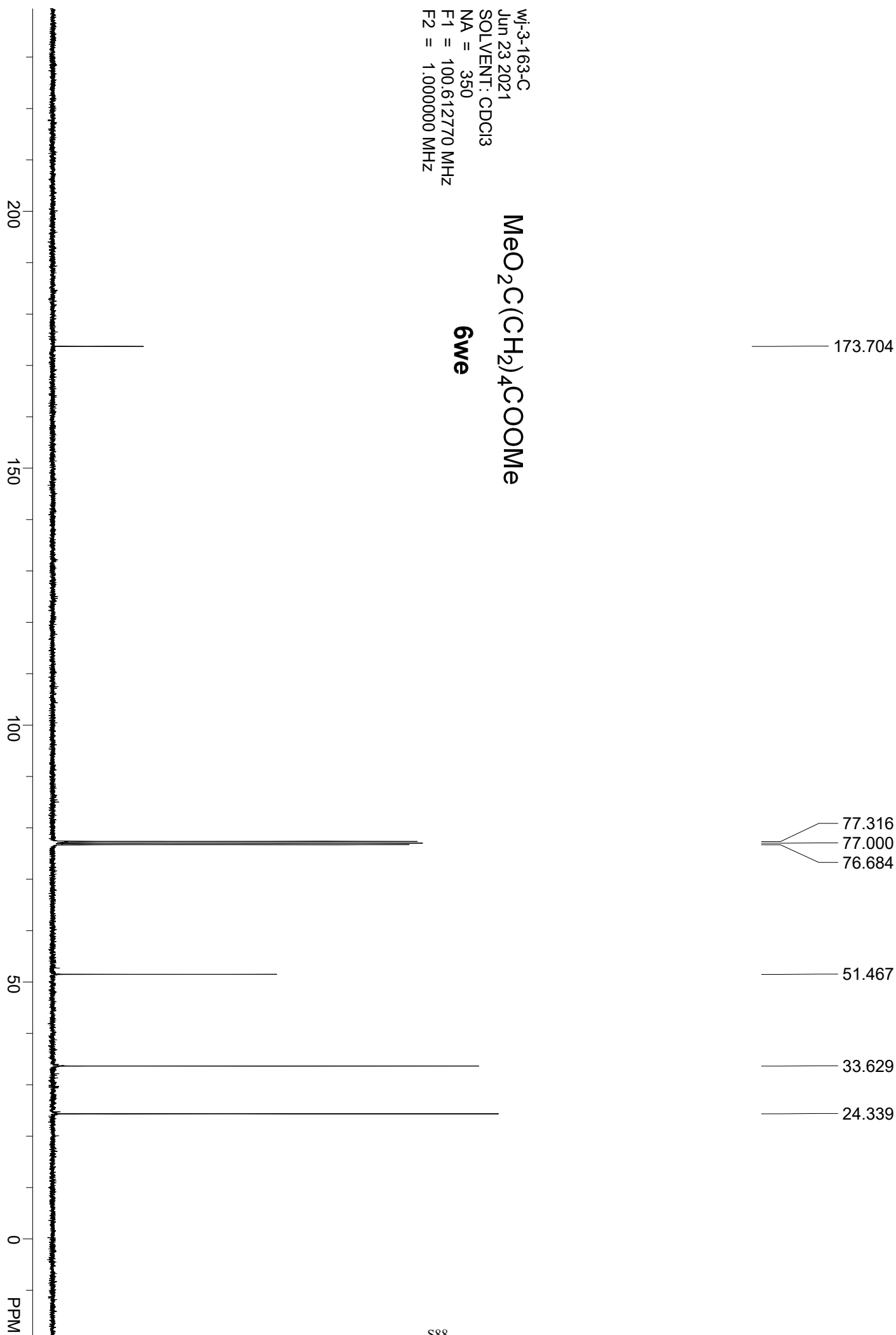
wj-3-163-H
Jun 23 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz
F2 = 1.000000 MHz



w/-3-163-C
Jun 23 2021
SOLVENT: CDCl3
NA = 350
F1 = 100.612770 MHz
F2 = 1.000000 MHz

MeO₂C(CH₂)₄COOMe

6we



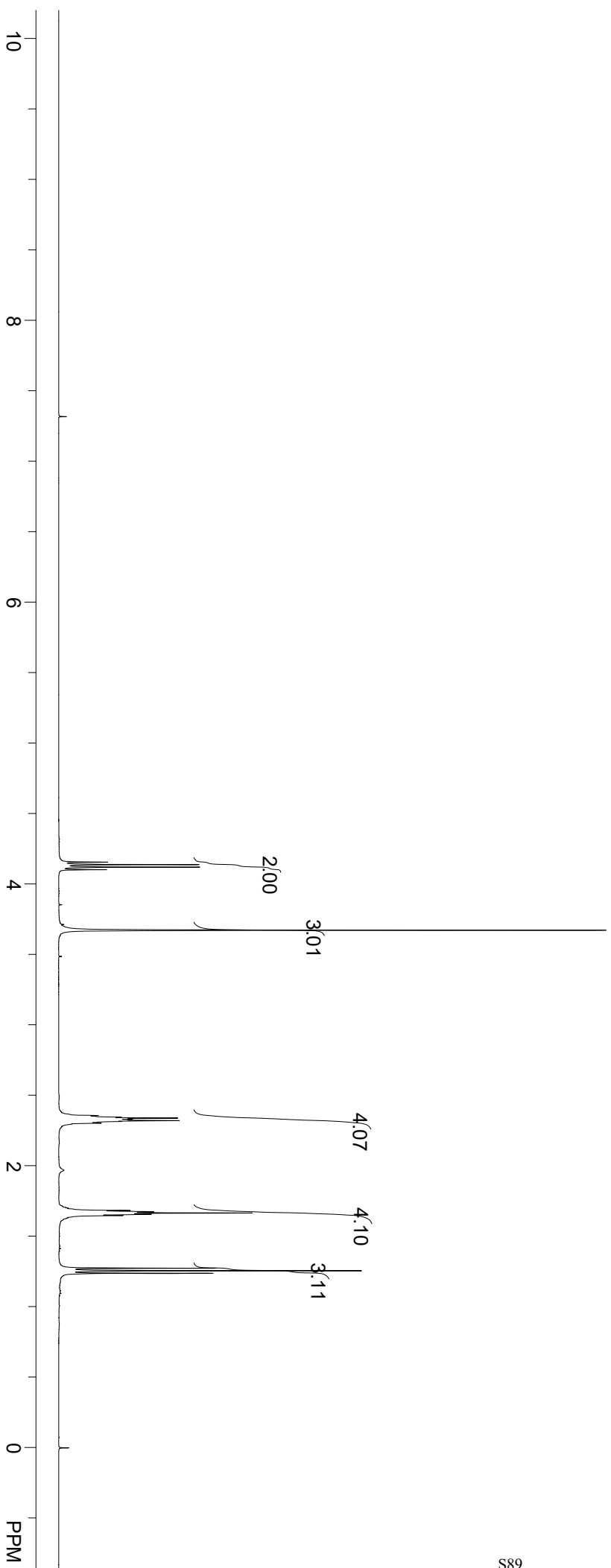
yyb-7-039-H
Oct 27 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.129974 MHz

EtO₂C(CH₂)₄COOMe

6xe

4.155
4.137
4.119
4.102
3.672

2.357
2.346
2.339
2.333
2.328
2.321
2.316
2.303
1.684
1.675
1.666
1.657
1.648
1.275
1.257
1.239
0.000



173.656
173.206

77.316
77.000
76.676

60.165

51.388

33.795
33.542

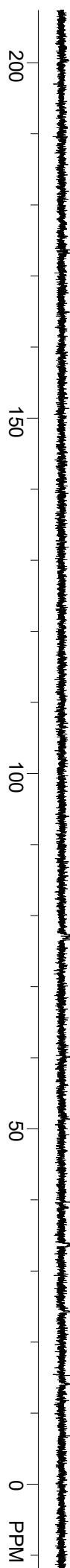
24.260
24.236

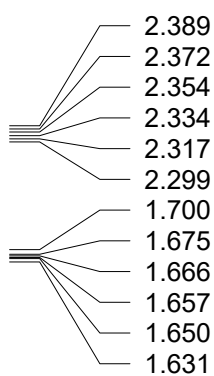
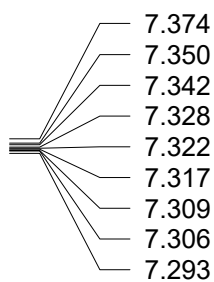
14.084

yyb-7-039-C
Oct 27 2021
SOLVENT: CDCl₃
NA = 61
F1 = 100.612770 MHz



6xe

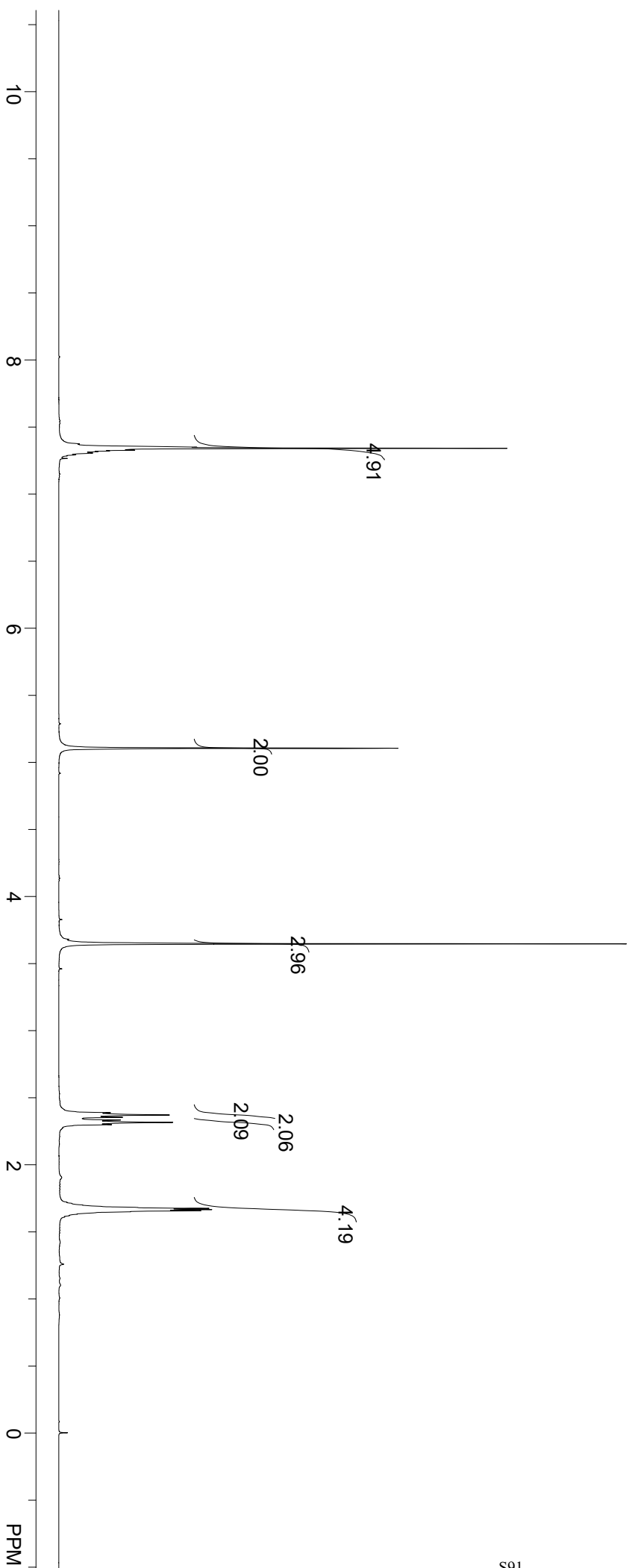




yyb-7-036-H
Oct 25 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz

BnO₂C(CH₂)₄COOMe

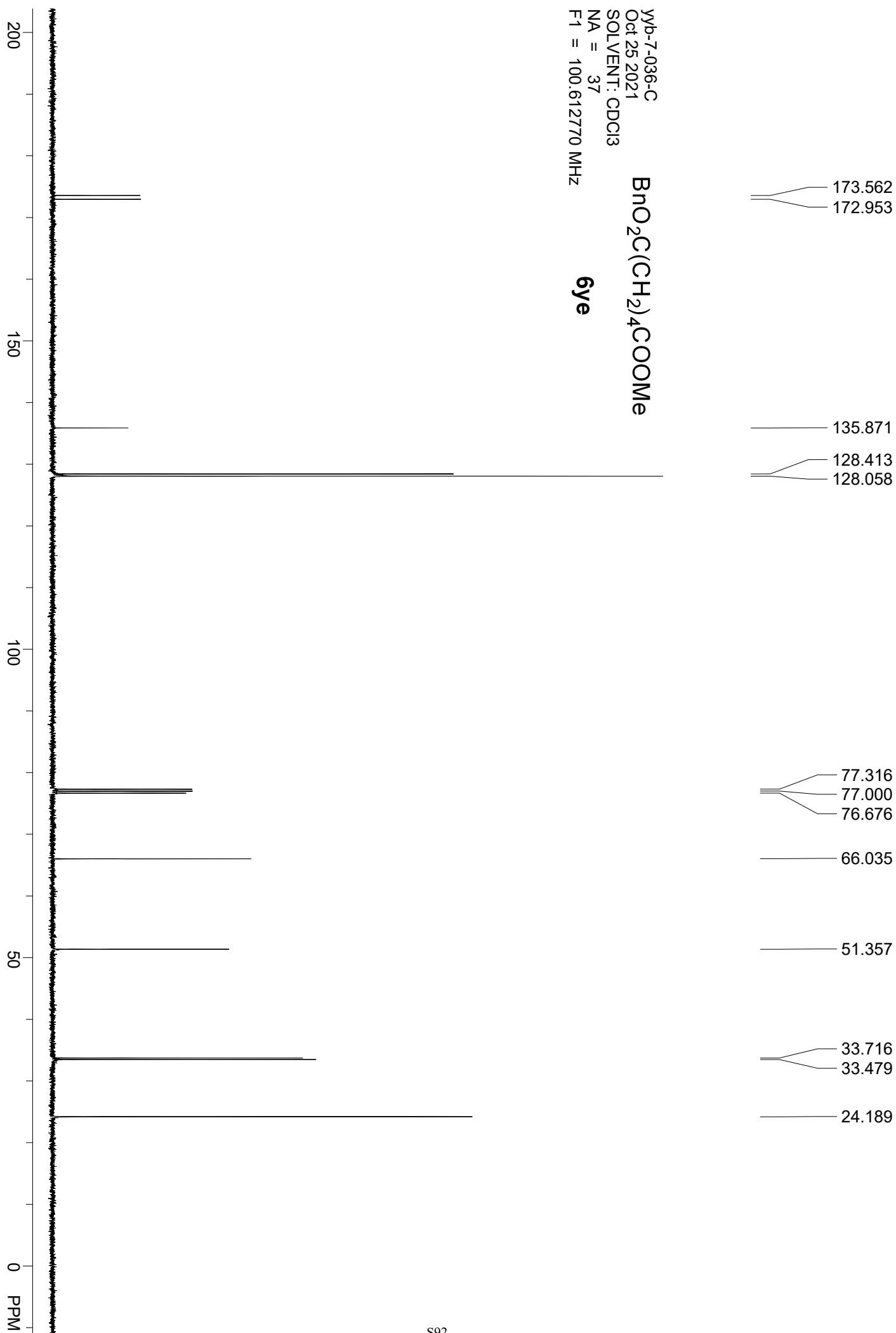
6ye



yyb-7-036-C
Oct 25 2021
SOLVENT: CDCl3
NA = 37
F1 = 100.612770 MHz

$\text{BnO}_2\text{C}(\text{CH}_2)_4\text{COOMe}$

6ye



7.845
7.837
7.831
7.824
7.814
7.740
7.731
7.723
7.718
7.710

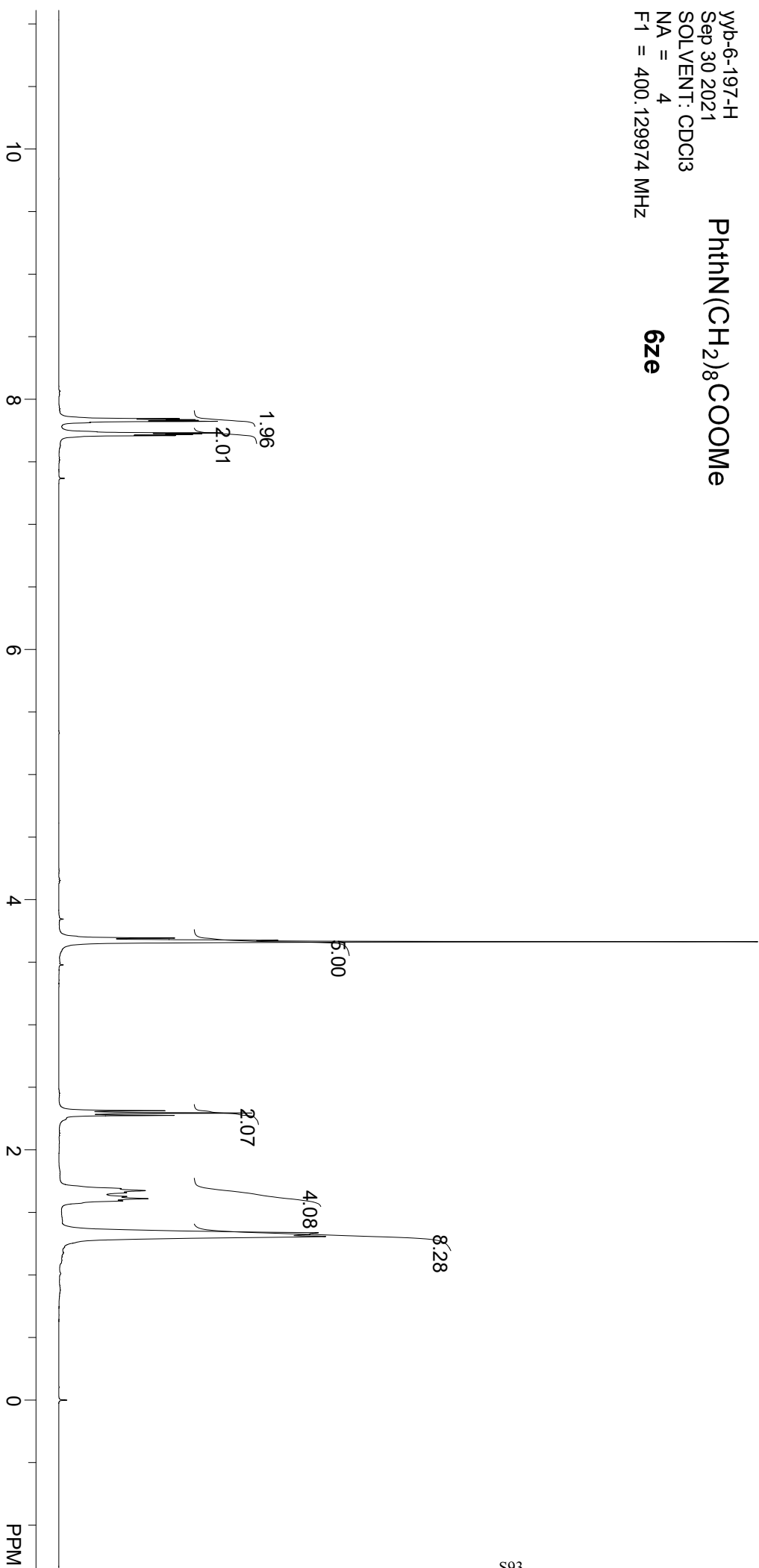
3.691
3.673
3.662

2.311
2.293
2.274
1.690
1.673
1.657
1.627
1.609
1.591
1.336
1.323
1.307

-0.000

yrb-6-197-H
Sep 30 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.129974 MHz

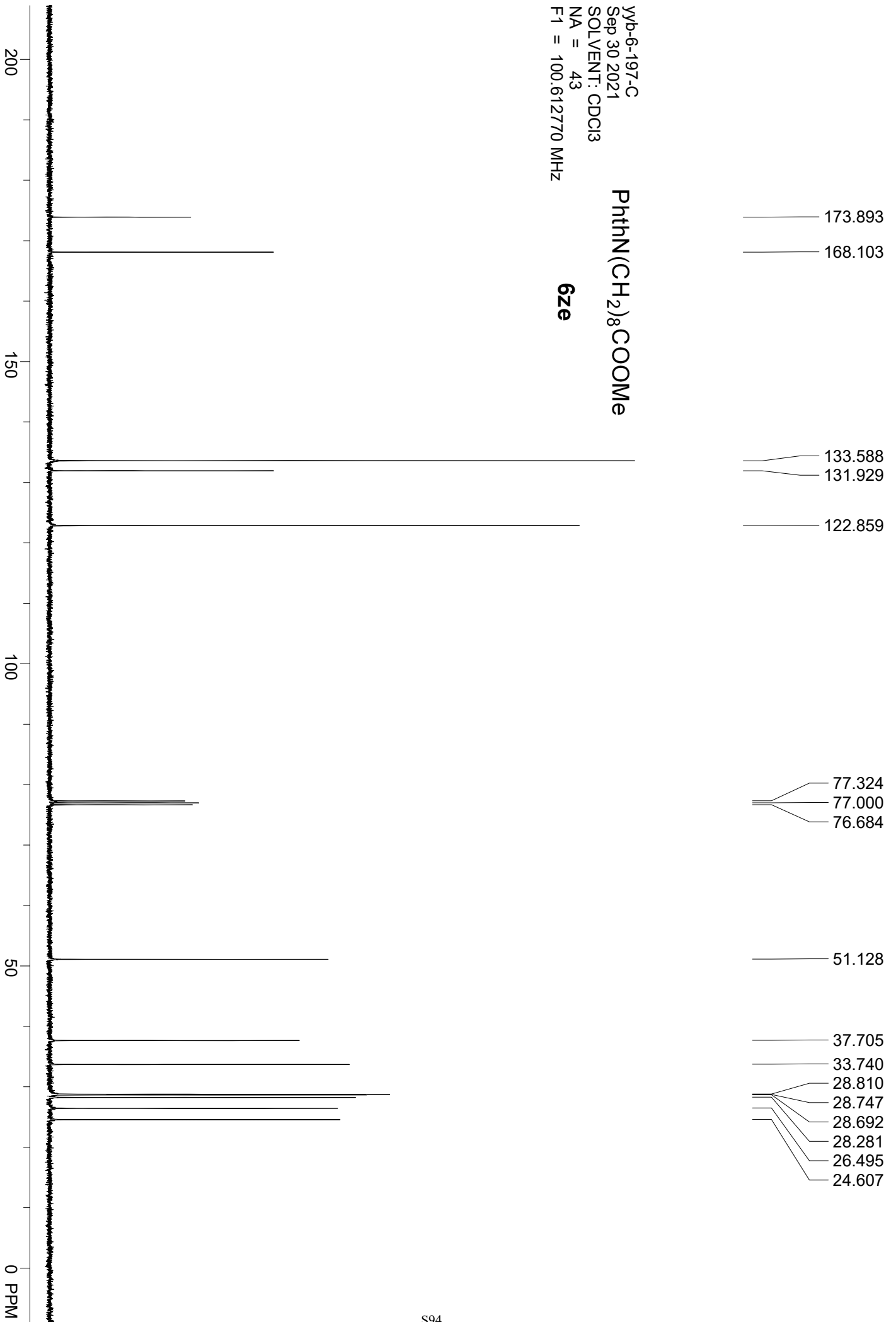
6ze

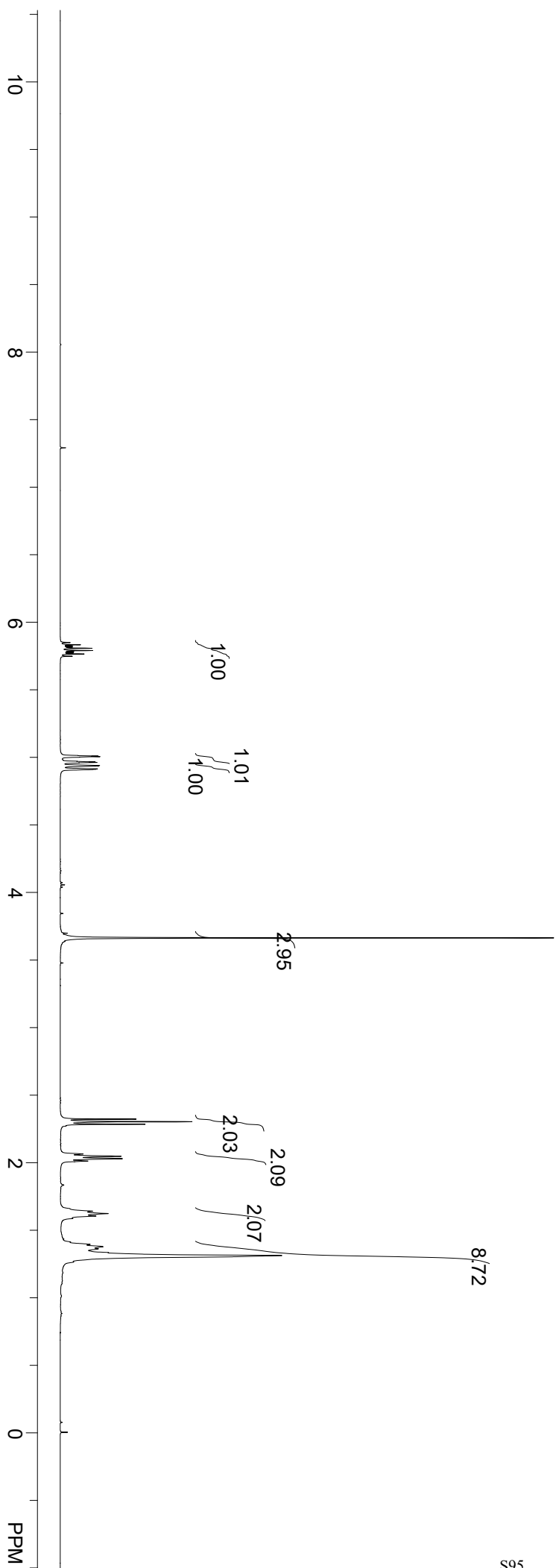


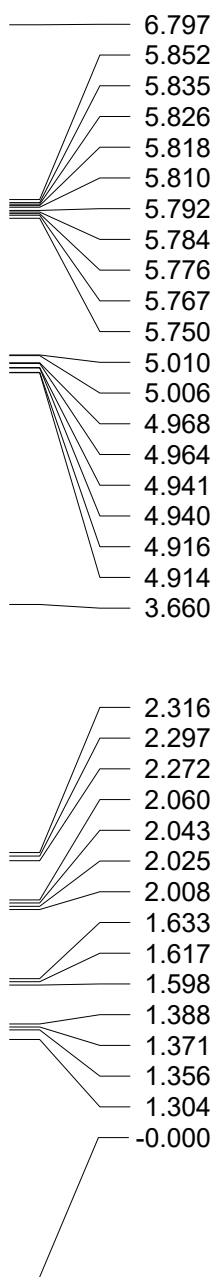
yyb-6-197-C
Sep 30 2021
SOLVENT: CDCl3
NA = 43
F1 = 100.612770 MHz

Phthn(CH₂)₈COOMe

6ze



COOC(=O)CCCCC/C=C/

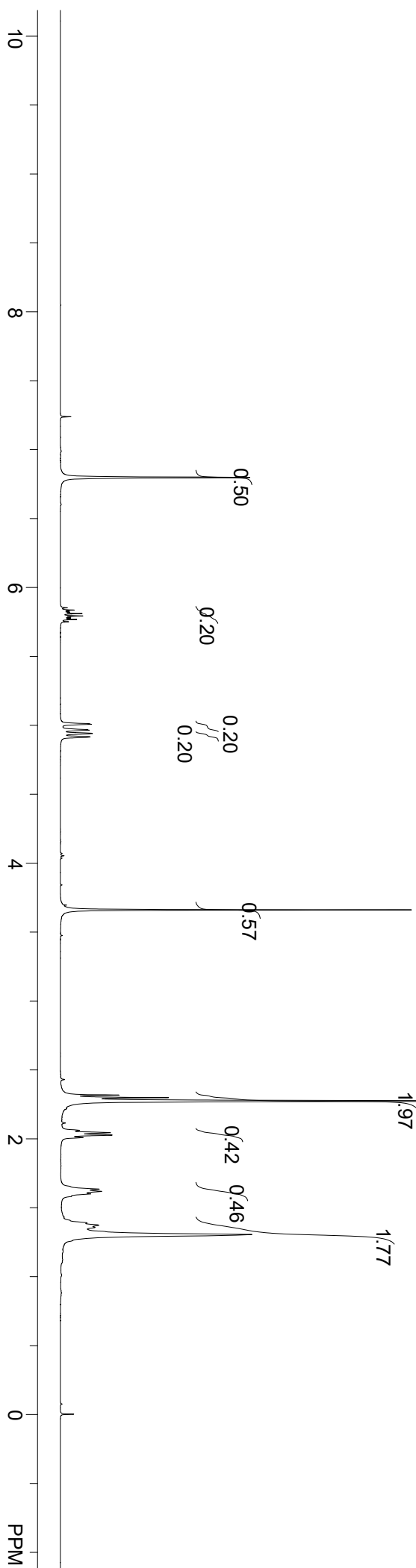


Wb-7-075-p
 Nov 19 2021
 SOLVENT: CDCl₃
 NA = 4
 F1 = 400.130005 MHz

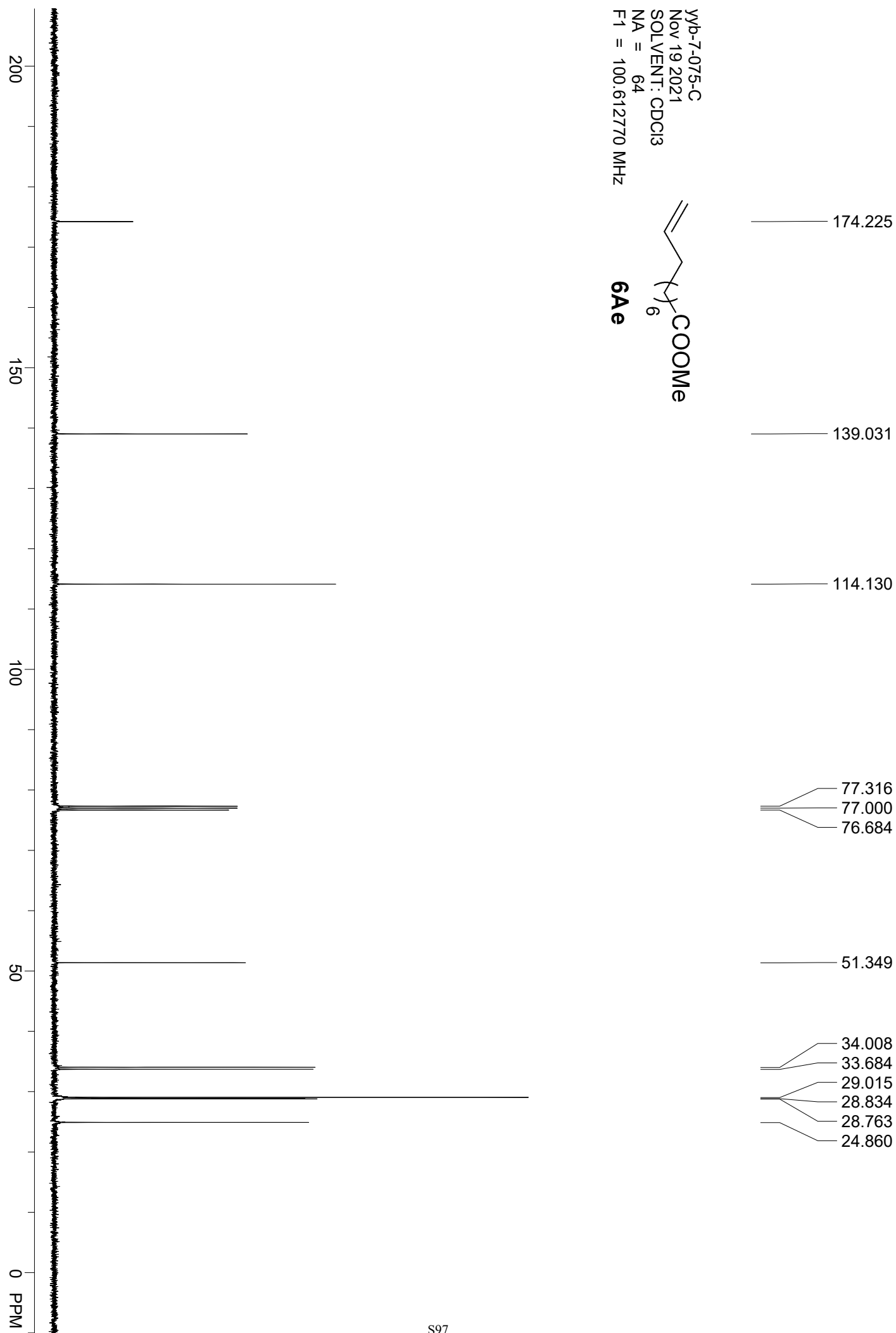
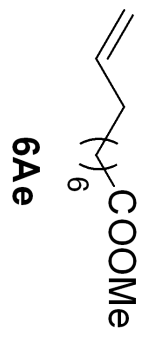


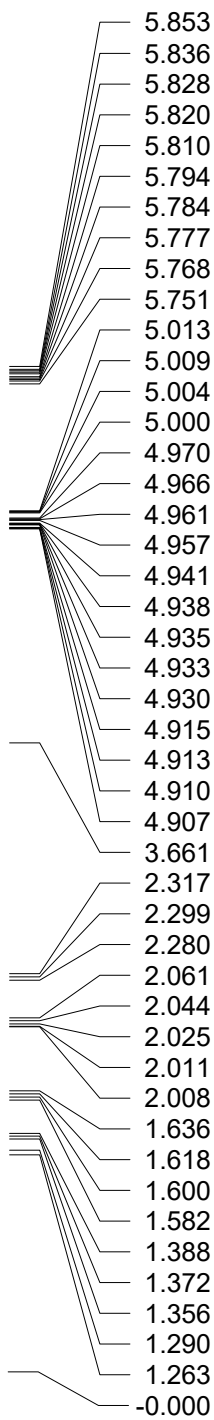
6Ae

Purity (98%) determined by using mesitylene (23.0 uL, 0.5 mmol Ar-H)
 as the internal standard in 35.7 mg of sample.

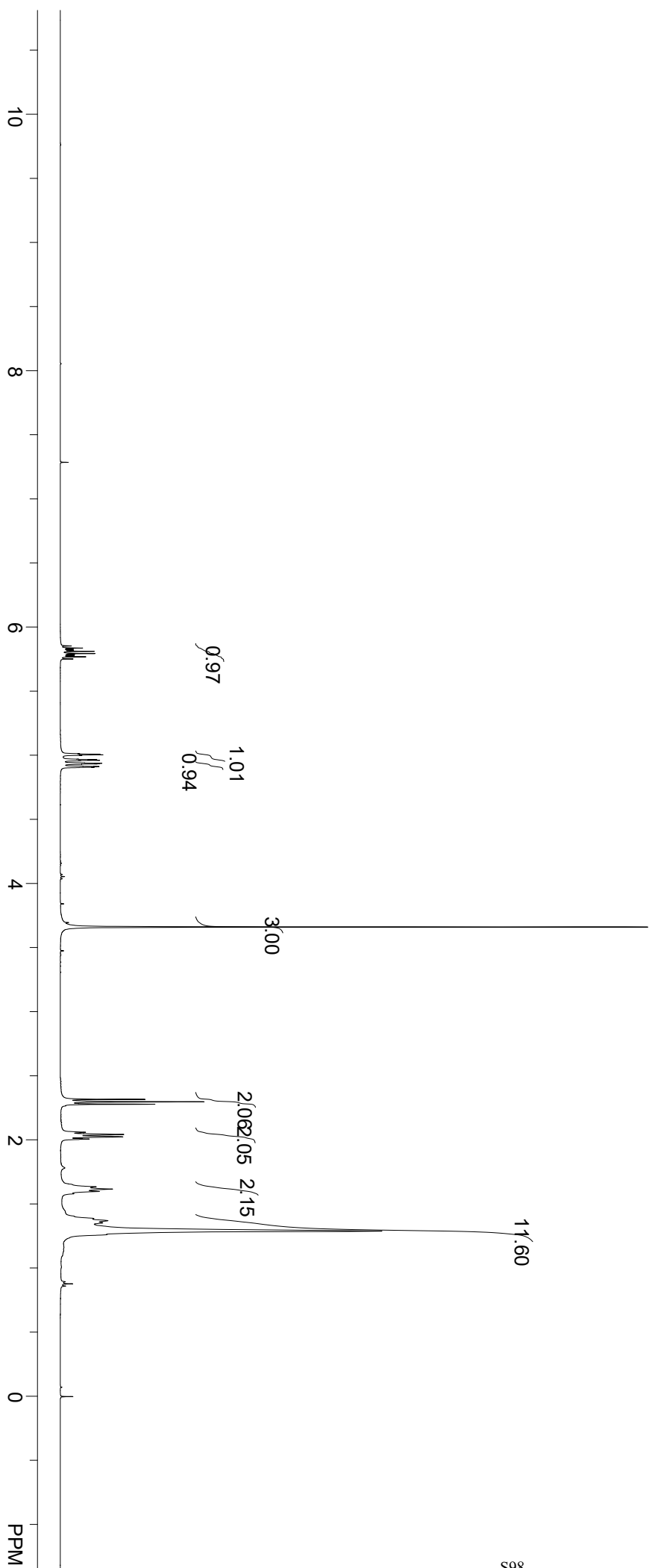
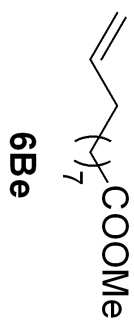


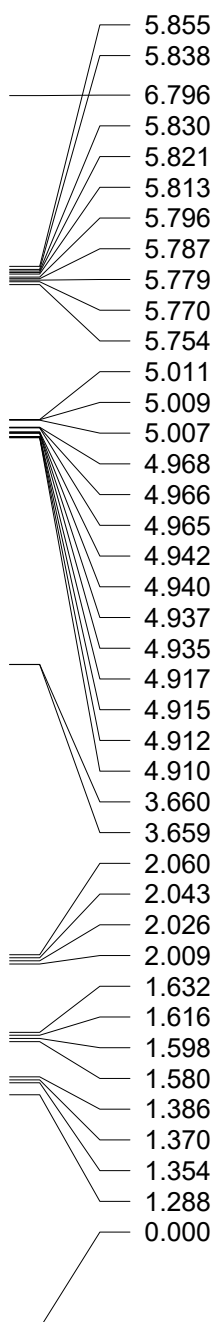
ywb-7-075-C
Nov 19 2021
SOLVENT: CDCl₃
NA = 64
F1 = 100.612770 MHz



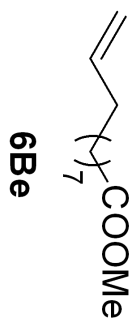


yyb-6-147-H
Sep 07 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz



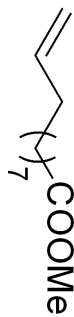


YYB-6-147-P
 Sep 07 2021
 SOLVENT: CDCl₃
 NA = 4
 F1 = 400.130005 MHz

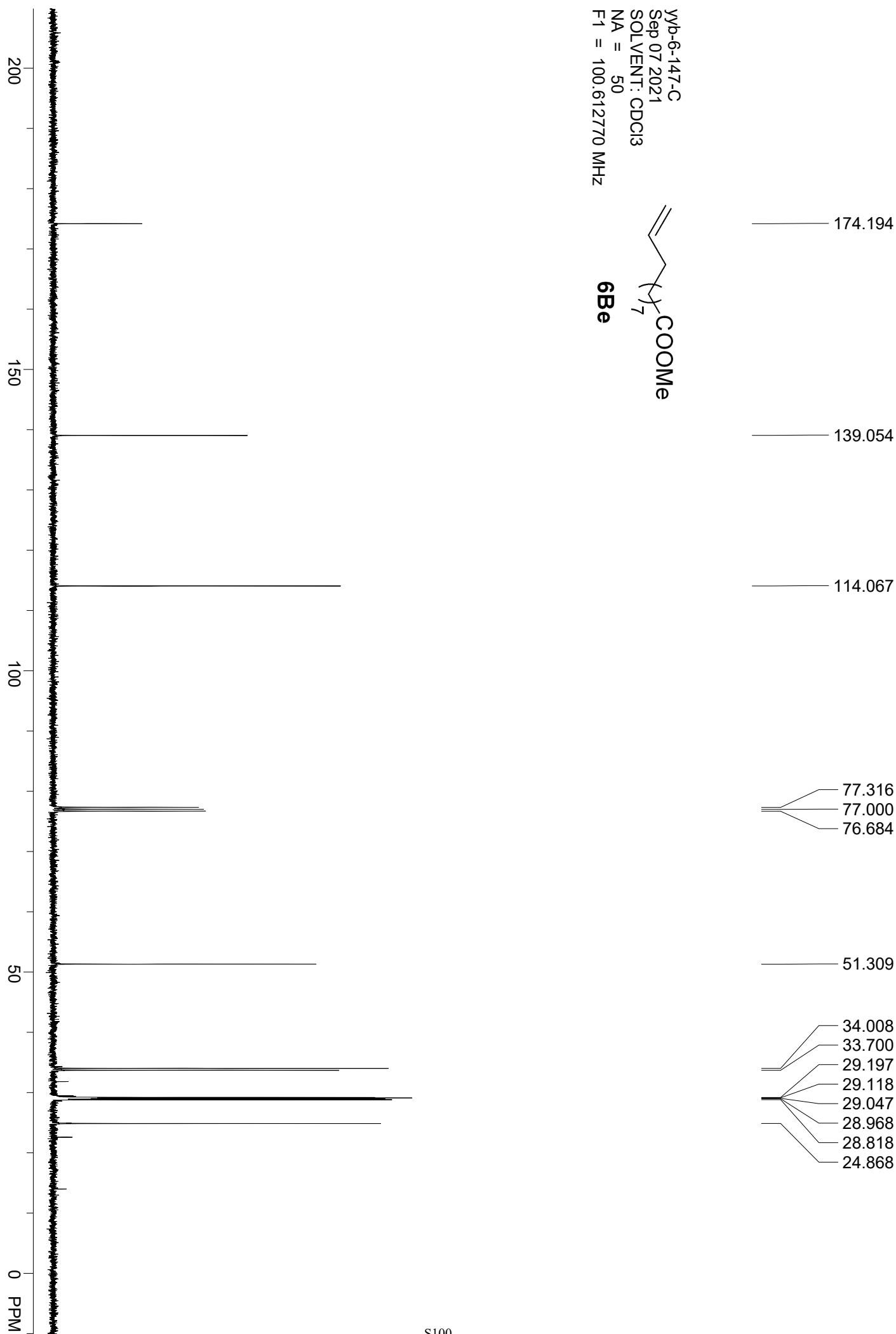


Purity (96%) determined by using mesitylene (23.0 uL, 0.5 mmol Ar-H)
 as the internal standard in 40.5 mg of sample.

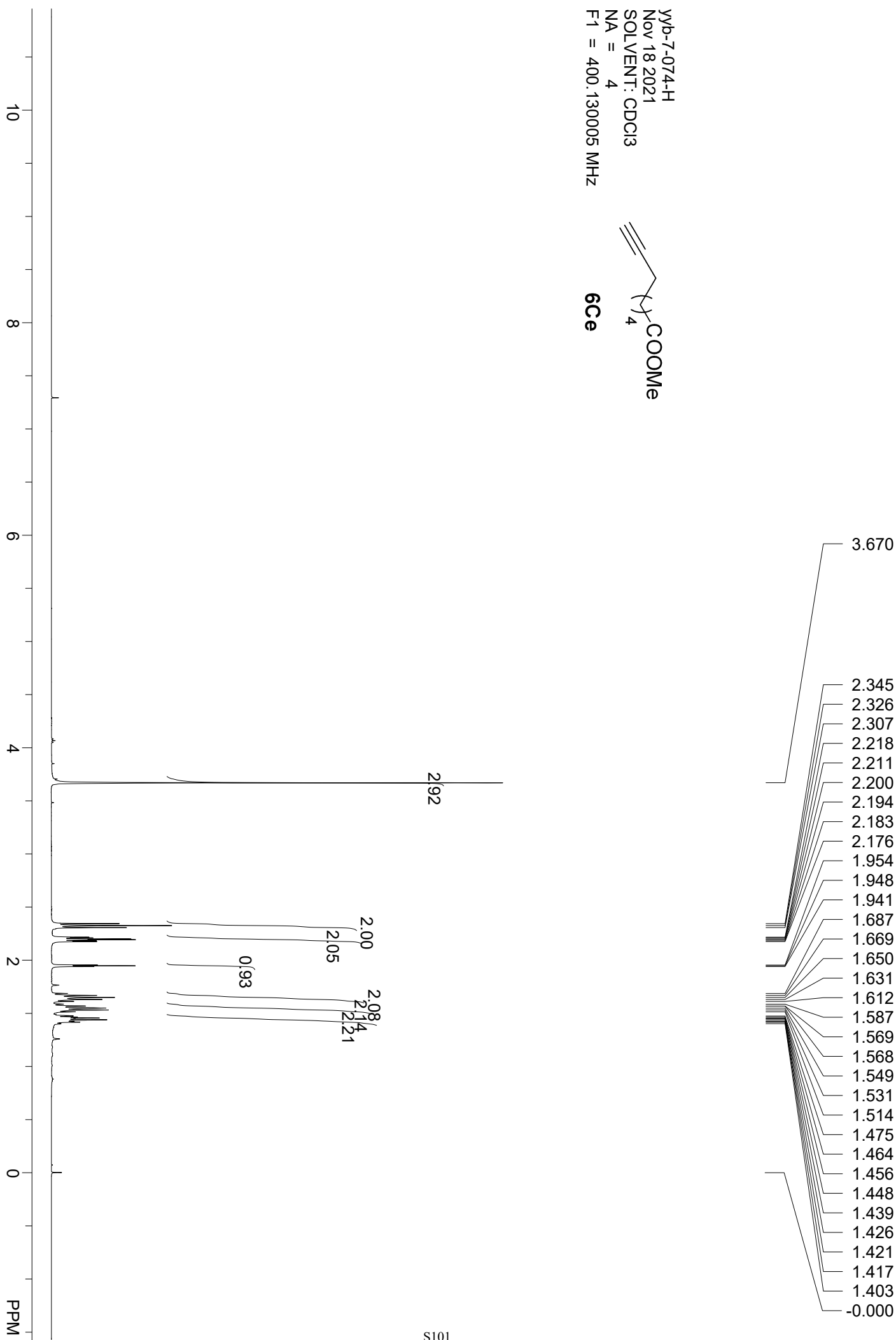
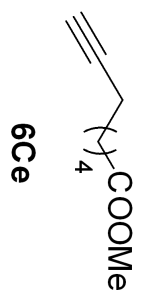
yyb-6-147-C
Sep 07 2021
SOLVENT: CDCl3
NA = 50
F1 = 100.612770 MHz



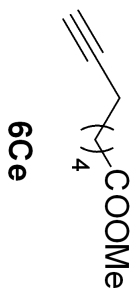
6Be



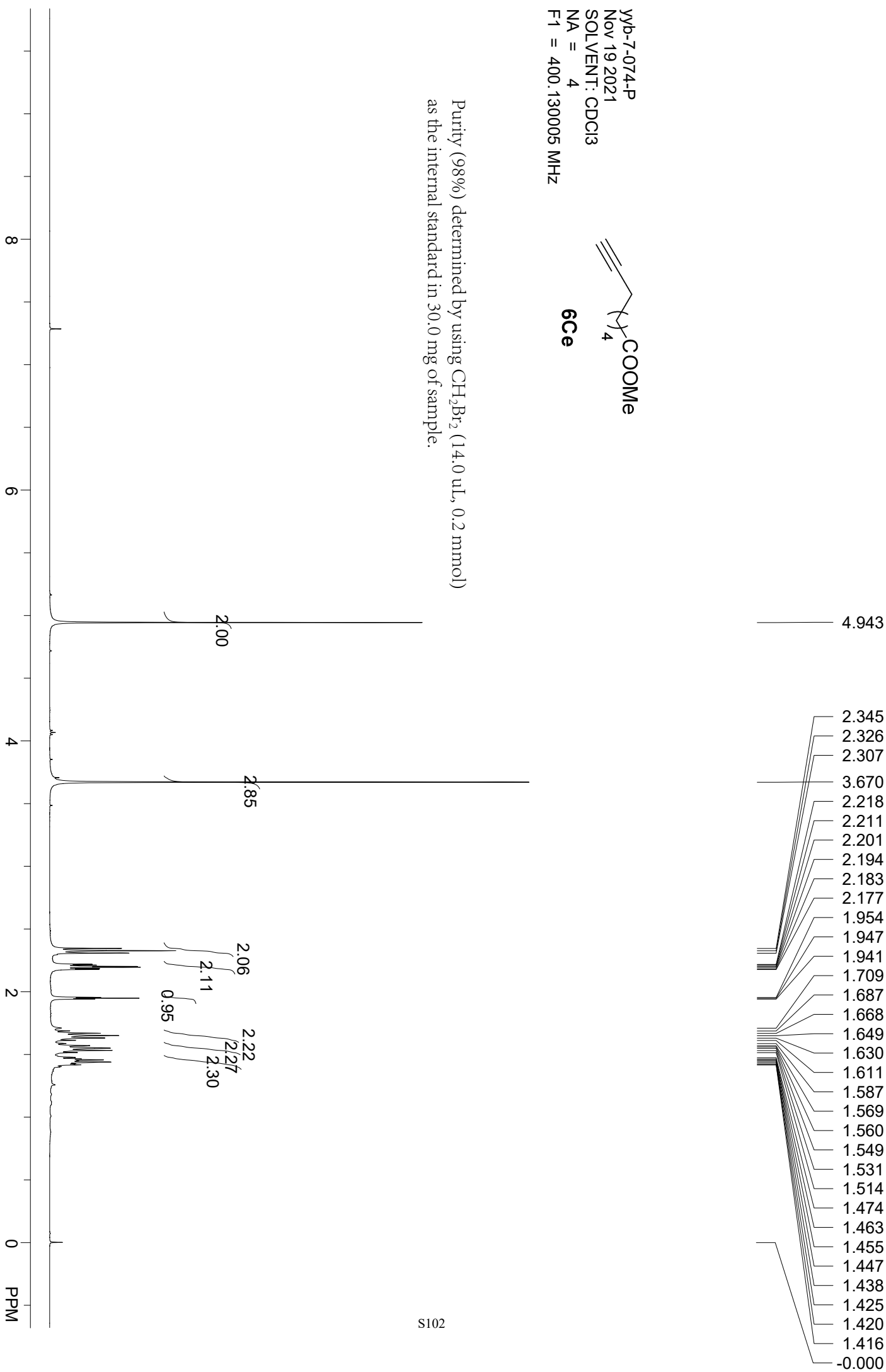
yyb-7-074-H
Nov 18 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



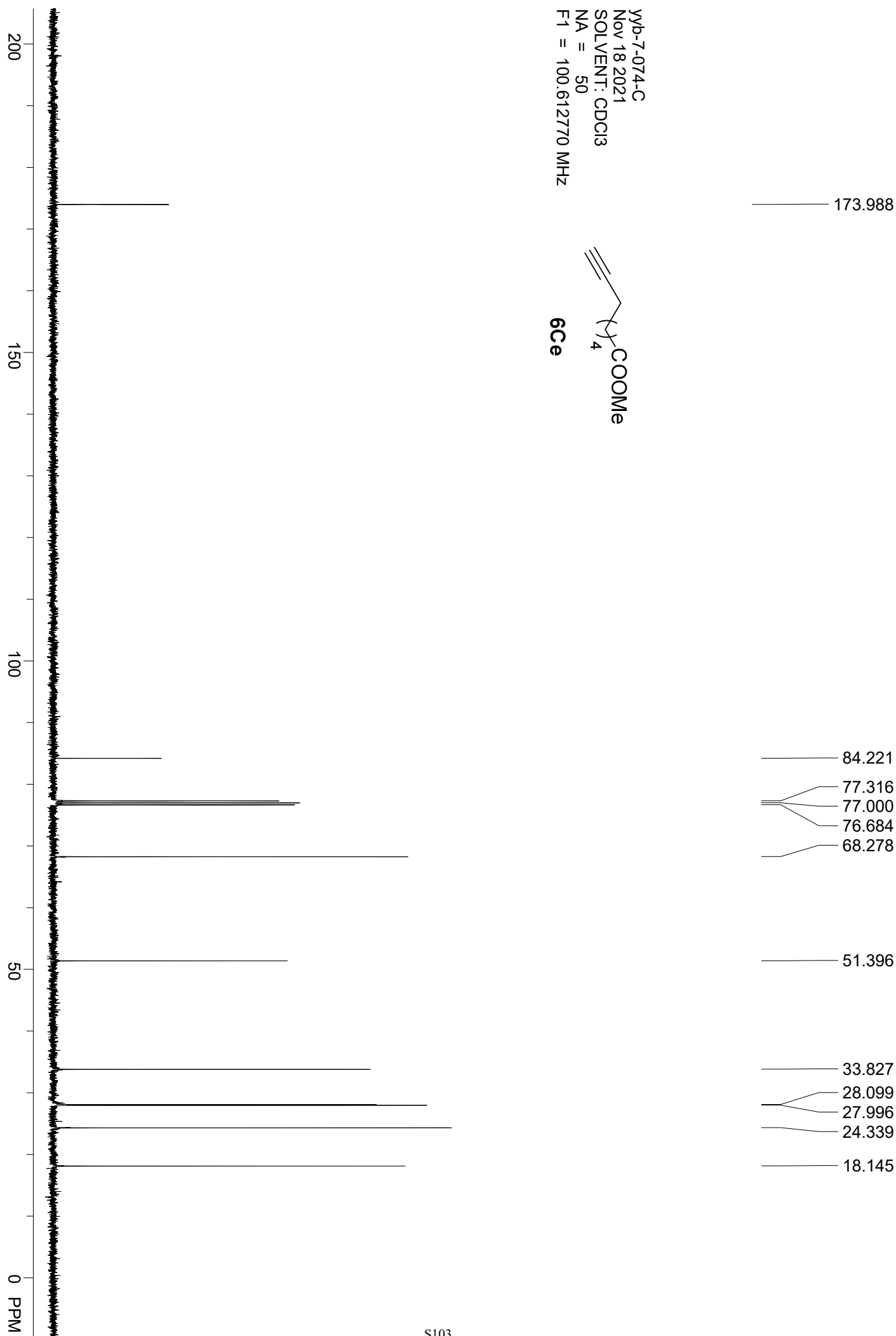
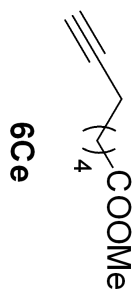
Yb-7-074-P
 Nov 19 2021
 SOLVENT: CDCl₃
 NA = 4
 F1 = 400.130005 MHz



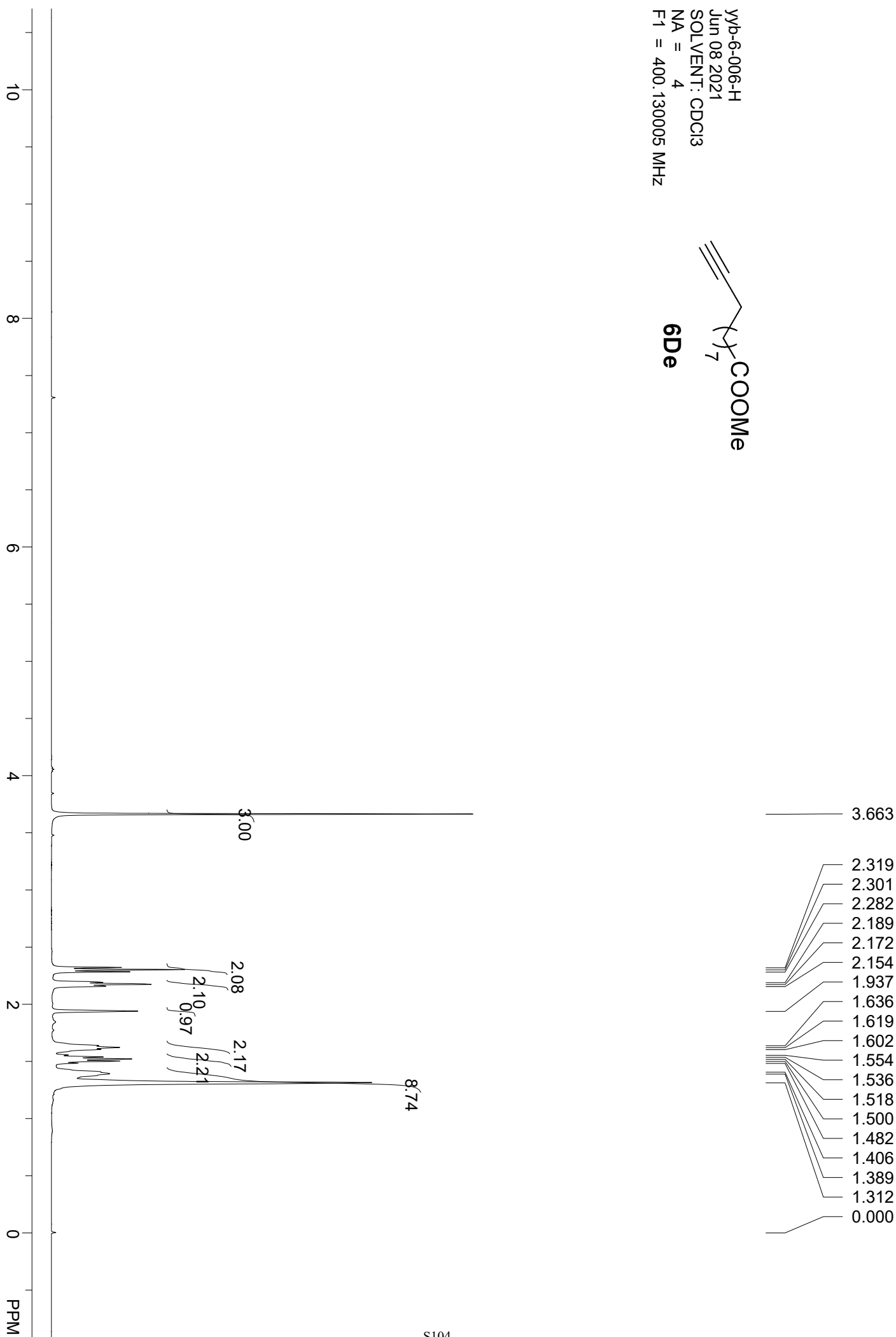
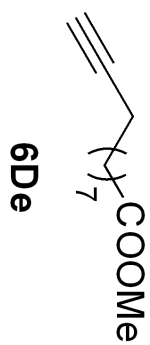
Purity (98%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
 as the internal standard in 30.0 mg of sample.



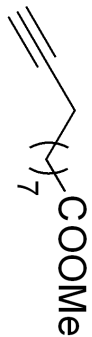
yyb-7-074-C
Nov 18 2021
SOLVENT: CDCl₃
NA = 50
F1 = 100.612770 MHz



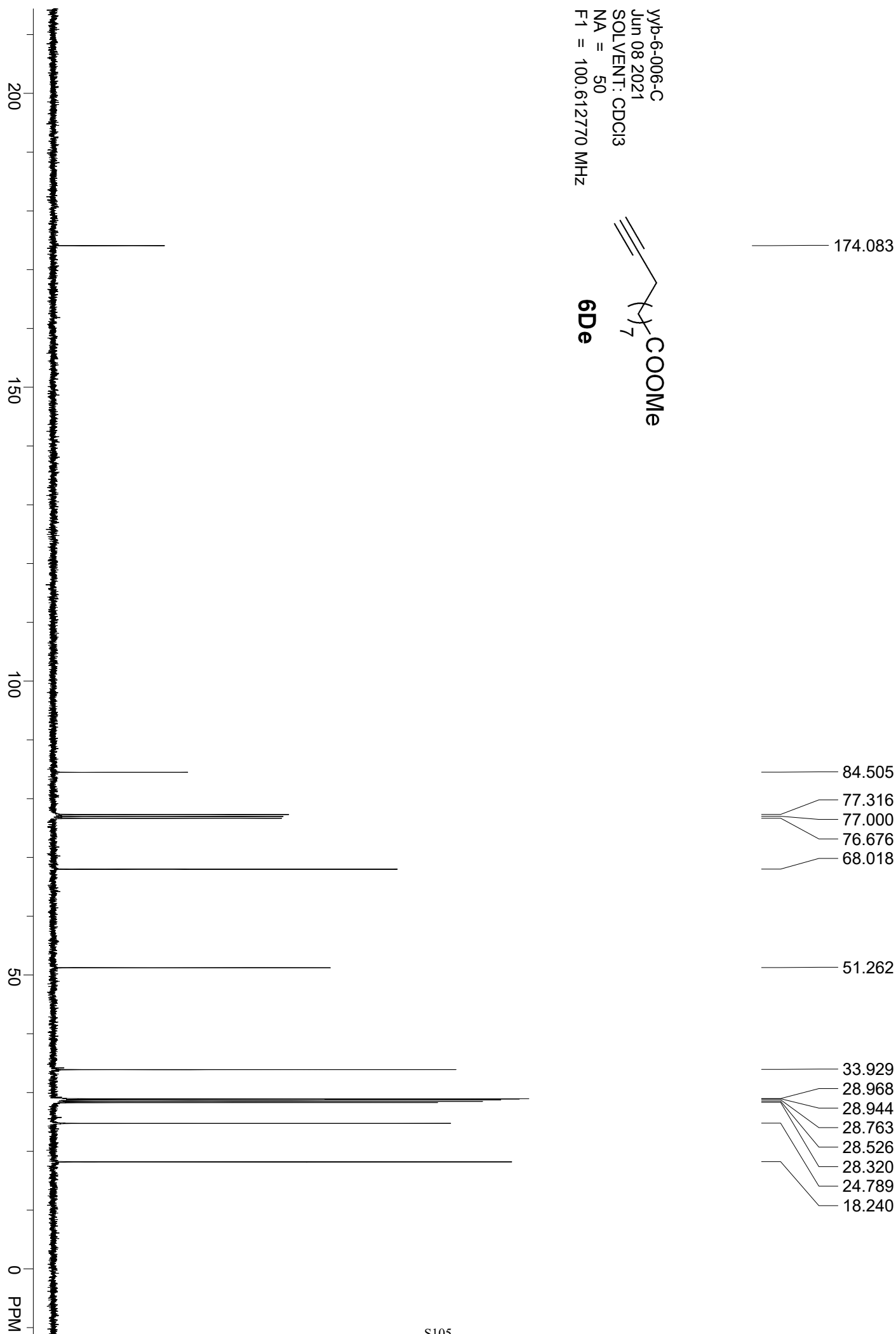
yyb-6-006-H
Jun 08 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz



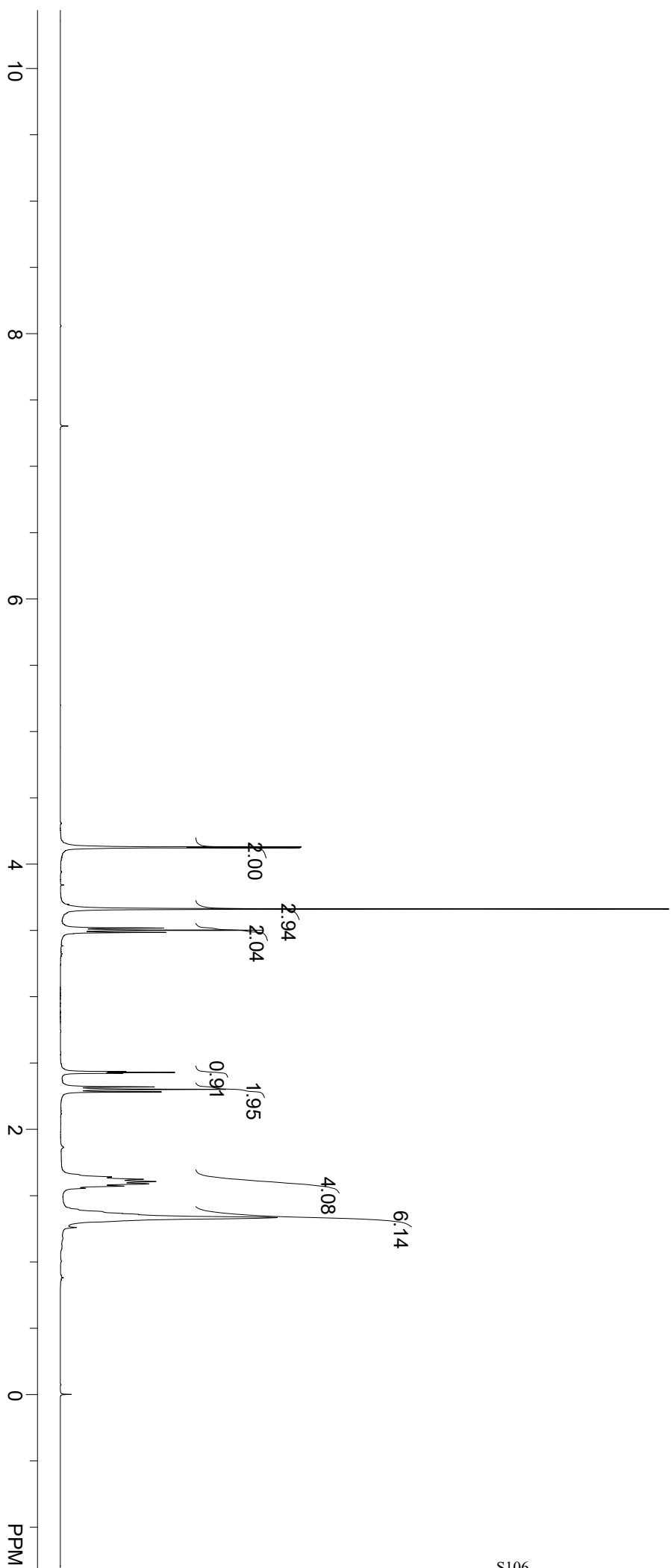
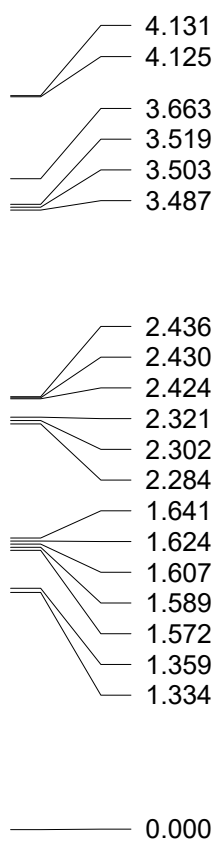
yyb-6-006-C
Jun 08 2021
SOLVENT: CDCl3
NA = 50
F1 = 100.612770 MHz



6De



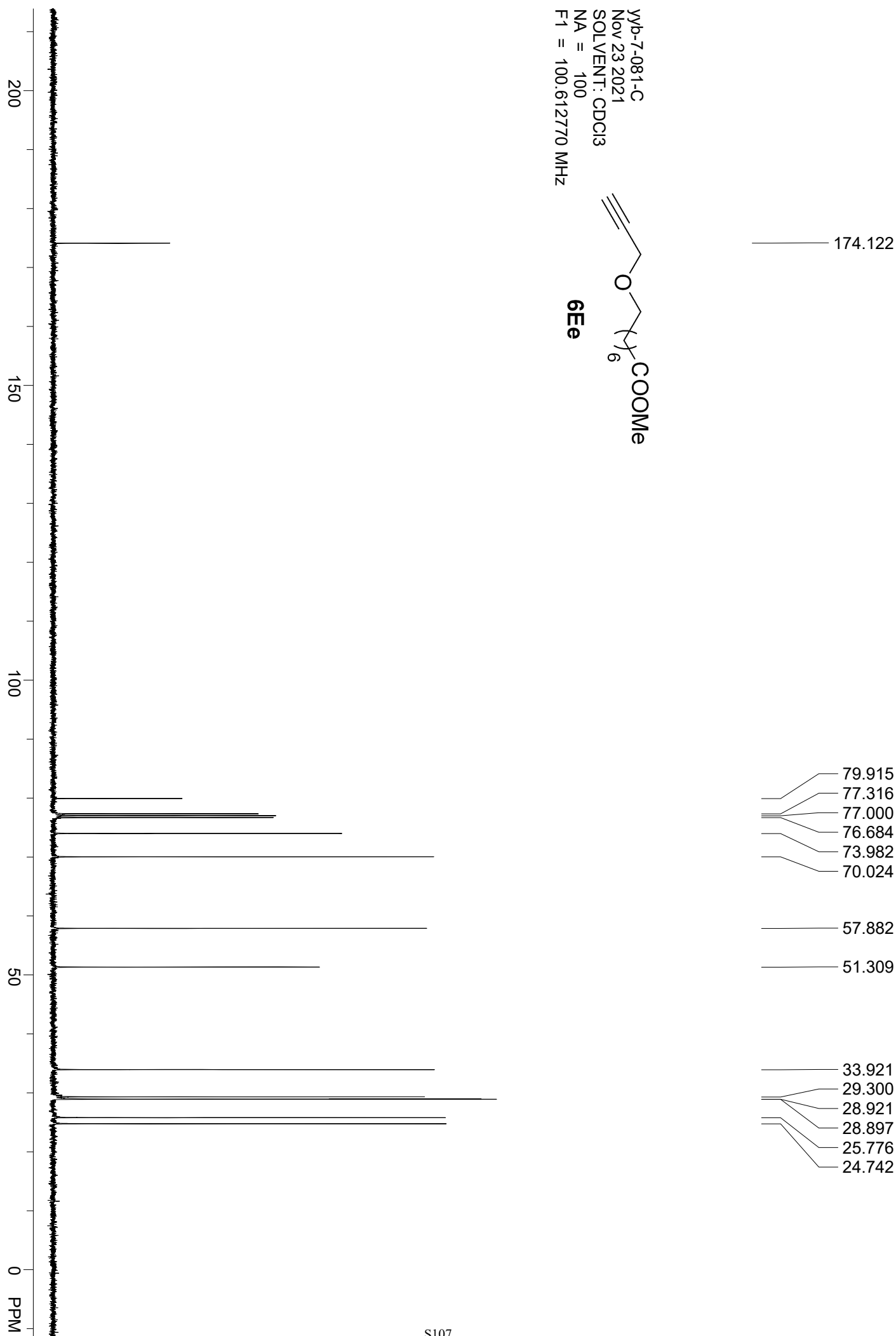
yyb-7-081-H
 Nov 23 2021
 SOLVENT: CDCl₃
 NA = 4
 F1 = 400.130005 MHz



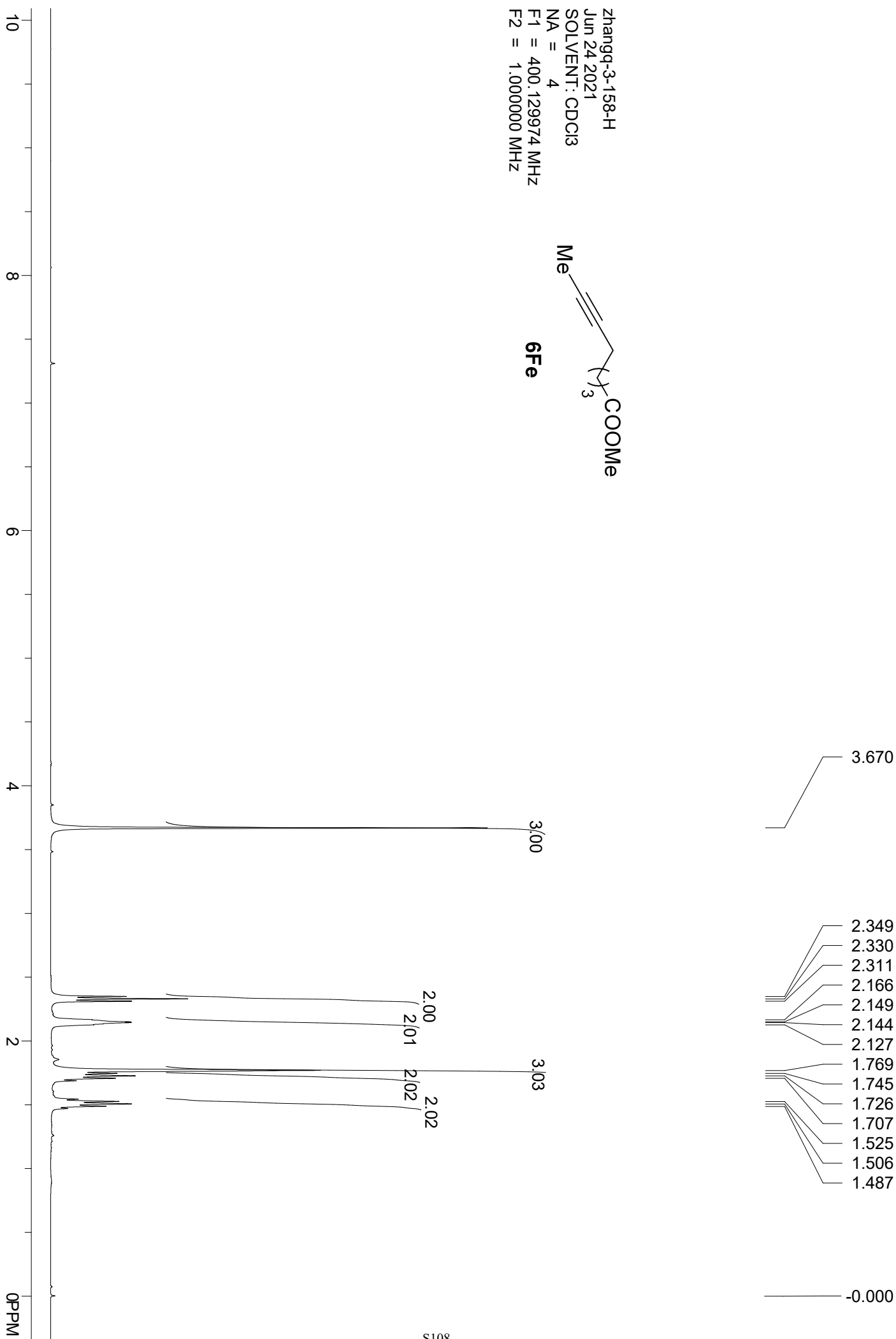
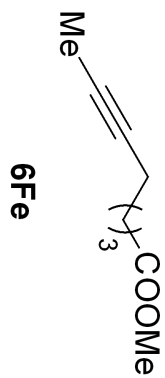
Wb-7-081-C
Nov 23 2021
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz



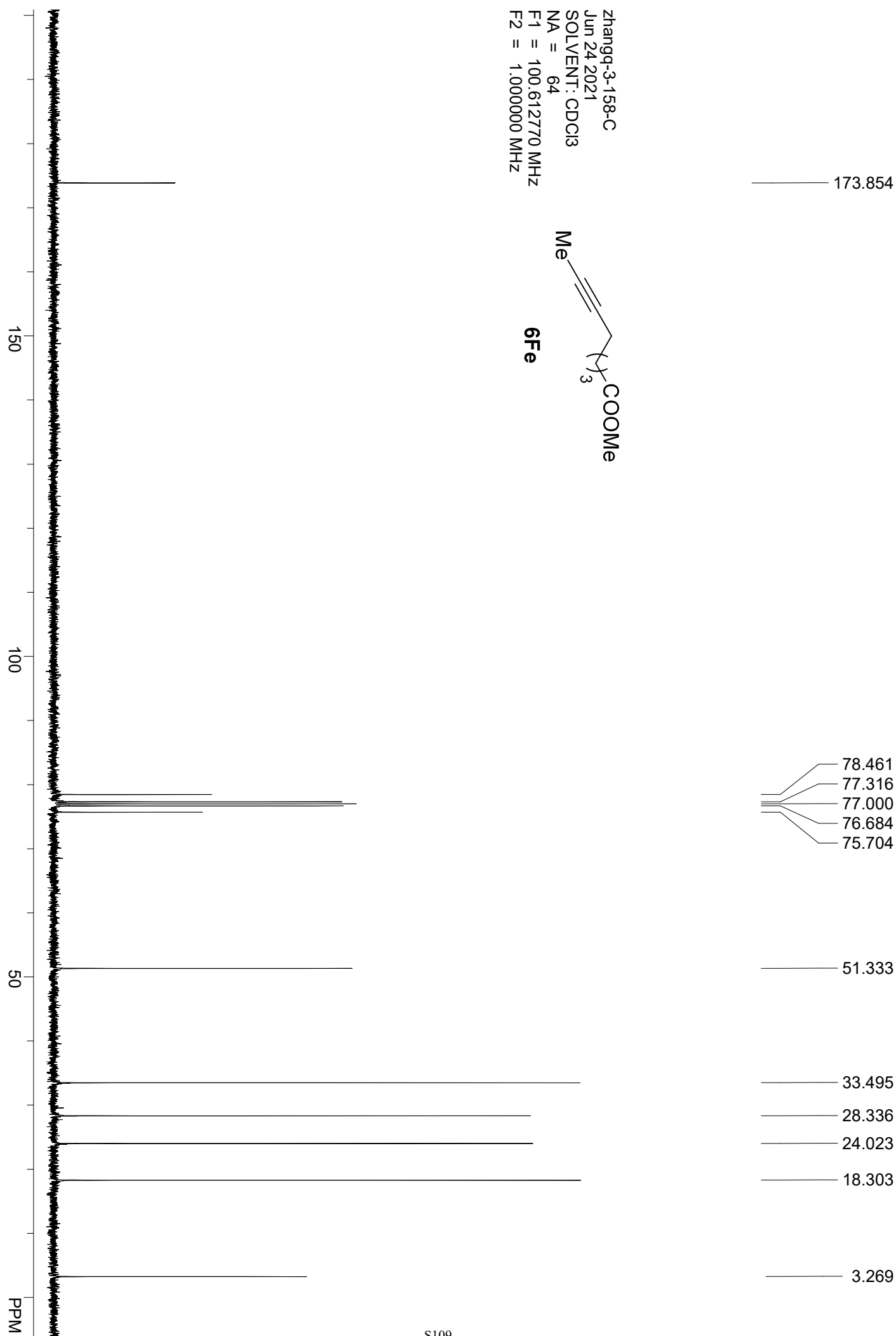
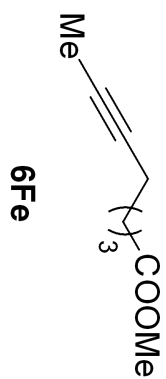
6Ee

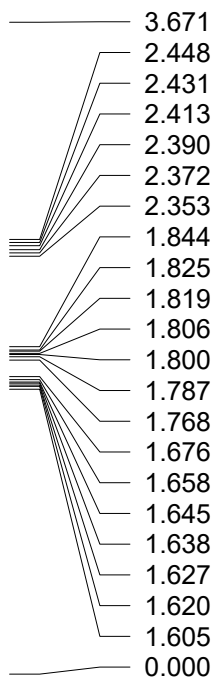
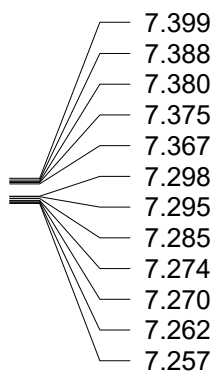


zhangq-3-158-H
Jun 24 2021
SOLVENT: CDCl₃
NA = 4
F1 = 400.129974 MHz
F2 = 1.000000 MHz

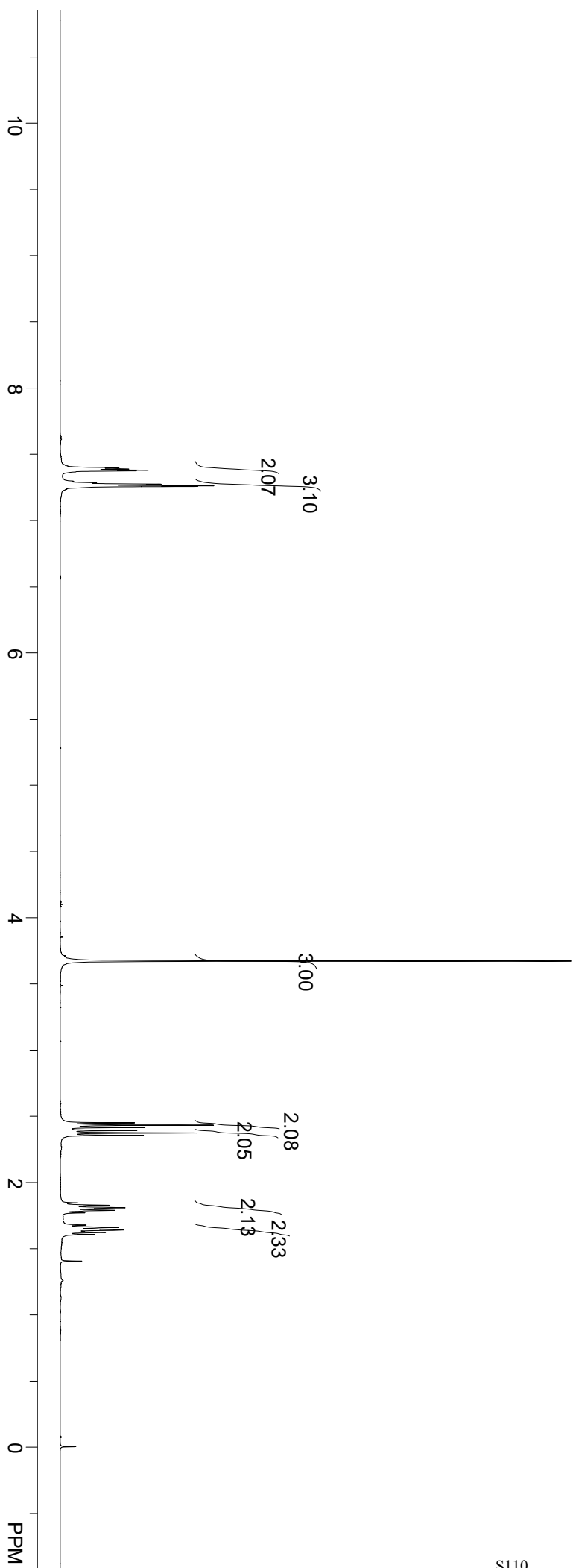
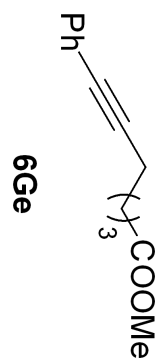


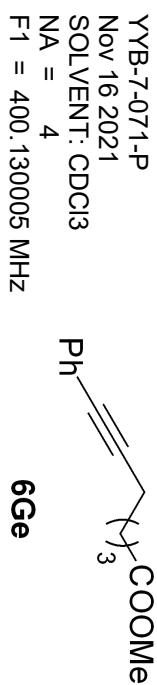
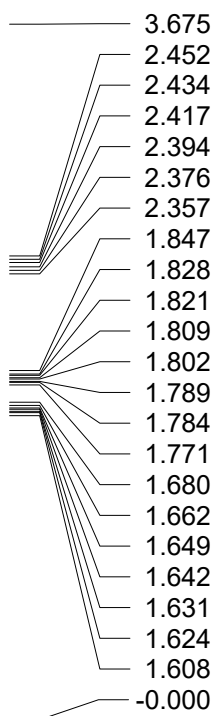
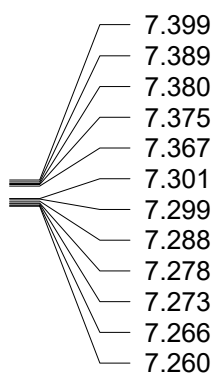
zhangq-3-158-C
Jun 24 2021
SOLVENT: CDCl₃
NA = 64
F1 = 100.612770 MHz
F2 = 1.000000 MHz



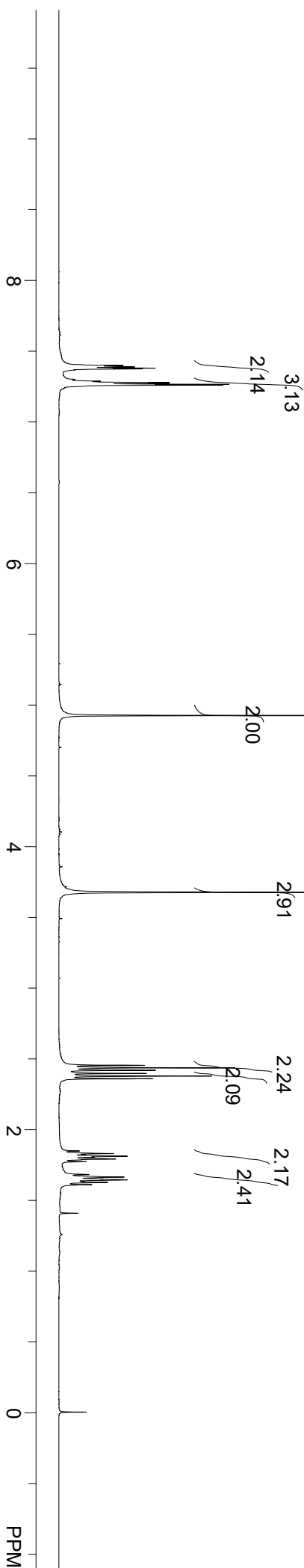


yyb-7-071-H
Nov 16 2021
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz

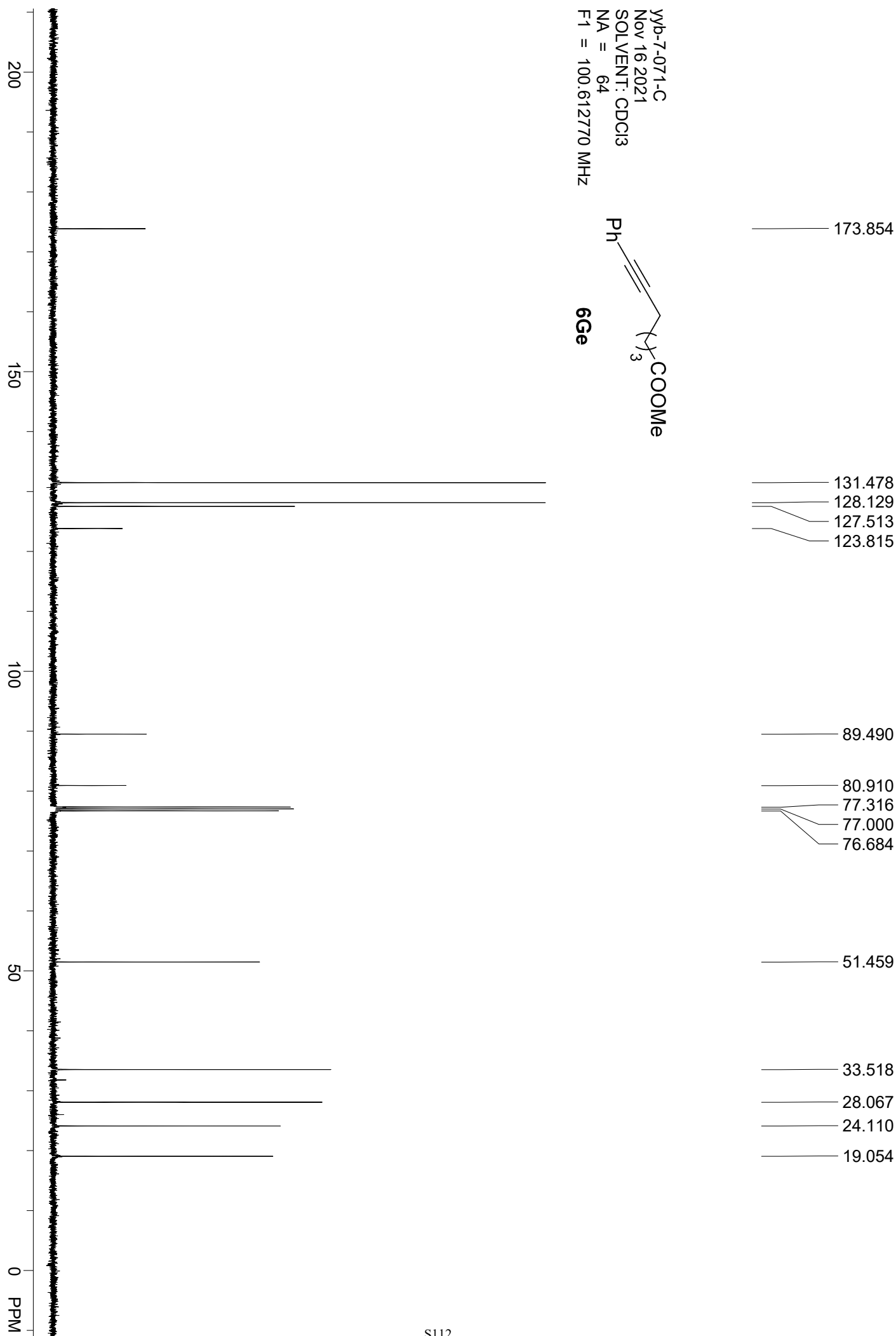
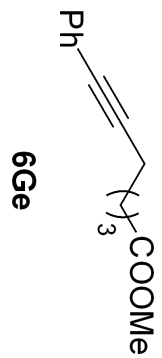




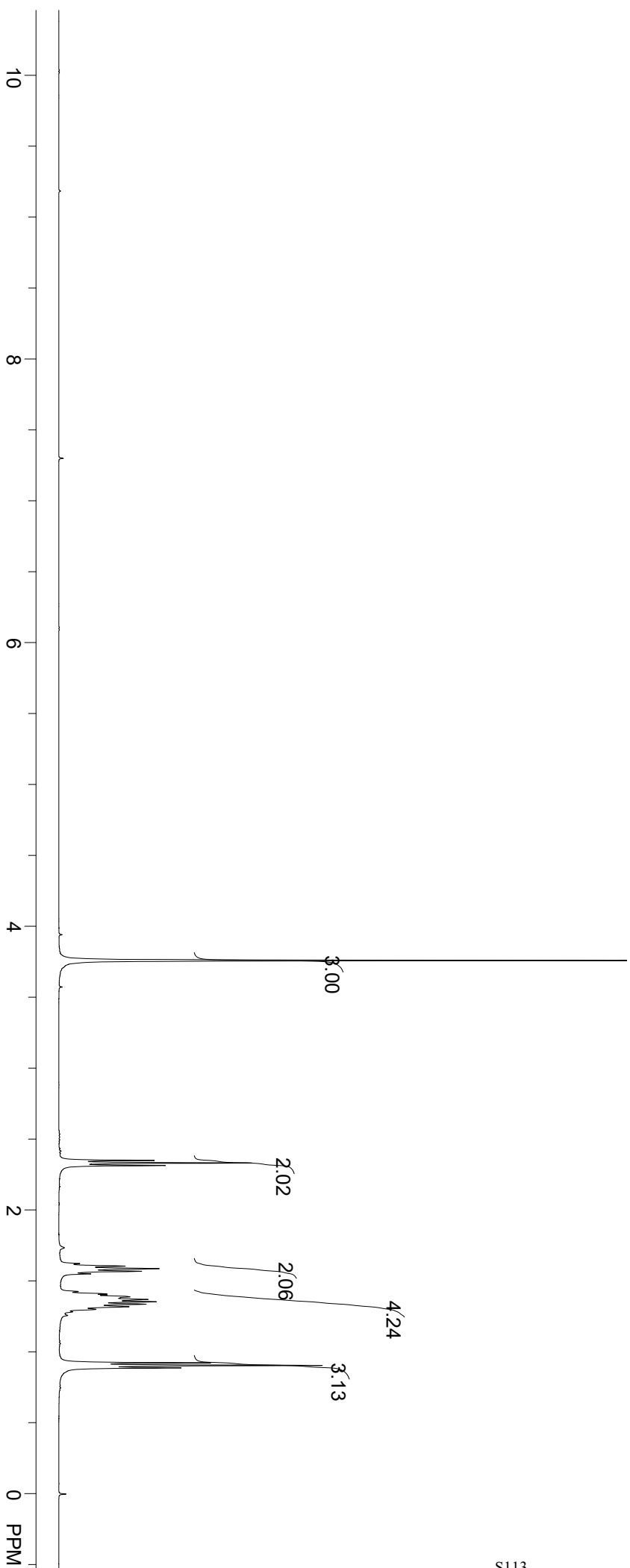
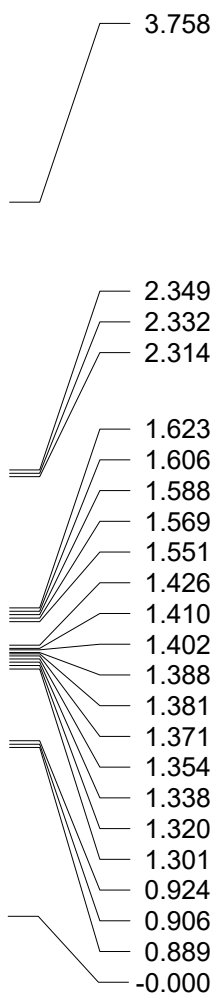
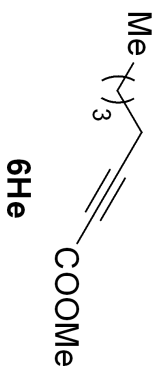
Purity (98%) determined by using CH₂Br₂ (14.0 uL, 0.2 mmol)
as the internal standard in 42.7 mg of sample.



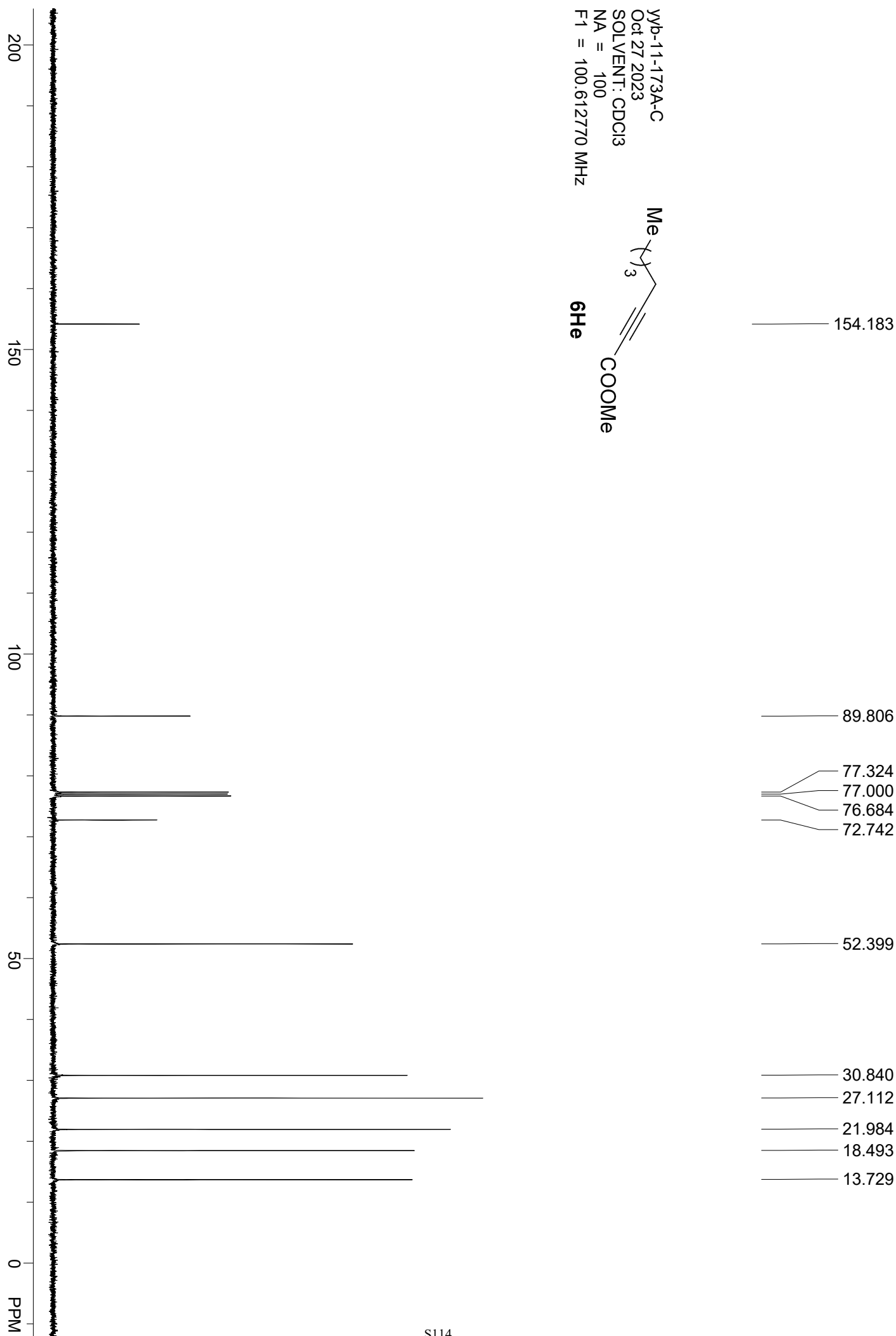
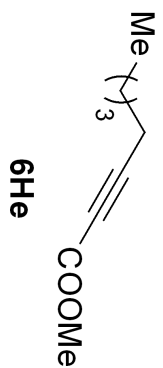
yyb-7-071-C
Nov 16 2021
SOLVENT: CDCl3
NA = 64
F1 = 100.612770 MHz



yyb-11-173A-H
Oct 27 2023
SOLVENT: CDCl3
NA = 4
F1 = 400.130005 MHz



Wb-11-173A-C
Oct 27 2023
SOLVENT: CDCl₃
NA = 100
F1 = 100.612770 MHz

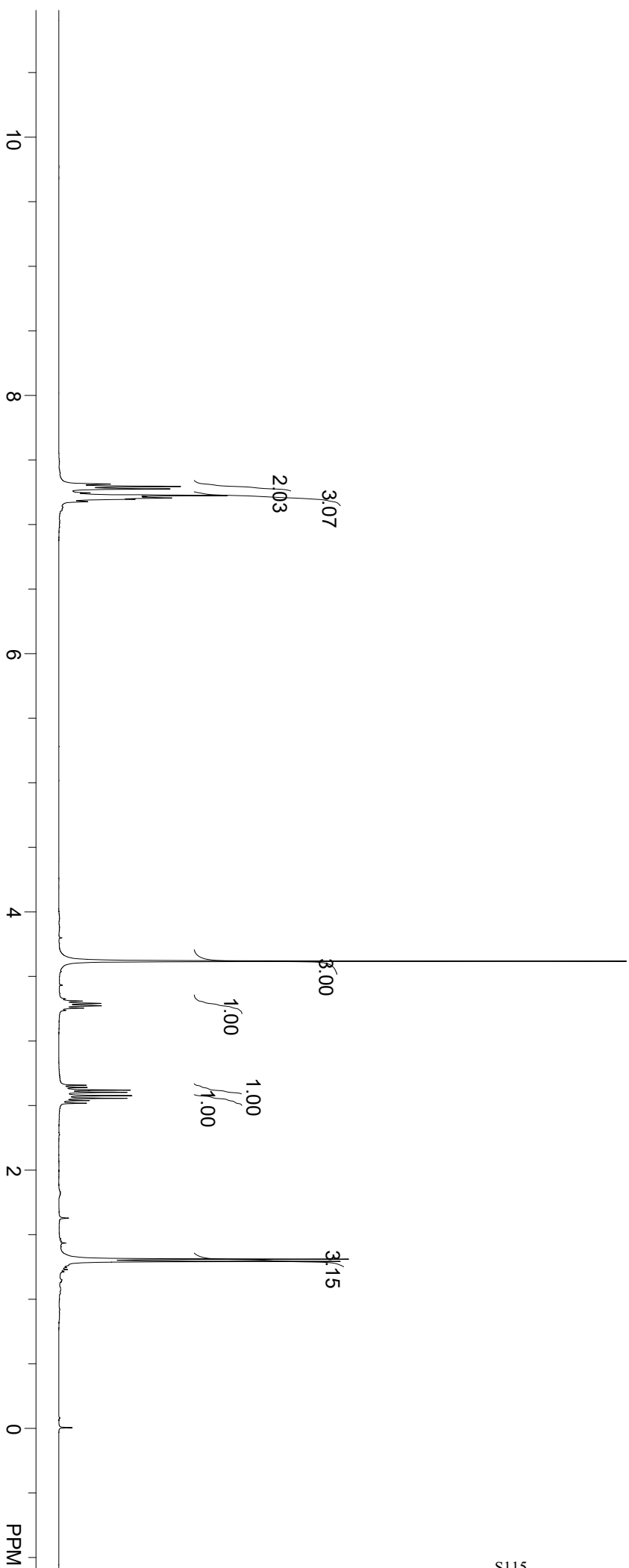
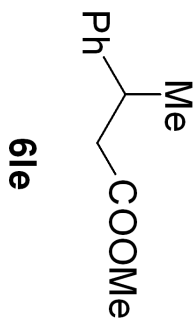


7.313
7.294
7.276
7.245
7.224
7.216
7.206
7.196
7.177

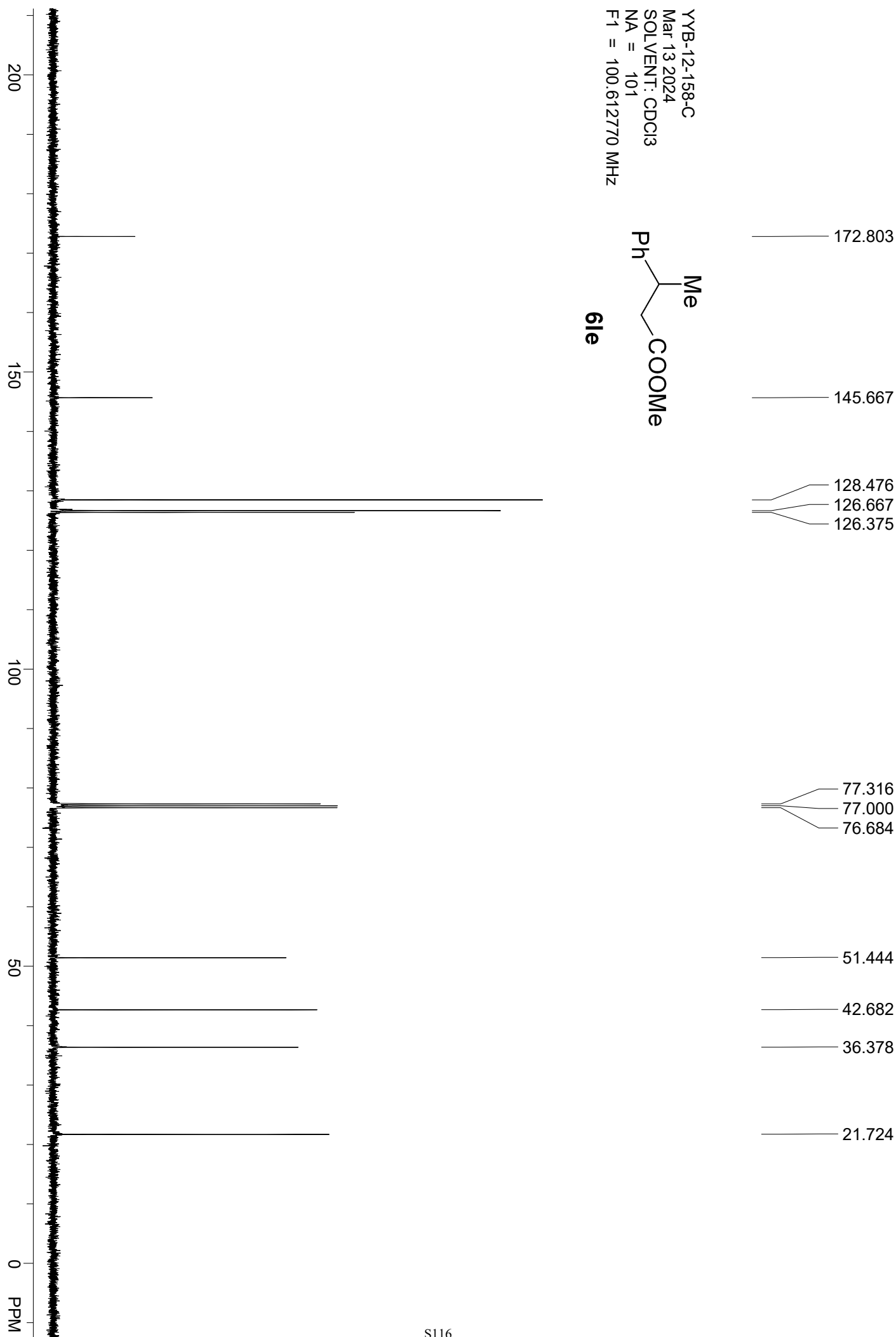
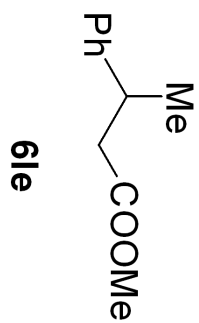
3.324
3.307
3.615
3.289
3.270
3.252
3.234
2.654
2.637
2.616
2.599
2.573
2.552
2.535
2.515
1.306
1.289

-0.000

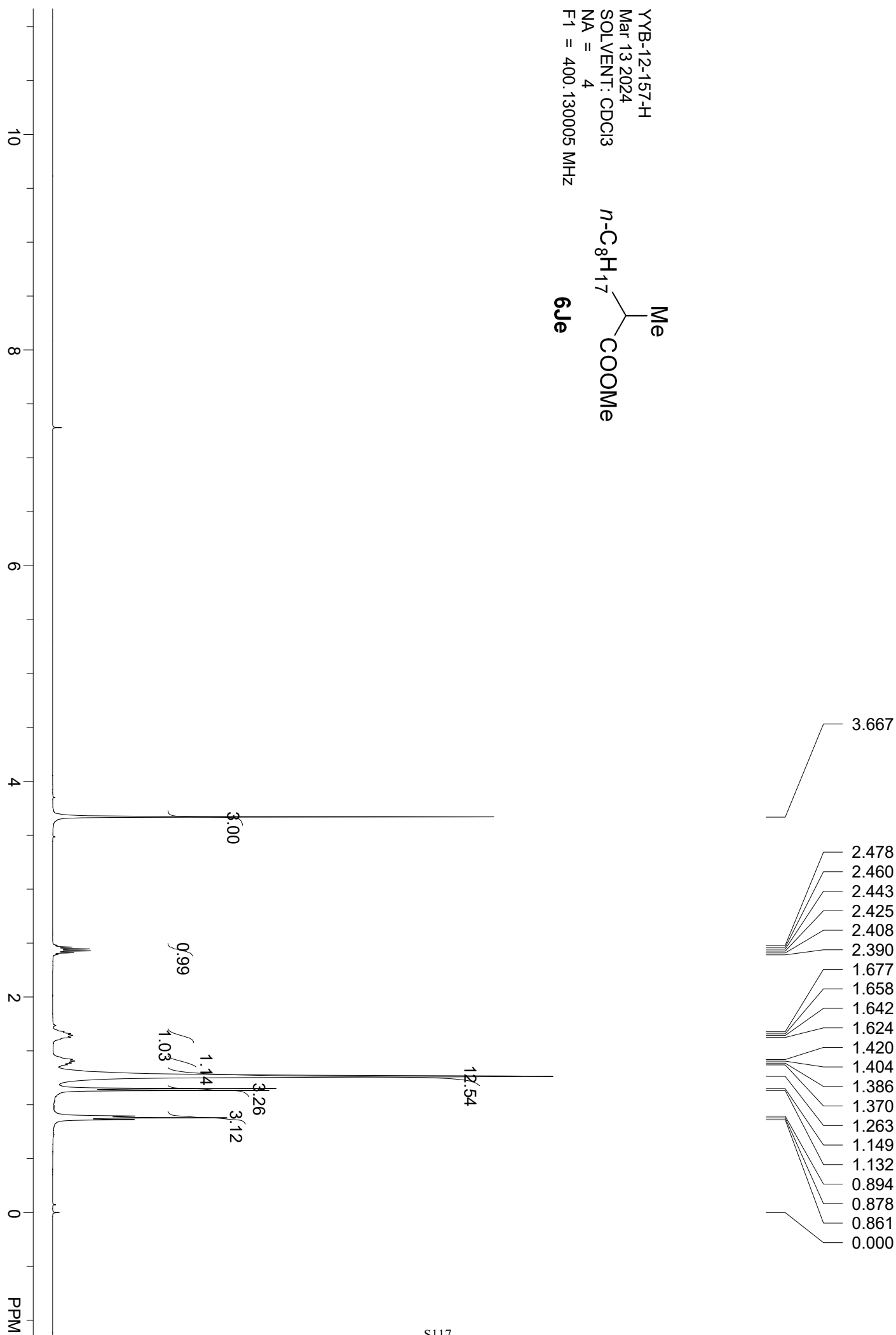
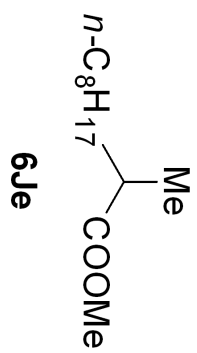
YYB-12-158-H
Mar 13 2024
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



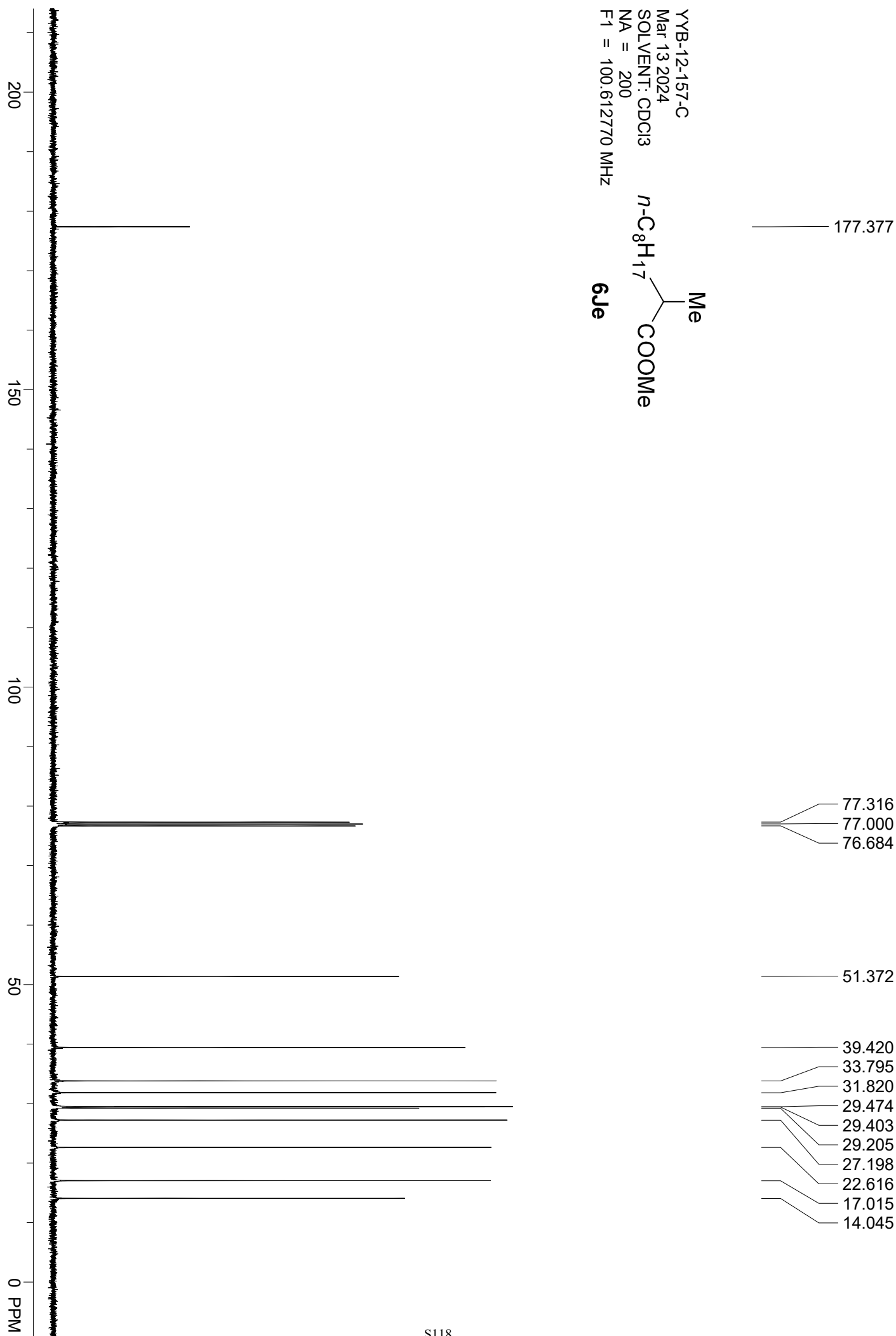
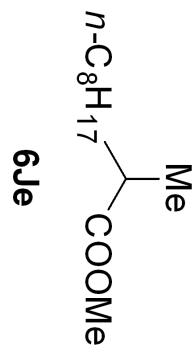
YYB-12-158-C
Mar 13 2024
SOLVENT: CDCl₃
NA = 101
F1 = 100.612770 MHz



YYB-12-157-H
Mar 13 2024
SOLVENT: CDCl₃
NA = 4
F1 = 400.130005 MHz



YYB-12-157-C
 Mar 13 2024
 SOLVENT: CDCl₃
 NA = 200
 F1 = 100.612770 MHz



203.850

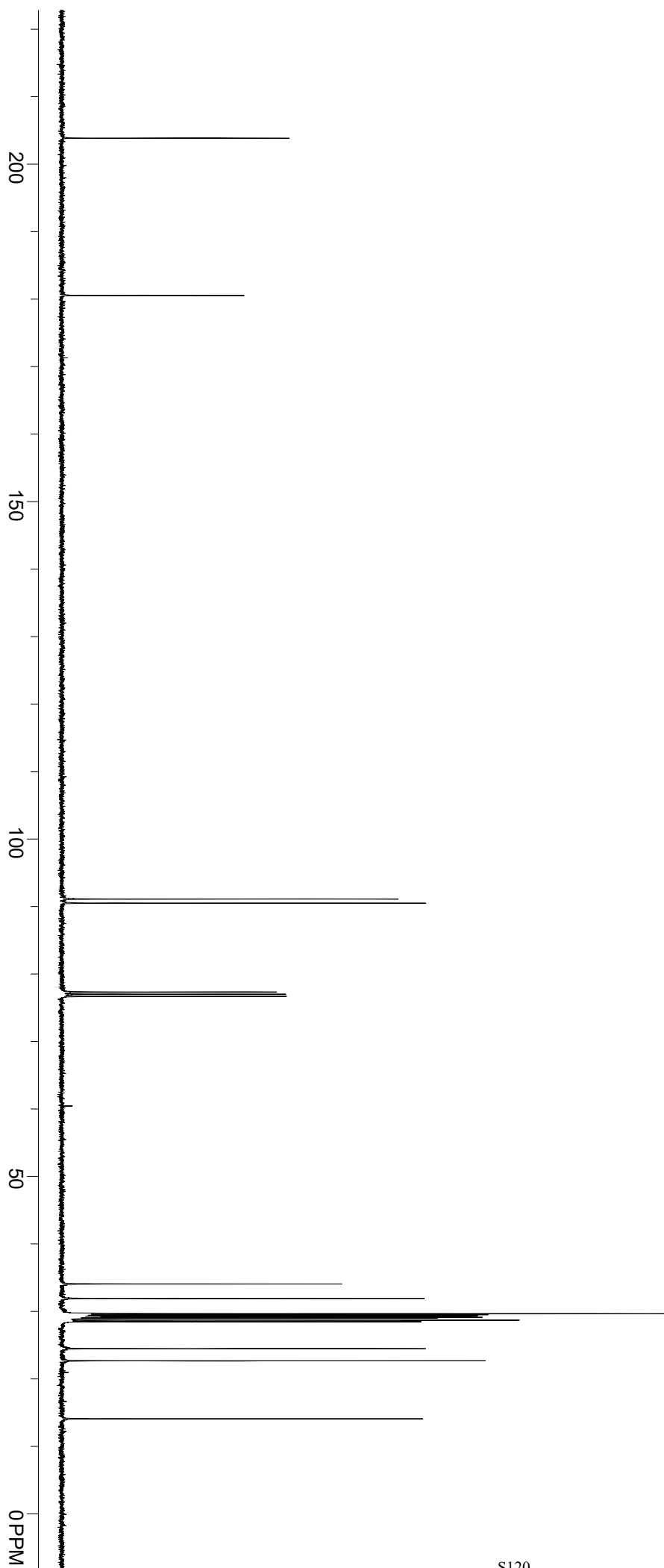
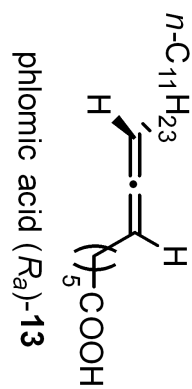
180.537

91.094
90.493

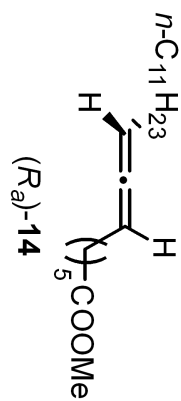
77.316
77.000
76.684

34.056
31.907
29.647
29.489
29.355
29.213
29.110
28.960
28.707
28.684
28.462
24.465
22.672
14.069

YYB-7-160-C
Jan 14 2022
SOLVENT: CDCl3
NA = 150
F1 = 100.612770 MHz

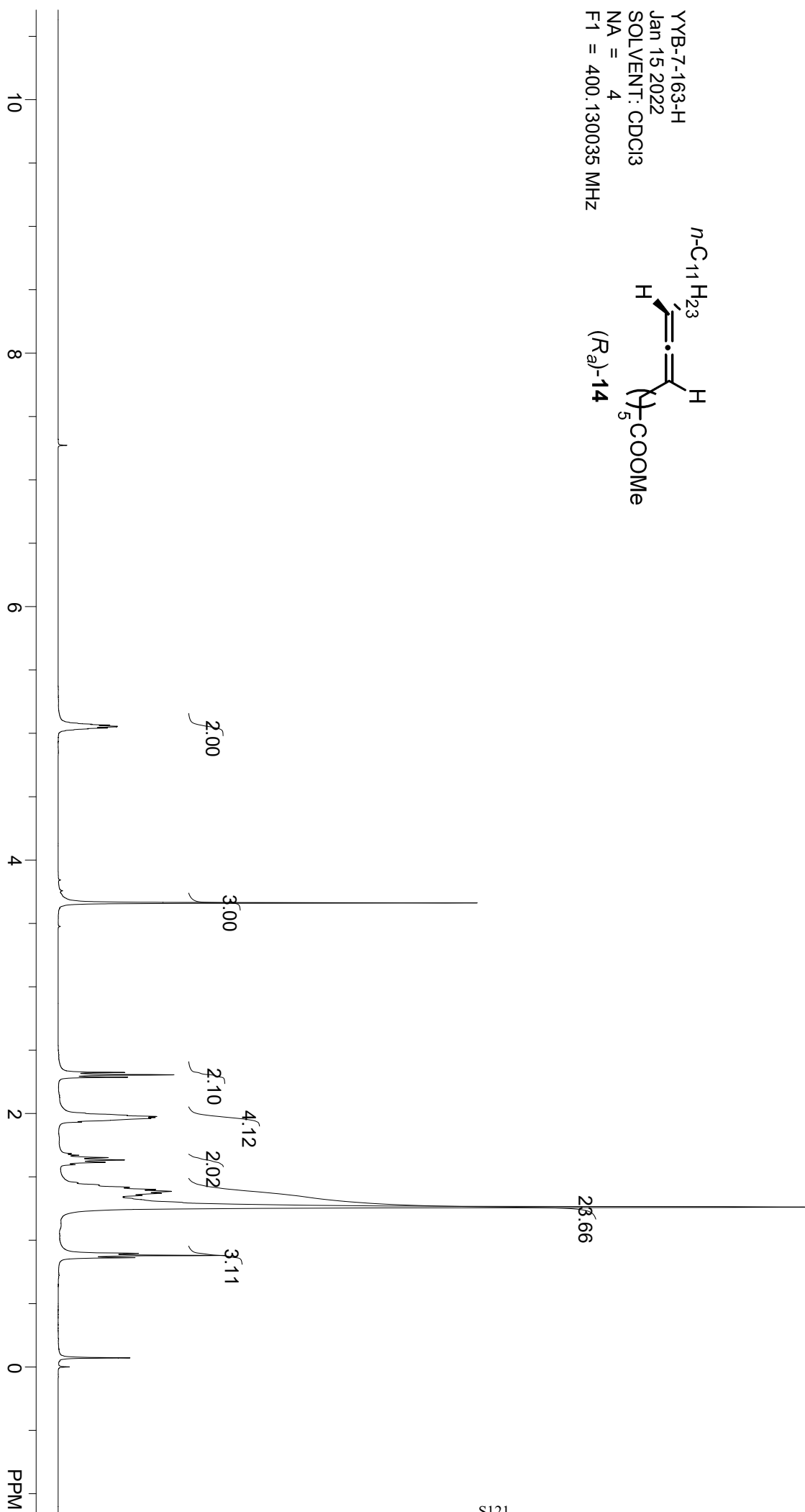


YYB-7-163-H
 Jan 15 2022
 SOLVENT: CDCl3
 NA = 4
 F1 = 400.130035 MHz



5.072
 5.066
 5.057
 5.052
 5.043
 5.036

3.664
 2.325
 2.306
 2.288
 1.987
 1.977
 1.969
 1.962
 1.672
 1.653
 1.634
 1.616
 1.597
 1.437
 1.418
 1.401
 1.386
 1.371
 1.262
 0.896
 0.880
 0.863
 0.000

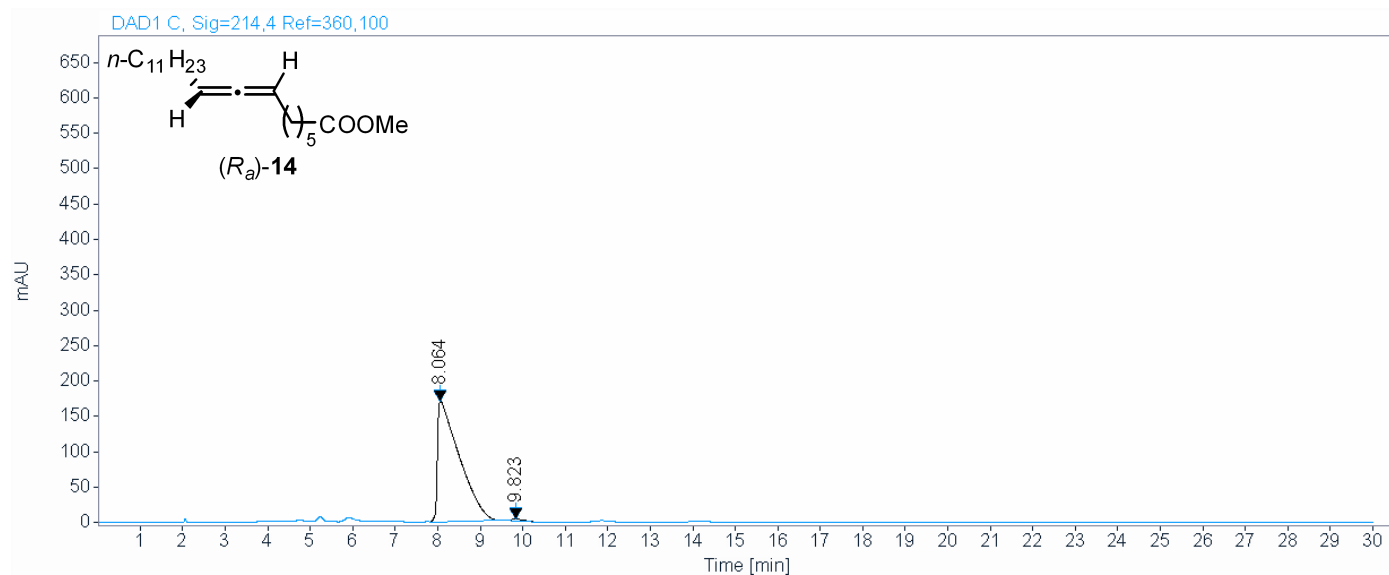


Area Percent Report

sample yyb-7-163-LC-100-0-1.5-214

Data file: C:\Users\Public\Documents\ChemStation\1\Data\zhangq 2022-01-18 08-52-28\016-P2-E2-yyb-7-163.D

Acquisition Data:



Signal: DAD1 C, Sig=214,4 Ref=360,100

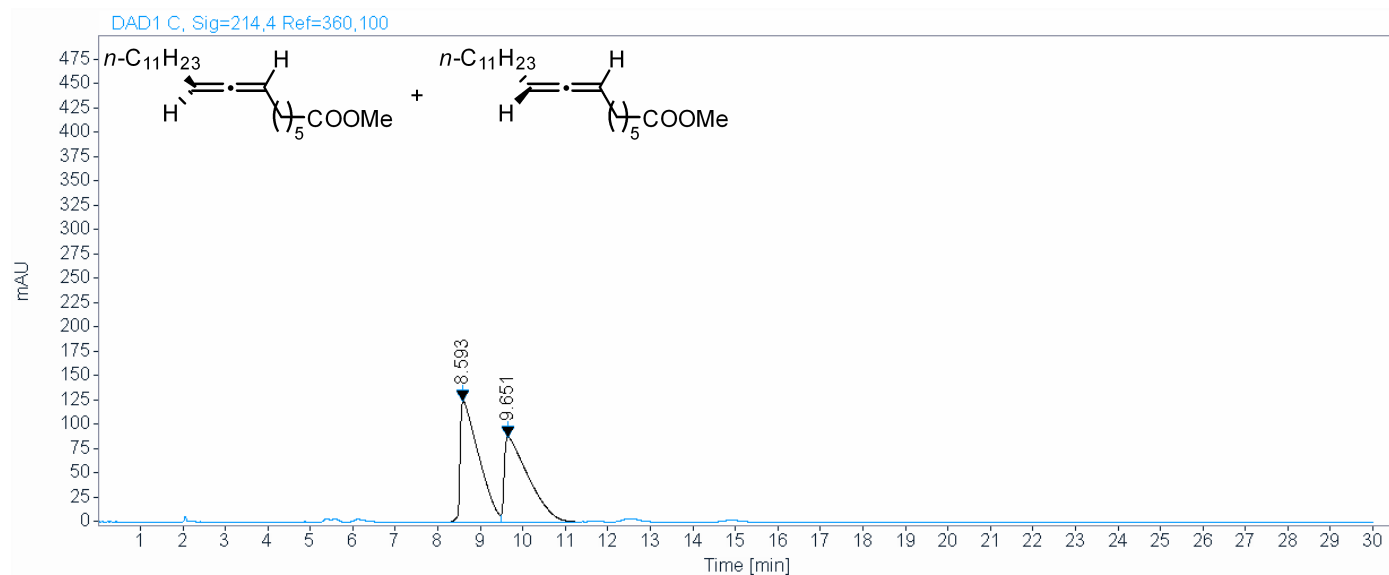
RT [min]	Width [min]	Height	Area	Area%
8.064	0.4733	170.6460	6073.9258	98.7344
9.823	0.3083	3.0251	77.8547	1.2656
Sum			6151.7805	100.0000

Area Percent Report

sample yyb-7-163-rac-LC-100-0-1.5-214

Data file: C:\Users\Public\Documents\ChemStation\1\Data\zhangq 2022-01-18 08-52-28\015-P2-E1-yyb-7-163-rac.D

Acquisition Data:



Signal: DAD1 C, Sig=214,4 Ref=360,100

RT [min]	Width [min]	Height	Area	Area%
8.593	0.4101	124.8337	3788.8767	52.3920
9.651	0.5235	87.4124	3442.9060	47.6080
Sum			7231.7827	100.0000